Continuous-Emission Hidden Markov Models for Equity Returns: Heavy-Tail Emission Families and Regime-Conditional Value-at-Risk

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Abstract

Low-state-count hidden Markov models with Gaussian emissions were long held to fail at reproducing the slow autocorrelation decay of absolute daily equity returns. We revisit that limitation with continuous-emission hidden Markov models that place Gaussian, Student-t, Laplace, and generalized-error emissions under one expectation-maximization framework with quantile-based initialization, plus a spectral identity that writes the absolute-return autocorrelation as finitely many geometric decay modes. The main finding, on SPY and a sector-balanced 30-ticker panel, is that at three states the class is not what limits the fitted autocorrelation: valid three-state chains optimized directly on the autocorrelation track it roughly three times more closely than the published converged multistart likelihood fits do (median error 0.016 against 0.050), converged fits at $K = 18$ do not materially improve the match over three states (better at 7 of 31 tickers in sample), and a weighted-objective sweep finds valid three-state fits whose autocorrelation error is close to the best achieved in the sweep while their distribution-body fit reaches the likelihood fit's level under an in-sample distribution-function criterion, so the autocorrelation and the bulk marginal do not compete at three states under that criterion. What these autocorrelation-competitive fits still lack is the deep tail; whether one three-state chain can carry the tail and the autocorrelation together remains open, and a heavy-tailed emission family substantially improves the fitted marginal itself. The boundary is equally sharp: the fitted models do not reproduce long-horizon persistence (lags beyond roughly three months), and genuinely non-geometric asymptotics still require a different duration law or long-memory latent process. Two applications exercise the fitted model: a regime-conditional Value-at-Risk that is not rejected by joint conditional-coverage tests at the 5% level on held-out data and on most walk-forward folds, and a multi-asset elliptical-copula composition in which a suggestive 2×2 experiment on identical quarter-level targets finds that quarterly refit of the dependence layer reduces out-of-sample dependence error by an order of magnitude more than the Gaussian-versus-Student-t family choice. The evaluation covers daily US equities on a single out-of-sample window containing a vendor feed change; periodic refitting is recommended when regimes drift.

Keywords: Continuous-Emission Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Value-at-Risk, Synthetic-Data Generator.

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1 Introduction

Synthetic generators of equity-return time series produce plausible sample paths that match the statistical structure of real markets without replaying the single realized history. This supports stress testing, regulatory backtesting, and scenario design against conditions absent from the observed record [1–3]. A useful generator should reproduce the three symmetric stylized facts of daily returns targeted here at once: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [4, 5]. Generalised autoregressive conditional heteroskedasticity (GARCH) models [6, 7] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators [8] match the marginal but lose temporal persistence; deep generators [9–11] reproduce some stylized facts at the cost of interpretability and reproducibility. One negative result has shaped two decades of work on regime-switching models. Rydén et al. [12] fitted Gaussian-emission hidden Markov models (HMMs) at low state counts and concluded that the slow decay of the absolute-return ACF was the one stylized fact the model could not generate, "at least not easily by ML estimation": a finite-state chain produces only exponential ACF decay, and the joint fit, forced to match the remaining properties of the data, did not land on a slow one. Bulla and Bulla [13] attributed the failure to the geometric sojourn-time distribution of the standard Markov chain and recovered the ACF fit by replacing the chain with a hidden semi-Markov chain, at the cost of much greater complexity. We separate the low-K Gaussian limitation into two possible failure channels: a temporal one (the absolute growth-rate ACF must allow slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed excess kurtosis). Rydén's own outlier-reduced subseries broke on the temporal channel; on our non-outlier-reduced data, the likelihood fits improve the marginal tail while leaving finite-band ACF capacity unused, the marginal body is jointly attainable with the ACF under the frontier criterion introduced below, and tail-and-ACF joint attainability remains open. A standard closed-form identity for the absolute growth-rate ACF, written as a sum over the non-unit eigenvalues of the transition matrix [14–16], bounds the number of decay modes by the rank of the centred transition matrix. We ask whether that bound actually matters on equity-return data once enough latent states are allowed.

We answer this with an established model class, the continuous-emission hidden Markov model (CHMM): a hidden chain of market regimes in which each regime emits the day's growth rate from its own probability distribution. The regime chain controls the geometric modes available to the absolute growth-rate ACF, while the per-regime distributions control the heavy-tailed marginal, allowing the temporal and distributional constraints of the low-state Gaussian fit to be examined separately. Our contribution is not the CHMM itself. It is a unified comparison of Gaussian, Student-t, Laplace, and generalized-error emissions under the same forward-backward framework; quantile-based initialization; an empirical spectral diagnosis of when the finite-mode bound is active, paired with a realizable capacity experiment that separates what the state budget permits from what the estimation objective selects; and extensions to regime-conditional Value-at-Risk (VaR) and multi-asset copula composition. We start each fit from quantile splits of the observed growth rates, seeding each regime with its own slice of the distribution, which keeps the starting regimes from overlapping and helps the fit converge with every regime still in use. On the fitted model we then build a regime-conditional VaR that forecasts tail risk from the model's running estimate of which regime the day belongs to, and a multi-asset version that couples several single-asset CHMMs into one joint generator while leaving each asset's fitted distribution untouched.

In this study we put the model through four complementary evaluation settings: a six-fold rolling-origin walk-forward on a decade of SPY, a sector-balanced 30-ticker panel across ten Global Industry Classification Standard (GICS) sectors, a Center for Research in Security Prices (CRSP) cross-decade transfer from a ten-year window to a later two-year slice, and a six-asset US-equity basket, which together reduce reliance on a single window or asset universe. With enough latent states and a heavy-tailed emission, the model substantially improved the heavy-tailed marginal fit (narrowing, not closing, the excess kurtosis gap left by the Gaussian baseline), left the raw growth rate serially uncorrelated, and reduced the absolute growth-rate ACF error relative to i.i.d. baselines at lags 1–63 (the long-horizon band beyond roughly three months is not reproduced). Under the diagnostics used here, the rank bound from the ACF identity was not empirically active at the typical ticker, although the result does not establish a general power-law or multi-scale approximation. The regime-conditional VaR was not rejected by the Christoffersen joint conditional-coverage test [17] on the main window or on most walk-forward folds, and the multi-asset composition, a cross-sectional construction that preserves each asset's marginal while modelling contemporaneous dependence, reproduced cross-asset correlations better than the single-index alternative in the tested basket. We therefore interpret the Rydén low-state-count failure more narrowly: on these daily US-equity data and metrics, increasing marginal flexibility resolved more of the observed fit gap than adding ACF modes beyond the selected state count. A realizable capacity experiment sharpens this: valid three-state chains optimized directly on the autocorrelation track it about three times more closely than the likelihood fits do, and a frontier sweep finds valid fits that carry the autocorrelation and the marginal body simultaneously under an in-sample distribution-body criterion; the unresolved gap is the deep tail. An i.i.d. bootstrap beats the CHMM on the raw single-window marginal fit, and a multistart local-EM explicit-duration hidden semi-Markov model (HSMM) leads the panel on the volatility-clustering axis at its declared duration-truncation specification, at a weaker marginal, but the fitted CHMM additionally supports the regime-conditional risk forecast and multi-asset coupling exercised here. We make no formal privacy guarantee for its synthetic series. The scope is daily US equities: a non-equity stress test on the GLD exchange-traded fund (ETF) broke down under static fitting, and we recommend periodic refit within that scope.

2 Related Work

The discrete-state regime-switching tradition opens with Hamilton [18], who introduced Markov-switching to economics, and Schaller and Van Norden [19], who estimated regime-switching specifications on US monthly stock returns and documented the predictive content of the latent regime for return forecasting. Continuous-emission HMMs and non-Gaussian state densities are established parts of this tradition; the present paper contributes a common estimation and evaluation framework rather than a new model class. The main evaluation against the stylized facts later codified by Cont [5] comes from Rydén et al. [12], who fitted Gaussian-emission hidden Markov models at $K = 2$ or 3 zero-mean states to subseries of the 1928 to 1991 S&P 500 index and concluded that the slow decay of the absolute-return ACF was the one stylized fact the model could not generate, for the reasons summarised in the Introduction. Bulla and Bulla [13] recovered the fit with hidden semi-Markov models, replacing the geometric sojourn with an explicit-duration family at the cost of a duration-augmented forward-backward, and Nystrup et al. [20] extended the same line to long-memory volatility. Each of these works makes the same kind of structural choice we make here: the latent regime is a finite-state object with per-state emission densities, and what varies across the lineage is the sojourn distribution and the per-state family. The emission families that supply the per-state densities used in this work draw on the maximum-likelihood Student-t formulations of Peel and McLachlan [21] and Liu and Rubin [22], with the Generalised Error Distribution (GED) [23, 24] providing a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degree-of-freedom parameter ν_k plays in

CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [3], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism. The present work replaces that discretisation step with four continuous per-state emission families under a single estimation framework and tests empirically whether the finite collection of eigenvalue-driven ACF modes is adequate on the studied data.

Conditional-variance and stochastic-volatility models form a parallel line in which the latent state is the volatility process rather than a discrete regime. The basic Gaussian GARCH framework [6, 7] encodes volatility persistence through a parsimonious variance recursion; in our panel the tested GARCH(1,1) specification had a lower Kolmogorov-Smirnov pass rate than the CHMM variants (Table 2). Heavy-tailed innovations [25], asymmetric leverage variants (exponential GARCH, EGARCH, 26; Glosten-Jagannathan-Runkle GARCH, GJR-GARCH, 27; threshold GARCH, 28), regime-switching extensions (Markov-switching GARCH, MS-GARCH, 29, with the `MSGARCH` R package of 30 the reference implementation), and realised-variance frameworks [31, 32] add tail, leverage, regime, or multi-scale structure at the variance-process layer. The continuous-latent counterpart is the stochastic-volatility family of Taylor [33], which evolves log-volatility as a latent AR(1) and proceeds by simulation-based estimators [34, 35], hidden-Markov approximations on a discretised volatility grid [36, 37], or Markov-switching stochastic volatility [38] that interpolates between the continuous-volatility and finite-state formulations. These model classes can produce heavy-tailed predictive distributions and conditional VaR forecasts; unimodality alone does not impose an excess kurtosis ceiling. The distinction relevant to this study is narrower: the tested CHMM assigns a separate location, scale, and potentially shape parameter to each finite state, whereas the tested conditional-volatility baselines place most regime dependence in the variance recursion. The resulting empirical comparison separated those specifications on marginal fit and excess kurtosis (Table 2; extended baseline panel in the appendix), without establishing a structural impossibility for the broader GARCH or stochastic-volatility families.

Deep generative market models offer high-capacity distributional and temporal representations, often with less direct state-level interpretation. GAN-based generators require temporal structure in the architecture or training objective to reproduce volatility clustering [39, 40]; prominent examples include the TimeGAN family of Yoon et al. [9] and the convolutional Wasserstein GAN QuantGAN of Wiese et al. [10]. Path-signature generators [41–43] and score-based or diffusion approaches [11, 44, 45] provide alternative mechanisms for representing time-series dependence. Some models in these broad families can be conditional, can support likelihood or score-based prediction, or can be combined with multivariate dependence models; these capabilities are architecture-specific rather than absent from deep generators in general. We included one in-house QuantGAN implementation as the deep-generative row in the main generator panel, read as a negative control on whether that architecture recovered the same stylized facts at our sample size rather than as a representative forecasting baseline for the entire family. In that comparison, the CHMM's explicit finite-state semantics and closed-form state-conditional mixture made the regime-conditional VaR and copula composition used in this paper directly available, while the evaluated QuantGAN was not designed to provide those outputs. Reports of tail mode collapse in GAN-based financial generators [39, 40] motivate the heavy-tail diagnostic, but do not establish that failure for every deep generative formulation.

Cross-asset synthesis, synthetic-data evaluation, and non-stationarity handling round out the literature this paper draws on. The Single Index Model (SIM) [46] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal, and Sklar's theorem [47, 48] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [49, 50] the usual family choices and rank-reordering [51] the simulation step that preserves marginals exactly. Dynamic-conditional-correlation models [52] and composite-likelihood estimators for vast covariance matrices [53] are the standard alternatives at higher dimensions, and our six-asset construction sits well below the dimensions where those estimators become essential while the rank-based copula architecture composes with either at scale. Synthetic-data evaluation in this work is anchored on the framework of Stenger et al. [54], the proper-scoring-rule literature [55–57], kernel two-sample tests [58], and signature-based metrics [41, 59], which together fix the comparison metrics that the main panel uses: marginal goodness-of-fit, autocorrelation fidelity, excess kurtosis, proper scores, and signature distance. The non-stationarity literature documents structural breaks in single-name equity returns [60–62] and responds with window selection [63] and online or change-point updating [64–66]; the walk-forward evaluation and the periodic-refit recommendation in this work draw directly on that line.

3 Methods

3.1 Data and model

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024, a ten-year in-sample (IS hereafter) window of length $T_{\text{IS}} = 2{,}516$ trading days. The period from January 4, 2024 through April 20, 2026 is held out for out-of-sample (OoS hereafter) evaluation. Differencing the held-out prices among themselves gives $T_{\text{OoS}} = 572$ growth rates, the first dated January 5, 2024; the generator diagnostics use this 572-observation vector. The VaR back-tests additionally prepend the boundary growth rate from the last in-sample session into January 4, 2024, so their forecast count is 573 and the first forecast target is the return into the first held-out session. Prices are split-adjusted but not dividend-adjusted; the data sources are described in the Data Availability Statement. The price series is stitched from two vendors whose raw files are date-disjoint: Polygon.io consolidated aggregates (via Massive) end on December 31, 2024, and an Alpaca Markets IEX-feed extension begins on January 3, 2025, so the feed switch falls inside the OoS window and no same-date cross-vendor sample exists from which a genuine overlap check could be run. A descriptive pre/post-switch comparison of the two OoS segments is reported in Appendix A.5; because the segments cover different calendar periods, feed effects and genuine time variation cannot be separated there, and the change from consolidated to IEX-only aggregates is a scope limitation of the OoS results. For ticker i and trading day j, with $P_{i,j}$ the session volume-weighted average price (VWAP), $\Delta t = 1/252$ the annualised trading-day step, and r_f the risk-free rate (set to zero here), the annualised excess growth rate is

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f. \tag{1}$$

We use daily data and the matching annualised growth rates. Since Δt is fixed, the growth rate is a constant positive multiple of the day's log return. The excess kurtosis and the autocorrelations that define the stylized facts are therefore invariant to the rescaling and take the same values on returns and on growth rates; the absolute growth-rate ACF measures the same slow-decay stylized fact the literature reports for absolute returns. The probabilistic framework does not depend on the time step, although application at other frequencies requires frequency-specific preprocessing, state calibration, and evaluation horizons. Two reading notes attach to this definition. First, because prices are split- but not dividend-adjusted and r_f is set to zero, G_t is an annualised log *price*-growth rate: it omits the dividend component of total return (ex-dividend price drops enter as negative growth), and the "excess" qualifier is nominal at $r_f = 0$. Shape and dependence diagnostics are insensitive to the small daily dividend increments, but levels are not, and a total-return (dividend-adjusted) sensitivity, which would matter most for the high-dividend names of the sector panel, is left to future work. Second, location- and scale-dependent quantities (per-state means and scales, VaR, ES) are reported on the annualised scale throughout; the corresponding daily-unit value is the reported value divided by 252 (e.g. a reported median 1% VaR of -4.56 corresponds to a daily log return of about -1.8%).

The model is fit one ticker at a time. We write G_t for the daily excess growth rate on trading day t for the ticker being fit (equation (1) with the ticker index dropped and the day index renamed from j to t). The sequence $\{G_t\}$ is the observed output of a continuous hidden Markov model with K latent states. We write the model as $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{F}, \boldsymbol{\pi})$ [67], where $\mathcal{S} = \{1, \ldots, K\}$ is the finite state space and $s_t \in \mathcal{S}$ is the (unobserved) latent state at time t. The transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$ has its (i, j) entry equal to the probability of moving from state i at time t to state j at time $t + 1$, $T_{ij} = \mathbb{P}(s_{t+1} = j \mid s_t = i)$; the rows therefore sum to one (each row is a valid probability distribution over next states). The initial-state distribution $\boldsymbol{\pi} \in \mathbb{R}^K$ has kth entry $\pi_k = \mathbb{P}(s_1 = k)$, the probability that the chain starts in state k. The per-state emission densities are collected in $\mathbf{F} = \{f_k(\,\cdot\,; \boldsymbol{\theta}_k)\}_{k=1}^K$, where $f_k(x; \boldsymbol{\theta}_k)$ is the density of the day's growth rate G_t when the chain is in state k, evaluated at $G_t = x$; the parameter vector $\boldsymbol{\theta}_k$ holds that state's location, scale, and any shape parameters. The per-state cumulative distribution function (CDF) is $F_k(x; \boldsymbol{\theta}_k) = \int_{-\infty}^{x} f_k(u; \boldsymbol{\theta}_k)\, du$, used in the regime-conditional VaR construction.[1]

Over many days the chain settles into a stable long-run distribution over its states. Write $\bar{\boldsymbol{\pi}} = (\bar{\pi}_1, \ldots, \bar{\pi}_K) \in \mathbb{R}^K$ for it, the one state distribution that a transition step leaves unchanged ($\bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}$), so $\bar{\pi}_k$ is the long-run fraction of days the chain spends in state k. Collect every growth rate the model produces and ignore which state generated each one; the density of that pooled set of growth rates is a weighted average of the K per-state densities, each weighted by how often its state is visited,

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k\, f_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}. \tag{2}$$

We call $f(x)$ the marginal mixture: *marginal* because it averages over the hidden state to describe one day's growth rate on its own, a *mixture* because it is a weighted sum of the per-state densities. A unique $\bar{\boldsymbol{\pi}}$ exists when $\mathbf{T}$ is irreducible (every state can eventually be reached from every other) and aperiodic (returns to a state are not locked to a fixed period); the formal conditions are given in Assumption 1. We learn $\mathbf{T}$, the per-state parameter vectors $\boldsymbol{\theta}_{1:K} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K)$, and $\boldsymbol{\pi}$ jointly by expectation-maximisation (EM; Figure 1). Unconditional simulation initialises the chain from the stationary distribution $\bar{\boldsymbol{\pi}}$ rather than from the fitted $\boldsymbol{\pi}$, which removes transient dependence on the first observed day. The CHMM-N variant uses Gaussian emissions, $f_k = \mathcal{N}(\mu_k, \sigma_k^2)$ with $\sigma_k > 0$. The CHMM-t variant uses Student-t, $f_k = t_{\nu_k}(\mu_k, \sigma_k)$ with $\sigma_k > 0$ and $\nu_k \in [\nu_{\min}, \nu_{\max}]$; an optional exponential penalty on $1/\nu_k$ is used only in the $\lambda = 20$ sensitivity specification. The CHMM-L variant uses Laplace, $f_k = \text{Laplace}(\mu_k, b_k)$ with $b_k > 0$. The CHMM-GED variant uses the generalised error distribution, $f_k = \text{GED}(\mu_k, \alpha_k, p_k)$ with $\alpha_k > 0$ and $p_k \in [p_{\min}, p_{\max}]$; the Gaussian and Laplace variants are nested special cases of CHMM-GED, recovered at the interior values $p_k = 2$ and $p_k = 1$ respectively.

[1] "Conditional" here means conditioning on the latent regime: the VaR is a quantile of the one-step-ahead state-mixture predictive and moves as the regime belief updates. It is distinct from the conditional Value-at-Risk (CVaR), equivalently the expected shortfall (ES), which is a tail average beyond that quantile and which we report only unconditionally (Table S16), and from the Christoffersen conditional-coverage property, which is a backtest criterion.

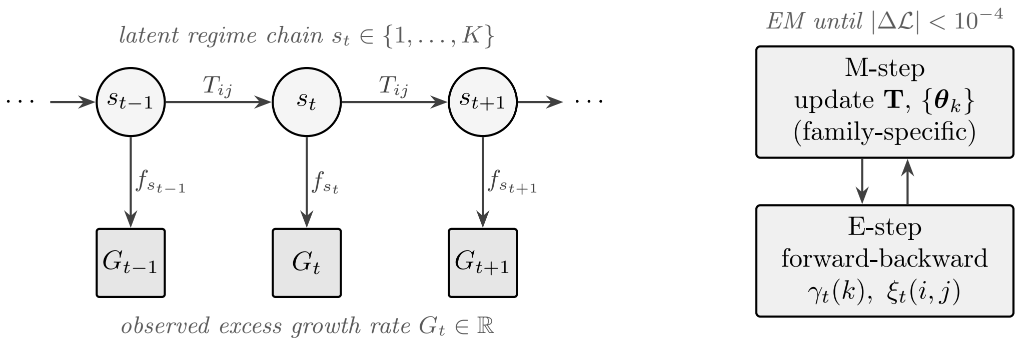

Figure 1. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t emits the observed excess growth rate G_t through a state-specific density f_{s_t}. The four CHMM variants share everything except the per-state emission family (Gaussian, Student-t, Laplace, GED). The EM procedure alternates a log-space forward-backward E-step and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

3.2 Estimation by EM

With the model specified, we estimate the parameters $(\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi})$ by EM; the forward-backward recursions, and likewise the one-step-ahead forward filters of the risk heads, evaluate emission log-densities and normalise by logsumexp in log-space for numerical stability, with a prior fallback retained in the filters as a safeguard against fully underflowed rows. From here on we write $O_t \equiv G_t$ for the observation at time t (O is the standard symbol for an HMM observation; here it just renames G_t), $O_{1:T} = (O_1, O_2, \ldots, O_T)$ for the full observed sequence over the fitting window of length T, and $\gamma_t(k) = \mathbb{P}(s_t = k \mid O_{1:T})$ for the smoothed posterior probability that the chain was in state k at time t, given the entire observed sequence and the current parameter iterate ("smoothed" means computed using both past and future observations, not just past). The pair posterior $\xi_t(i,j) = \mathbb{P}(s_t = i, s_{t+1} = j \mid O_{1:T})$ is the smoothed probability of an $i \to j$ transition at time t, used in the transition update. The collection $\{\gamma_t(k)\}_{t=1,\ldots,T;\, k=1,\ldots,K}$ is computed at every E-step by the standard forward-backward recursion (one forward pass and one backward pass through the data). We also use the superscript (n) to mark the value of any quantity at EM iteration n, so (0) is the initial value. All four emission families share the same log-space forward-backward recursions [68] and the same quantile-based initialisation: observations are sorted, split into K equal-sized chunks, and state k's initial location and scale are seeded from chunk k, with the transition matrix and initial distribution started uniform, $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. The initial distribution is treated differently across families: the Gaussian fitter holds $\boldsymbol{\pi}$ fixed at its uniform initial value throughout (and $\boldsymbol{\pi}$ is accordingly excluded from CHMM-N parameter counts), while the Student-t, Laplace, and GED fitters update $\pi_k \leftarrow \gamma_1(k)$ at each iteration; on a single long training sequence the initial distribution is weakly identified either way, entering the likelihood through the first observation only. The EM loop runs until the observed-data log-likelihood $\mathcal{L}^{(n)} = \log p(O_{1:T} \mid \mathbf{T}^{(n)}, \boldsymbol{\theta}_{1:K}^{(n)}, \boldsymbol{\pi}^{(n)})$, the likelihood of the growth rates alone with the latent states summed out, stops changing meaningfully between iterations, $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < \epsilon$ with $\epsilon = 10^{-4}$, or a maximum iteration cap is reached; the iteration cap, bracket widths, one-dimensional-search settings, and the full forward-backward recursion are reported in the algorithmic appendix. Quantile-based initialisation is chosen to reduce overlap among starting state components, with the aim of improving convergence to non-degenerate $K \geq 3$ fits. Each EM iteration costs $\mathcal{O}(K^2 T)$ time for the forward-backward recursions and the transition update, the standard cost of HMM inference [67], plus the per-family emission updates: closed form for Gaussian and Laplace, bracketed one-dimensional searches for Student-t and GED.

The four emission families share the latent-state forward-backward recursion, the transition update, and the quantile-based initialisation; they differ in the emission density evaluated at each E-step and in the M-step update for the per-state parameter vector $\boldsymbol{\theta}_k$. Given the smoothed posteriors $\gamma_t(k)$, each family applies its own parameter update. For CHMM-N, the per-state parameters are the Gaussian mean and variance, $\boldsymbol{\theta}_k = (\mu_k, \sigma_k^2)$, and the classical Baum-Welch M-step [69] updates them in closed form using weighted sample-mean and sample-variance formulas, where each observation is weighted by the posterior probability $\gamma_t(k)$ that state k generated it (all sums $\sum_t$ below run over $t = 1, \ldots, T$):

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k)\, (O_t - \mu_k)^2}{\sum_t \gamma_t(k)}.$$

For CHMM-t, the per-state parameters are the Student-t location, scale, and degrees of freedom, $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$. We use the expectation conditional maximisation (ECM) construction of Peel and McLachlan [21] and Liu and Rubin [22], which replaces the single M-step with a sequence of conditional maximisation (CM) sub-updates that each move one block of parameters at a time. The location and scale use the augmented-data closed-form CM updates, while the degrees of freedom use a bracketed one-dimensional numerical update on a posterior-weighted marginal objective, adapted from the expectation/conditional-maximisation-either (ECME) variant of Liu and Rubin [22]; this is the same heavy-tailed innovation choice used by Abanto-Valle et al. [37] for stochastic-volatility-in-mean models. The construction uses a standard representation of the Student-t as a Gaussian with random variance: conditional on a latent positive scalar u, the observation is Gaussian, $G_t \mid s_t = k,\, u \sim \mathcal{N}(\mu_k, \sigma_k^2/u)$, and u itself is drawn from a Gamma distribution, $u \sim \Gamma(\nu_k/2,\, \nu_k/2)$. Averaging over u recovers the Student-t. In ECM, u is treated as an additional latent variable, and the E-step adds its conditional expectation under the current parameter iterate,

$$u_{t,k} \equiv \mathbb{E}[u \mid O_t, s_t = k] = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{3}$$

and the M-step then updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{4}$$

for the location and scale, both in closed form given ν_k. The degrees of freedom ν_k are then updated by a finite one-dimensional search on the bounded interval $[\nu_{\min}, \nu_{\max}]$ for a high-value point of the state-k objective $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$, the posterior-weighted marginal Student-t log-likelihood at the current smoothed posteriors, used in place of the augmented-data CM update in $\mathbb{E}[\log u]$. In the single-component i.i.d. setting of Liu and Rubin [22], where all $\gamma_t(k) = 1$, this objective is the observed-data likelihood and its direct maximisation is the ECME step; already for i.i.d. mixtures the posterior-weighted objective differs from the observed-data likelihood, with Peel and McLachlan [21] noting that extending ECME to mixtures is not straightforward, and in the HMM the observed-data likelihood additionally depends on ν_k through the forward recursion. With γ held fixed the update is therefore a hybrid generalised block-coordinate step on a surrogate objective rather than an ECME observed-likelihood maximisation, and monotone observed-data ascent is not guaranteed by the ECM theory [70, 71]. We therefore assess numerical convergence from the observed-data log-likelihood trace and stop at the stated tolerance or iteration cap; every parameter iterate is scored on the observed-data likelihood before the next update can mutate it, and the fitter returns the best evaluated finite iterate (also restored if an update degenerates to a non-finite likelihood), so the reported final likelihood, the smoothed posteriors, and the returned parameters always correspond to one another.

For CHMM-L, the per-state parameters are the Laplace location and scale, $\boldsymbol{\theta}_k = (\mu_k, b_k)$, and both have closed-form updates. The location μ_k is a weighted median of the observations $\{O_t\}_{t=1}^{T}$, where each observation is weighted by the posterior probability $\gamma_t(k)$ that state k generated it. The scale $b_k \leftarrow \sum_t \gamma_t(k)\,|O_t - \mu_k| / \sum_t \gamma_t(k)$ is the weighted mean absolute deviation of the observations from μ_k, using the same weights.

For CHMM-GED, the per-state parameters are the GED location, scale, and shape, $\boldsymbol{\theta}_k = (\mu_k, \alpha_k, p_k)$, with conditional density $f_k(x; \mu_k, \alpha_k, p_k) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$. The three parameters cannot all be updated together in closed form, so the M-step uses three sequential block updates. The location update searches a bounded interval for a low-value point of $\sum_t \gamma_t(k)\,|O_t - \mu|^{p_k}$. This objective is convex for $p_k \geq 1$ but need not be convex for $p_k < 1$, so the finite search is a numerical update rather than a certified global conditional minimisation. The scale then has a closed-form update at the just-updated location, $\alpha_k \leftarrow \left[(p_k/W_k) \sum_t \gamma_t(k)\,|O_t - \mu_k|^{p_k}\right]^{1/p_k}$, where $W_k = \sum_t \gamma_t(k)$ is the total posterior weight on state k. The shape is updated last by a bounded one-dimensional search over $[p_{\min}, p_{\max}]$ at the just-updated location and scale. We therefore describe CHMM-GED as an ECM-style numerical block-coordinate procedure and diagnose convergence from its observed-data log-likelihood trace; the exact ECM monotonicity theorem applies only when every block is conditionally maximised.

3.3 Evaluation

With the model fitted, we evaluate the CHMM at two levels, single-asset and multi-asset, across four datasets: the SPY in-sample and out-of-sample windows, a sector-balanced 30-ticker panel spanning ten GICS sectors with three large-cap representatives each, a CRSP cross-decade transfer between a 1994 to 2004 in-sample window and a 2004 to 2006 out-of-sample slice, and a six-asset US-equity basket (SPY, NVDA, JNJ, JPM, AAPL, QQQ). The *single-asset* evaluation fits one CHMM per ticker and covers the stylized-fact diagnostics, K selection, main generator comparison, cross-ticker generalisation panel, CRSP cross-decade transfer, walk-forward stress test, and Value-at-Risk panel. The *multi-asset* evaluation keeps each asset's fitted CHMM marginal fixed and models only how the assets move together. A copula makes this split possible: Sklar's theorem [47] says any joint distribution factors into its one-asset marginals and a copula that carries the dependence between them, so the marginals and the dependence can be fitted separately. We use a Student-t copula [49, 50] on the six-asset basket, with its correlation matrix fixed from Kendall's τ via $\rho = \sin(\pi\tau/2)$; its degrees of freedom $\nu^\star$ are selected by a grid search over the profile likelihood. To draw joint samples we use the rank-reordering scheme of Iman and Conover [51], which reorders the per-asset CHMM draws into the copula's dependence pattern without altering any asset's own values, so the reordering preserves each asset's simulated marginal exactly. The reordering has a cost that we state up front: because the copula sample is drawn independently across time, assigning each asset's sorted CHMM values to the copula sample's time ranks permutes that asset's simulated path in time, so the composed paths carry the CHMM marginals and the contemporaneous cross-asset dependence but not the per-asset serial dependence (volatility clustering, regime residence) of the underlying CHMM paths. The multi-asset construction is therefore a cross-sectional distribution and dependence experiment, evaluated on per-asset marginal fit and cross-asset correlation reproduction, not a multivariate time-series generator; temporal diagnostics apply to the single-asset generator only. A single-index model (SIM), a Gaussian copula on the same CHMM marginals, and a truncated C-vine are reported in the appendix as multi-asset comparators.

The single-asset CHMM is benchmarked against one representative from each model class, all fit on the same IS window: a plain i.i.d. empirical bootstrap [8] as the strongest non-parametric baseline; Gaussian and Laplace i.i.d. generators as marginal baselines; GARCH(1,1) with Gaussian [7] and

Student-t [25] innovations as conditional-variance baselines; and Markov-switching GARCH (MS-GARCH) [29] as the closest regime-switching comparator. Extended GARCH-family, semi-Markov, stochastic-volatility, multifractal, and jump-diffusion baselines are reported in the appendix; a deep-generative QuantGAN row [10] appears in the main comparison table as a negative control, with its construction detailed in the appendix. For every model in the panel we simulate P independent paths (reported as N_{paths} in the result tables) matching the window being scored: T_{IS} in sample and $T_{\text{OoS}} = 572$ out of sample [54]; the path count and the deterministic seed used in each run are reported in the corresponding result table. We score the panel on three primary metrics. The two-sample Kolmogorov-Smirnov (KS) pass rate at significance level $\alpha = 0.05$ [72, 73] counts the fraction of simulated paths whose marginal distribution is not rejected against the observed window, measuring distributional fidelity at the marginal level. The asymptotic two-sample KS null assumes i.i.d. observations, whereas the observed and simulated paths carry serial dependence, so we read this pass rate as a descriptive marginal-fidelity score rather than a calibrated test and recalibrate it with the stationary block bootstrap in the appendix. Mean simulated excess kurtosis, read against the observed excess kurtosis on the same window, measures how well the heavy-tailed marginal is reproduced. The mean absolute error (MAE) between the observed and simulated autocorrelation functions (ACF) of absolute growth rates $|G_t|$ over $L = 252$ lags (one trading year), computed per simulated path and averaged over the P paths, is

$$\text{ACF-MAE} = \frac{1}{P}\sum_{p=1}^{P}\frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}_{|G|}^{\text{obs}}(\tau) - \hat{\rho}_{|G|}^{\text{sim},(p)}(\tau)\right|, \tag{5}$$

where τ is the lag index, $|G| = \{|G_t|\}_{t=1}^{T}$ is the absolute growth-rate sequence, $\hat{\rho}_{|G|}^{\text{obs}}(\tau)$ is the sample autocorrelation of $|G|$ at lag τ on the observed window, and $\hat{\rho}_{|G|}^{\text{sim},(p)}(\tau)$ is the same quantity on simulated path p; this captures volatility clustering, the third Cont stylized fact [5]. The Value-at-Risk panel additionally reports the unconditional Kupiec coverage test [74], the Christoffersen joint conditional-coverage test [17] as our primary VaR backtest, and an expected-shortfall (ES) envelope (the $[5\%, 95\%]$ band of simulated ES across paths). Auxiliary distributional metrics confirm robustness of the cross-generator ranking: the block-bootstrap KS recalibration and Anderson-Darling (AD) panels are reported in the appendix, the continuous ranked probability score (CRPS) is reported in the main text with its estimator defined in the appendix, and the Wasserstein-1 (W_1) and Hellinger panels are companion artefacts in the code repository. The models and evaluation pipeline were implemented in Julia in the companion repository `CHMM-Model`.

Regime-conditional VaR construction. The regime-conditional VaR back-tested in the Results is defined explicitly as follows. With parameters $(\mathbf{T}, \boldsymbol{\theta}_{1:K})$ held fixed at their in-sample estimates, the one-step forward filter produces the filtered state distribution $\boldsymbol{\alpha}_{t|t}$, with $\alpha_{t|t,k} = \mathbb{P}(s_t = k \mid G_{1:t})$; the filter is run through the in-sample window and continued through the held-out observations, so only the state belief updates out of sample. The one-step-ahead state forecast and predictive CDF are

$$\mathbf{q}_{t+1|t} = \boldsymbol{\alpha}_{t|t}\,\mathbf{T}, \qquad F_{t+1|t}(x) = \sum_{k=1}^{K} q_{t+1|t,k}\, F_k(x; \boldsymbol{\theta}_k), \tag{6}$$

with F_k the per-state CDFs of Section 3.1, and the level-α regime-conditional VaR is the predictive quantile

$$\widehat{\text{VaR}}_{t+1|t}(\alpha) = \inf\{x \in \mathbb{R} : F_{t+1|t}(x) \geq \alpha\}, \tag{7}$$

solved numerically on the mixture CDF. The first forecast targets the boundary return into the first held-out session (the 573-forecast convention of Section 3.1), VaR is reported in the annualised units of G_t with losses negative, and a breach is recorded when the realised growth rate falls below the forecast quantile.

3.4 Spectral mechanism

The route from the transition matrix to slow ACF decay runs through the spectrum of $\mathbf{T}$. Let $\mathbf{v}_k$ and $\mathbf{w}_k$ denote the right and left eigenvectors of $\mathbf{T}$ at non-unit eigenvalue λ_k, normalised biorthonormally so that $\mathbf{w}_k^\top \mathbf{v}_l = \delta_{kl}$, and let $\sigma^2_{|G|}$ denote the stationary variance of $|G_t|$ under the marginal mixture (2). The closed-form bilinear identity for the absolute growth-rate ACF of a stationary CHMM, $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ with $\mathbf{m}_k = \mathbb{E}[|G_t| \mid s_t = k]$, and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are standard in the regime-switching literature [14–16]. Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1, with $\bar{\boldsymbol{\pi}}$ the unique stationary distribution; 75), finite per-state second moments (Assumption 2), and diagonalizability of $\mathbf{T}$, the spectral decomposition $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top$ subtracts the marginal product to give the ACF

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} a_k\, \lambda_k^\tau, \qquad a_k = \big(\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\big(\mathbf{w}_k^\top \mathbf{m}\big)/\sigma^2_{|G|}. \tag{8}$$

This is a real-valued sum of geometric modes at the non-unit eigenvalues of $\mathbf{T}$; complex-conjugate λ_k pairs combine into damped oscillatory modes. For a non-diagonalisable transition matrix, the corresponding Jordan expansion contains polynomial-times-geometric terms instead of equation (8). A self-contained derivation, the lag-zero behaviour, and the additional fourth-moment condition needed for a squared growth-rate ACF are reported in the appendix. The appendix also states the scope of the identifiability and numerical-convergence claims; neither is required for the algebraic ACF identity. Our contribution is purely empirical. We apply equation (8) to recast the Rydén et al. [12] low-K failure on the SPY 2014 to 2024 instance as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$. At $K = 2$, the ACF reduces to the mono-exponential form $\rho_{|G|}(\tau) = a_2\lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$, so its persistence can be tuned through $T_{11} + T_{22}$ but its shape remains restricted to one geometric mode. The fitted SPY $K = 2$ chain confirms this empirically: a single non-unit eigenvalue carries 100% of the lag-1 $|G|$ ACF, and it is strongly persistent ($\lambda_2 = 0.94$ for Gaussian emissions, 0.95 for the shared-ν Student-t; Appendix A.7.6), so the slow decay is already present at the lower bound. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture admits more components. The empirical results test whether those finite-mode and finite-mixture restrictions are active on the studied data; they do not imply that a finite-state HMM reproduces general power-law or multi-scale decay.

4 Results

4.1 Evaluation Protocol and State-Count Selection

This paper is a family-comparison platform rather than an advocate for one CHMM configuration: each application below uses the variant whose properties fit that use case, and no single variant is designated primary across applications. Table 1 is an ex-post analysis map: it records which variant each application used and on what basis, so the reader can audit the choices in one place. It was written after the evaluation, not pre-registered, and where a selection basis draws on out-of-sample performance (the heavy-tail recommendation row) the recommendation is exploratory: no untouched test window is reserved behind the OoS evaluation window. The use-case-level variant guide is Appendix Table S3.

Table 1. Specification map (ex post). CHMM variant and state count used by each empirical application, with the selection basis, recorded after the evaluation for auditability. All marginal fits use the quantile-initialised EM of Section 3.1.

Application	Variant (state count)	Selection basis
Generator comparison (Table 2)	all five CHMM rows ($K^\star = 3$)	family comparison is the object of study
Heavy-tail recommendation (Appendix Table S3)	CHMM-t shared-ν ($K^\star = 3$)	best CHMM OoS KS with a reduced kurtosis gap and no penalty hyperparameter; selected on the OoS evaluation window, hence exploratory
Regime-conditional VaR (Table 5)	CHMM-N and CHMM-t pen. ($K \in \{3, 18\}$); all four families in Appendix Table S34	Gaussian filter as base case; the family panel shows the coverage result is not family-specific
Cross-asset copula (Table 4)	CHMM-N marginals ($K^\star = 3$)	simplest adequate marginal for the PIT layer (all four families expose per-state CDFs); marginal adequacy per Table 4
Cross-ticker panel (Table S4)	CHMM-t pen. $\lambda = 20$ ($K^\star = 3$)	headline panel and quarterly-refit result at $K^\star = 3$; the $K = 18$ configuration (where λ was tuned) is the sensitivity check

We computed sample stylized-fact diagnostics on the SPY in-sample and out-of-sample windows and observed all three Cont stylized facts on both windows (Figure S3, Table 2 Observed row). The marginal was heavy-tailed (in-sample excess kurtosis 7.68, out-of-sample 5.29), the linear autocorrelation in the raw growth rate G_t was negligible, and the volatility clustering in $|G_t|$ was persistent. Because the observed tail index sits below 4 (next subsection), raw excess kurtosis has no stable population value, so its stationary-block-bootstrap bands are descriptive displays of resampling spread (wide: IS $[2.17, 12.40]$, OoS $[0.90, 8.26]$ at $L = 20$; Table S8); the inferential comparison uses winsorized excess kurtosis (trim $q = 0.01$ per tail, a functional with finite moments at any tail index), whose IS-minus-OoS difference has a 95% CI of $[-1.26, 1.83]$ at $L = 10$, covering zero at every tested block length (the assumptions of within-window stationarity and clamp-quantile regularity are stated in the appendix): the data do not exclude equal IS and OoS winsorized excess kurtosis, so excess kurtosis targets should be read against this uncertainty rather than the point estimates. Having confirmed the three stylized facts on both windows, we next swept $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and observed that the $K = 3$ versus $K = 6$ comparison could not be separated under four-fold and six-fold rolling-origin cross-validation (CV; full K-sweep CV panel in Table S31). The mean held-out log-likelihood differences were small and flipped sign between fold designs (0.007 nats per observation favouring $K = 3$ at the full-year cadence, 0.027 favouring $K = 6$ at the half-year cadence); with only four or six fold differences we treat these comparisons as descriptive and avoid formal significance language for them. The Bayesian information criterion (BIC), computed on the estimation window of the full-window design, selected $K = 3$. We chose $K^\star = 3$ as the default state count as a parsimony choice: BIC and CAIC selected it, the held-out scores for $K = 3$ and $K = 6$ were empirically similar (the rolling-origin designs did not separate the two), so we picked the smaller model, while $K = 18$ had a substantially lower mean held-out log-likelihood at both fold cadences. Absolute growth-rate ACF-MAE was nearly flat across the whole sweep, consistent with Eq. (8); the spectral mechanism behind that flatness, and what it does and does not establish, is examined next.

4.2 What the Fitted Spectra Do and Do Not Show

A direct effective-rank diagnostic on *converged* fits (complex-conjugate eigenvalue pairs grouped into real components, shares on the absolute component-magnitude budget) shows the SPY $K = 18$ chain concentrating its lag-1 ACF budget on two persistent components ($|\lambda| = 0.931$ and 0.921, jointly 94.0%; Table S6), and the cross-ticker version at both $K^\star = 3$ and $K = 18$ (Table S7) tells the same story under a horizon-aware norm: integrated over lags 1–252, the median number of components carrying 95% of the budget is 2 even at $K = 18$ (against a median of 5 under the lag-1 norm; the extra lag-1 components are fast-decaying). Every fit behind these numbers is a converged multistart estimate (tolerance 10^{-4}, no iteration cap reached, three to five starts, per-ticker optimizer evidence retained in the artifact); the earlier fixed 60-iteration convention left high-state fits materially short of convergence and is not used for the spectral and cross-ticker ACF comparisons of this subsection.

ACF attainability. A realizable experiment turns these descriptions into a statement about the decay-mode budget: for each ticker we directly optimize an actual valid K-state Gaussian-emission HMM (row-stochastic transitions, exact stationary law) so that its population $|G|$ ACF matches the observed sample ACF, by multistart gradient descent with one start seeded at a likelihood fit (a fresh single-start Baum-Welch refit, the internal single-start reference, not the published converged multistart fit; its own population-ACF error ships in the artifact, and the seeded start can only improve on it). Every reported error is attained by a valid HMM whose full parameters, fitted curve, and per-start diagnostics are persisted and reload-verified: an attainability certificate, not a certified global optimum. The result: a valid three-state chain tracks the sample ACF roughly three times more closely than the converged likelihood fit at the same state count (cross-ticker median near-band MAE 0.0162 against 0.0497; far band 0.0162 against 0.0245). The class is therefore not what limits the fitted ACF at $K^\star = 3$. The $K = 18$ arm of the same experiment attains 0.0140/0.0154, a small achieved gap to $K^\star = 3$ (paired median +0.0019 near band); this is *achieved accuracy under the declared optimizer*, with every $K = 18$ winner stopping at the iteration cap with near-degenerate stationary mass (Appendix A.7.5), not a class-level equivalence statement. Consistent with the achieved picture, the converged $K = 18$ likelihood fits do not materially improve the model-vs-sample ACF over $K^\star = 3$: in-sample, $K = 18$ is better at 7/31 tickers (median $\Delta = -0.005$ in $K^\star = 3$'s favour); on the held-out OoS window the difference is small in the other direction (22/31 tickers, median $\Delta = +0.002$), a sign reversal that reads as instability of a small difference rather than corroboration.

Joint attainability. The best achieved ACF-only fit above has a poor marginal (median mixture excess kurtosis 0.32 against the observed 8.86), which by itself cannot distinguish a genuine trade-off from an artifact of single-criterion optimization. A frontier experiment answers this directly: for each ticker we sweep the weighted objective ACF-SSE $+ \lambda \cdot s \cdot$ CvM (Cramér–von Mises distance of the stationary mixture CDF to the empirical CDF on a 500-point in-sample quantile grid; s a fixed balance scale) over eight λ values plus a pure-marginal arm, with the reading rule declared in the runner before the run; the comparator throughout is the published converged multistart likelihood fit, recomputed deterministically and re-evaluated on the same metrics in the artifact. The outcome is *joint attainability of the ACF and the marginal body under this in-sample CDF criterion*: the $\lambda = 0.1$ arm attains median near-band ACF MAE 0.0165 (within 1% of the best achieved ACF arm) while its CDF distance (7.2×10^{-6}) is *better* than the multistart likelihood fit's own (5.2×10^{-5}) and its stationary-mixture marginal log-likelihood is within 0.011 nats per observation of that fit's (-2.608 against -2.597). The joint behaviour holds at the ticker level, not merely axis-by-axis: the median per-ticker maximum of the ACF and CvM regrets is 1.17, and 24/31 tickers hold both regrets at or below 1.5 with one and the same chain. Under this criterion the ACF and the bulk marginal do not compete at three states. What the ACF-competitive fits still lack is the *deep tail*: their 1%/99% quantile errors are roughly three times the multistart likelihood fit's (0.83 against 0.23 at the 99% quantile, improving on it at only 5/31 and 6/31 tickers, against 21/31 at the central 5%/95% quantiles), with mixture excess kurtosis 2.3 against 7.0 as the descriptive counterpart (descriptive because, as noted above, raw kurtosis has no stable population target on these data); the likelihood criterion, which weights tail observations heavily, selects fits with better extreme tails and far worse ACF tracking.

Interpretation and limits. At $K^\star = 3$, then: the decay-mode budget does not bind the fitted ACF (attainability), the marginal body does not trade off against the ACF (frontier, under its in-sample CDF criterion), and the open question is specifically whether one three-state chain can carry the *deep tail* and the ACF simultaneously: no achieved fit did, but the frontier's marginal criterion underweights tails, so tail-and-ACF joint attainability is neither established nor excluded here. All frontier and capacity fits are valid achieved feasible points from capped multistart heuristic searches (most weighted-arm winners hit the iteration cap; per-arm stop counts are in the artifact, and the achieved points bound the attainable frontier from above rather than certify class optima), the analysis is descriptive across a dependent panel, and none of it establishes a general power-law or multi-scale approximation (the exploratory out-of-class curve-fit reference and all optimizer caveats are in Appendix A.7.5).

The held-out criterion also surfaces a scope limit. At the median single-name ticker, both fitted models sit *behind* the OoS zero-curve reference on the near band (0.043–0.044 against 0.0354): the 2024–2026 OoS sample ACF is much flatter than the IS one, so IS-fitted persistence does not transfer at the typical single name, the ACF-axis analogue of the regime-drift pattern in the cross-ticker KS panel. On SPY itself, the headline instance, the fitted models do beat the OoS zero curve (0.0502 at $K^\star = 3$ and 0.0378 at $K = 18$, against 0.0654). Finally, a compute disclosure: the $K = 18$ rows elsewhere in the paper (sector-panel, walk-forward, and VaR sensitivity configurations) retain the original fixed 60-iteration budget and should be read as fixed-compute sensitivity rows; the spectral and ACF claims of this subsection rest exclusively on the converged fits.

4.3 Single-Asset Generator Comparison

With the state count fixed, we fit the four CHMM variants on the SPY in-sample window at $K^\star = 3$ and observed out-of-sample performance using seven benchmark generators and an explicit-duration semi-Markov reference (Table 2). The CHMM-N row landed at 91.5% IS and 78.0% OoS, simulated excess kurtosis 3.83 and 3.62, and per-path $|G_t|$ ACF-MAE 0.0462: below the i.i.d. floor of ≈ 0.063, but not the best conditional-volatility figure. Under the like-for-like per-path estimand (all rows of Table 2; the pooled population-curve estimand of Table S23 is systematically smaller and not comparable), the one-regime GARCH rows sit slightly above CHMM-N (0.0474–0.0490 across GARCH(1,1), GJR, and GARCH-t against 0.0462), while the regime-switching MS-GARCH at $K = 3$ remains stronger at 0.0408; the CHMM's advantage over the GARCH family therefore lies on the marginal (KS) axis, with the volatility-clustering axis led by MS-GARCH rather than by one-regime GARCH fits. The shared-ν Student-t row provided the best marginal-fit and heavy-tail trade-off among the CHMM variants on this table's axes, the best CHMM OoS KS together with a smaller kurtosis gap than CHMM-N and no penalty hyperparameter (no composite joint score is defined; it is the heavy-tail recommendation of the variant guide, not a primary specification for the downstream applications, per Table 1), at 91.9% IS and 82.1% OoS KS with simulated excess kurtosis 4.68 and 4.46 against observed 7.68 and 5.29; on the excess kurtosis distance alone the penalty-free CHMM-L and CHMM-GED rows sit closer to observed, at lower KS. We include the penalised CHMM-t at $\lambda = 20$ as a sensitivity reference. Its IS aggregate excess kurtosis of 17.67 was a $1/\nu_k$-shrinkage outcome: λ was tuned at $K = 18$ and re-used at $K^\star = 3$, where the state pinned at the lower ν bracket carried a larger share of the three-state mixture, and the near-undefined fourth moment at that bracket left the aggregate excess kurtosis itself sensitive to the simulation draw. The remaining two variants, CHMM-GED and CHMM-L, sat between CHMM-N and the penalised CHMM-t, at IS and OoS excess kurtosis 5.45 and 5.09 for CHMM-GED and 5.24 and 5.20 for CHMM-L; CHMM-L lost roughly 15pp of OoS KS to CHMM-GED at $K = 3$.

The fitted CHMM reduced the lag-252 absolute growth-rate sample-ACF error relative to the i.i.d. baselines (per-path MAE 0.0462 for CHMM-N against the i.i.d. floor of ≈ 0.063; Table 2) on the SPY 2014 to 2024 instance, where the Rydén et al. [12] $K = 2$ to 3 Gaussian setup, fit jointly by ML, judged the ACF fit unsatisfactory, and the raw growth-rate ACF-MAE closed the third stylized fact at 0.0235 to 0.0240, matching the i.i.d. baselines to the reported precision (Table 2, G_t column). This is a horizon-bounded statement, not a claim of matched long-horizon persistence: Figure 2 plots the observed IS $|G_t|$ ACF (with an $L = 20$ block-bootstrap band shown only over lags 1–5, well inside the block length, because block-boundary contamination grows with the lag and reaches every resampled pair at $h = L$) against the CHMM-N simulated envelope and the fitted population ACF, and the horizon-banded errors show where the advantage lives. CHMM-N beats the i.i.d. floor decisively at lags 1–5 (0.042 against 0.289), 6–20 (0.067 against 0.221), and 21–63 (0.062 against 0.100), but not in the 64–252 band (0.041 against 0.036), where the fitted population ACF is effectively zero: the dominant eigenvalue 0.9534 has a half-life of 14.5 trading days, so the theoretical $\rho_{|G|}(252) \approx 2 \times 10^{-6}$. The lag-252 aggregate advantage is therefore earned at short and medium horizons, and the success criterion is relative-to-floor per band rather than a pre-registered numeric threshold.

On the OoS sample-CRPS, none of the ten pairwise equal-mean-loss nulls among the five CHMM variants, CHMM-N (1.0393), penalised CHMM-t (1.0382), CHMM-L (1.0432), CHMM-GED (1.0398), and shared-ν CHMM-t (1.0406), was rejected by a HAC Diebold-Mariano [57] panel after Holm correction at the 5% level; one pair (penalised CHMM-t vs. CHMM-L, mean difference -0.0050, HAC 95% CI $[-0.0099, -0.0002]$) is nominally significant before correction ($p = 0.042$, Holm $p = 0.42$). This is a non-rejection statement, not an equivalence claim: no equivalence margin was pre-specified, and the HAC CIs bound the plausible mean-CRPS differences at roughly ± 0.005–0.010, so the family choice was driven by the per-row excess kurtosis match rather than by CRPS (Table 2; the use-case decision guide is Table S3); the cross-generator ranking was robust to the distributional-metric choice across KS, Anderson-Darling, Wasserstein-1, and Hellinger. The MS-GARCH benchmark stayed well below the CHMM on KS under both point-estimate and Bayesian posterior-predictive variants (Table 2; full MS-GARCH panel in the appendix). A six-fold rolling-origin walk-forward (Table S33) reached median OoS KS 62.1% at $K^\star = 3$, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% at both tested state counts; the walk-forward panel covers CHMM-N at $K \in \{3, 18\}$, so the stress-fold statements in this paper are statements about those configurations rather than about every generator of the main comparison table. We treated the walk-forward median, not the single-window OoS pair, as the right summary for judging the model for live use. The i.i.d. bootstrap exceeded the CHMM on raw single-window OoS KS (Table 2), an advantage confined to the marginal axis: on volatility clustering the bootstrap row sits at the i.i.d. $|G_t|$ ACF-MAE floor (0.0628) while CHMM-N clusters below it at 0.0462. The explicit-duration HSMM-N benchmark, a multistart local-EM fit under the grid-bracketed censored truncated-discrete Pareto duration update, with the terminal segment right-censored so the estimator and its simulator describe the same finite-window model (five starts, all converged, cross-start spread 0.14 nats), splits the axes the other way: its heavy-tailed sojourns give the best volatility-clustering figure in the panel (0.0394, below MS-GARCH's 0.0408 and CHMM-N's 0.0462) at a weaker marginal (76.0% IS / 68.7% OoS KS against CHMM-N's 91.5/78.0). The direction matches the Bulla and Bulla [13] duration-law mechanism, but the figure is specification-scoped rather than established: the fitted

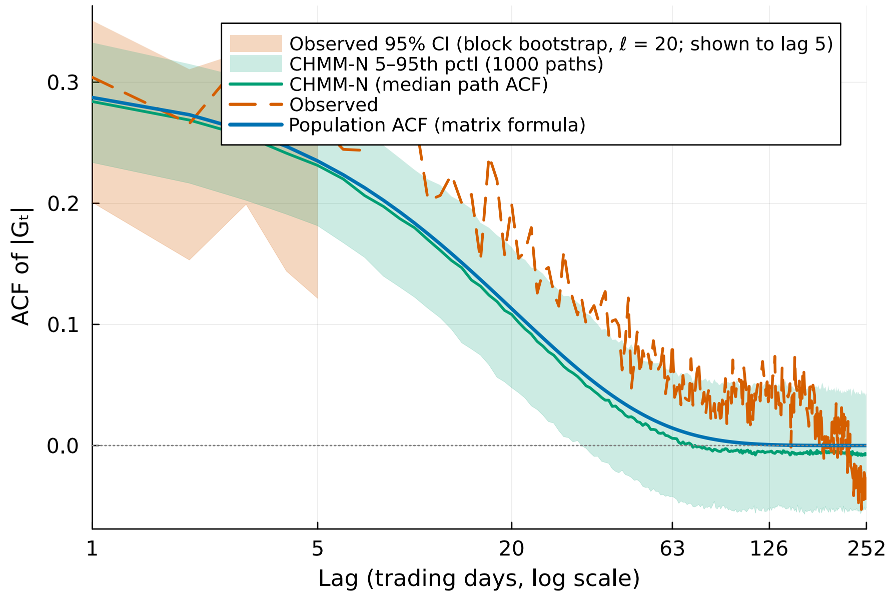

Figure 2. Observed versus simulated absolute growth-rate ACF, SPY IS, CHMM-N at $K^\star = 3$ (log-lag axis, lags 1–252; seed 20260420). The observed sample ACF carries a moving-block-bootstrap 95% band (block length $L = 20$, 1,000 replicates) shown only over lags 1–5, well inside the block length: block-boundary contamination is gradual (at lag h roughly h/L of resampled pairs cross an independently concatenated boundary, and at $h = L$ every pair does), so the band is displayed only where that fraction is small, and no longer-lag claim is made from it. The simulated band is the 5–95% per-path envelope with its median over 1,000 IS-length paths; the smooth curve is the fitted population ACF from the spectral identity (8), which is effectively zero beyond lag ≈ 126 (dominant-mode half-life 14.5 days). The horizon evidence is the banded per-path MAE comparison (CHMM-N vs. i.i.d. floor): lags 1–5: 0.042/0.289; 6–20: 0.067/0.221; 21–63: 0.062/0.100; 64–252: 0.041/0.036; the model beats the floor in the first three bands and is slightly worse than it in the far band.

persistent state's duration exponent sits at the declared search floor ($\alpha = 0.05$ at $D_{\max} = 200$), and a $D_{\max}\times$ exponent-floor sensitivity grid moves the clustering figure across 0.039–0.046 (best at $D_{\max} = 200$, worst at $D_{\max} = 100$) with the marginal moving oppositely (84.6% down to 60.9% IS KS as $D_{\max}$ rises from 100 to 400), so the panel-leading value holds at the declared $D_{\max} = 200$ specification (Appendix A.7.17). The CHMM is the one generator in the panel we equip with the regime-conditional VaR and multi-asset copula heads, where the i.i.d. bootstrap is structurally unable to produce a regime-conditional VaR (it carries no latent state) and HSMM-based heads are left as a companion-paper direction, with the full HSMM panel, state-assignment inspection, bootstrap discussion, and block-bootstrap KS recalibration reported in the appendix.

Because excess kurtosis is a fourth-moment functional that is fragile under heavy tails, we complemented it with a threshold-local Hill estimate [76] applied identically to the observed series and to every simulated family at $K^\star = 3$ (top 5% of $|G_t|$ exceedances; Table 3); for the light-tailed mixture families this quantity is a shape diagnostic of the estimated threshold range, not an asymptotic tail index (metric definition in the appendix). The observed tail index was $\hat{\alpha} = 3.15$ IS and 3.30 OoS, inside the $(2, 5)$ band reported by Cont [5] for daily equity returns, with a stationary block-bootstrap 95% CI of $[2.45, 4.25]$ on the IS estimate. The shared-ν CHMM-t landed at $\hat{\alpha} = 3.67$

Table 2. Main generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS $T = 2{,}516$, OoS $T = 572$. Columns: KS = Kolmogorov-Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute growth-rate / raw growth-rate ACF over 252 lags, under the *per-path* estimand throughout (the pooled population-curve estimand of appendix Table S23 is systematically smaller and not comparable: the same canonical grid-initialised ML GARCH(1,1) fit scores 0.0490 per-path and 0.0309 pooled). Bold marks the best entry on each axis *within block* (block = i.i.d. baselines, GARCH family, MS-GARCH, CHMM, co-main HSMM); for the excess kurtosis columns, best means closest to the observed value. Estimand conventions and the extended panels ($K = 6$ / $K = 18$ sensitivity, GARCH family, MS-GARCH, block-bootstrap KS recalibration) are in the appendix. † Canonical grid-initialised ML GARCH(1,1) fit (identical to the appendix fit); the MS-GARCH $K = 6$ per-path value was not recomputed (pooled value in the appendix). ¶ Reference Bayesian MS-GARCH re-run via the `MSGARCH` R package of Ardia et al. [30]; the lower KS pass rate reflects posterior-predictive parameter uncertainty. § QuantGAN: in-house WGAN re-implementation (spec in the appendix), the panel's deep-generative negative control. §§ Shared-ν Student-t ablation: single ν across all K states by aggregate-Q ECM, no penalty.

	KS (%) ↑		Exc. Kurt		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
Observed	–	–	7.68	5.29	–	–
Bootstrap	**99.8**	**91.8**	**7.55**	**6.86**	0.0628	**0.0235**
Gaussian i.i.d.	0.0	1.5	0.00	−0.02	0.0627	0.0235
Laplace i.i.d.	98.2	86.3	2.96	2.80	0.0628	0.0235
GARCH(1,1)†	27.4	59.6	**7.59**	**2.70**	0.0490	0.0244
GARCH(1,1)-t	**57.3**	**80.8**	15.13	–	0.0480	**0.0173**
MS-GARCH ($K = 2$)	27.7	**38.7**	4.73	–	0.0455	0.0173
MS-GARCH ($K = 3$)	**36.1**	33.1	4.10	–	**0.0408**	0.0173
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	–†	0.0173
MS-GARCH ref. Bayesian ($K = 2$)¶	0.0	5.8	4.00	2.46	0.0465	0.0240
MS-GARCH ref. Bayesian ($K = 3$)¶	0.1	5.1	4.52	2.53	0.0433	0.0241
MS-GARCH ref. Bayesian ($K = 4$)¶	0.0	5.3	**6.01**	**3.54**	0.0446	0.0241
QuantGAN TCN§	0.0	0.0	0.56	0.53	0.0617	0.0264
Main-paper choice: $K^\star = 3$, the BIC/CAIC parsimony choice; CV did not separate $K = 3$ vs $K = 6$.						
CHMM-N	91.5	78.0	3.83	3.62	**0.0462**	0.0240
CHMM-t pen. ($\lambda = 20$)	**91.9**	79.7	17.67	10.28	0.0538	0.0235
CHMM-L	80.5	63.6	5.24	**5.20**	0.0530	**0.0235**
CHMM-GED	90.3	78.4	**5.45**	5.09	0.0531	0.0236
CHMM-t shared-ν§§	**91.9**	**82.1**	4.68	4.46	0.0531	0.0236
HSMM-N (multistart local EM)	**76.0**	**68.7**	**3.37**	**2.95**	**0.0394**	**0.0244**

on both windows with an across-path 95% band of $[3.11, 4.38]$ overlapping the observed CI: the observed estimate is compatible with the simulated dispersion at this threshold under this descriptive diagnostic (the two ranges are different inferential objects, an observed-series block-bootstrap interval and an across-path simulation band, so their overlap is neither a difference CI nor an equivalence test), with the simulated tail sitting slightly thinner in the far tail rather than undershot. Every emission family produced an in-band threshold-local estimate at $K^\star = 3$ (CHMM-N 3.41, penalised CHMM-t 3.28, CHMM-L 3.26, CHMM-GED 3.24) even though simulated excess kurtosis spanned 3.83 to 17.67 across the same rows: at the depths the estimator sees, the regime mixture rather than the emission family carries the heavy tail, while the emission shape governs how far into the extreme tail the match extends, which is where the Gaussian row falls behind asymptotically. The same estimator applied per tail side reproduced the direction of the observed gain/loss asymmetry: the observed loss tail is heavier than the gain tail ($\hat{\alpha}_L = 2.50$ against $\hat{\alpha}_R = 3.05$ IS), and every symmetric family recovered a negative loss-minus-gain gap from the regime means alone (shared-ν -0.46, CHMM-L -0.44, against observed -0.55). This is the static asymmetry channel, distinct from the dynamic leverage effect (Table S25).

Table 3. Threshold-local Hill estimates $\hat{\alpha}$ on $|G_t|$ at $K^\star = 3$ (top 5% exceedances; 1,000 simulated paths, seed 20260420). Same estimator for every series. The observed band is a stationary block-bootstrap 95% CI (mean block length 10, $B = 2{,}000$); simulated bands are across-path 95% quantile bands on the IS window. The loss/gain columns apply the estimator to each tail side separately with equal exceedance counts; $L - R < 0$ means the loss tail is heavier. All rows sit inside the Cont [5] $(2, 5)$ band for daily equity returns.

	Pooled $\hat{\alpha}$		IS 95% band	IS by tail side		
Series	IS	OoS		Loss $\hat{\alpha}_L$	Gain $\hat{\alpha}_R$	$L - R$
Observed SPY	3.15	3.30	$[2.45, 4.25]$	2.50	3.05	-0.55
CHMM-N	3.41	3.54	$[2.90, 4.05]$	2.64	2.93	-0.29
CHMM-t shared-ν	3.67	3.67	$[3.11, 4.38]$	2.85	3.31	-0.46
CHMM-t pen. ($\lambda = 20$)	3.28	3.27	$[2.79, 3.91]$	2.58	3.36	-0.79
CHMM-L	3.26	3.26	$[2.84, 3.86]$	2.51	2.95	-0.44
CHMM-GED	3.24	3.24	$[2.78, 3.80]$	2.40	3.34	-0.93

Having established the main ranking on SPY, we next fit the penalised CHMM-t at $K^\star = 3$ and $\lambda = 20$ on each ticker of a sector-balanced 30-ticker panel (ten GICS sectors with three large-cap representatives each) and observed cross-ticker generalisation behaviour (Table S4). The in-sample distribution was concentrated (median 96.8%) and the out-of-sample distribution was wider, with median 69.1%, mean $66.2 \pm 28.2\%$, and 11/30 tickers below 60%. The deepest failures were tickers that introduced a new regime out of sample: LLY and UNH in Health Care, and NEM in Materials. The per-ticker rollup, the remaining eight sub-60% tickers, and the per-sector breakdown are reported in the appendix (Table S5); the 11/30 failure count and the ticker list hold at the $K = 18$ excess kurtosis-match check. An ANOVA on OoS KS by sector returned no significant sector effect on either the original $n = 3$ design or an $n = 6$ expansion to a 60-ticker panel (full ANOVA panel in the appendix), so the visible concentration of failures on regime-introducing tickers was a per-ticker observation rather than evidence about sector-level structure (a non-detection: no equivalence test or sector-variance interval is attached, so absence of a sector effect is not demonstrated). Periodic refit closed much of the gap: at $K^\star = 3$ a quarterly refit shifted the OoS KS median from 69.1% to 84.6% and reduced failures from 11/30 to 8/30 (Tables S38 and S14); a monthly cadence at $K = 18$ closed slightly more of the gap at roughly three times the compute. Sensitivity panels at $K = 6$ and $K = 18$ and the SPY-level shared-ν Student-t alternative are reported in the appendix.

4.4 Cross-Asset Composition

We then composed per-asset CHMM-N marginals at $K^\star = 3$ through a rank-based Student-t copula on a six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) and observed off-diagonal correlation reproduction across the four candidate dependence layers (Table 4; full comparator panel in the appendix). The rank reordering preserves each asset's simulated marginal exactly but permutes each asset's path in time, so the composed draws carry cross-sectional dependence only and this section makes no claim about joint temporal dynamics; the per-asset serial diagnostics of the earlier sections apply to the single-asset generator before composition.

On the in-sample window the Student-t copula gave off-diagonal correlation MAE of 0.027 over 200 simulated paths, against 0.029 for the Gaussian copula, 0.068 for the truncated C-vine, and 0.077 for the single-index model (SIM) (Table S36). Profile MLE selected $\nu^\star = 6$ on the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ with Wilks 95% profile-LL CI of $[6, 7]$ on the IS slice; an iterated pseudo-likelihood estimator alternating a likelihood step on ν with a moment refit of Σ confirmed the choice was robust to the two-step estimator (Fig. S5; Table S26), and a boundary-appropriate likelihood-ratio test against a simulated Gaussian-copula null separated the Student-t copula from its Gaussian limit on the IS window (LR = 697.2, Monte-Carlo $p = 0.002$; construction in the appendix). The universe contained overlapping ETFs and constituents (QQQ holds AAPL and NVDA; SPY holds all four single names), so the dependence structure was dominated by strong positive co-movement (mean absolute off-diagonal correlation 0.529) and the copula task was easier than on a non-overlapping basket. A non-overlapping comparison panel, six single names with one representative from each of six selected GICS sectors (MSFT, UNH, BAC, CAT, PG, XOM; mean absolute off-diagonal correlation 0.361), is reported in the appendix: profile MLE again selected $\nu^\star = 6$, the in-sample composition reproduced the dependence at off-diagonal MAE 0.031 for the Student-t copula against 0.032 for the Gaussian, and the static-fit OoS MAE degraded to 0.253, with the health-care name (UNH) showing the familiar single-name regime-introduction failure (OoS KS 21.5%). On OoS the off-diagonal MAE was 0.209 for Student-t and 0.204 for Gaussian (Table S36). A paired replication over 20 simulation seeds under strict common random numbers (appendix) put the mean paired difference at +0.0055 with a 95% t-interval of $[+0.0050, +0.0060]$: within this fixed-fit simulation design the small gap is stable across replicates rather than path-to-path noise, with the Gaussian copula's static OoS off-diagonal error consistently the lower of the two, reversing the in-sample ordering. That interval quantifies Monte-Carlo precision for the simulation experiment, not copula performance on new data.

Out-of-sample, we crossed the two design choices (static versus rolling-quarterly-refit, and Gaussian versus Student-t) in a 2×2 experiment with all four arms scored on the identical realised per-quarter correlation targets under the same fixed marginals (Table S28), which measures the refit and family effects on the same estimand for the first time. All four arms are simulated under strict common random numbers (identical base normals and marginal draws per quarter; the Student-t chi-square mixing stream is separate), so the paired contrasts isolate the design factors exactly. The refit effect is an order of magnitude larger: the quarterly refit reduced the mean per-quarter off-diagonal MAE from 0.264 to 0.186 within the Student-t family (paired $t(8) = 3.62$, $p = 0.007$) and from 0.261 to 0.179 within the Gaussian family ($t(8) = 3.63$, $p = 0.007$), while the family effect on the same targets is 0.003–0.007, precisely resolved by the common-random-number pairing, and slightly favours the Gaussian copula. With nine complete quarter blocks and overlapping estimation windows these paired results are suggestive rather than confirmatory, but the ordering (refit policy over family choice) is consistent across both families and 7–9 of 9 blocks (Table 4). The cross-asset OoS distribution was bimodal: NVDA and JPM failed OoS KS while the other four passed above 77%, driven by single-name regime shifts (NVDA on the 2024-2025 AI rally, JPM on the 2023 regional-bank stress). The quarterly-rolling refit of Table S28 refits the dependence layer only, holding the per-asset marginals fixed, and reports correlation reproduction rather than per-asset KS, so the per-asset marginal failures are not re-tested under refit; the refit benefit substantiated by that table is the off-diagonal MAE reduction quoted above.

Table 4. Cross-asset Student-t copula on CHMM-N marginals at $K^{\star} = 3$ (multi-asset construction; $\nu^{\star} = 6$; 200 paths; seed 20260422). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction computed per simulated path against the observed correlation matrix and then averaged over paths (full-matrix convention; the off-diagonal MAE averages the $d(d-1)/2$ unique upper-triangle entries). Main-paper $K^{\star} = 3$ matches the single-asset Table 2. Per-asset KS sits lower at $K^{\star} = 3$ than at higher state counts, but the cross-asset use case is dominated by the dependence layer rather than per-asset marginal fit. The comparison against SIM, Gaussian, and truncated C-vine is reported in the appendix.

	KS (%) ↑	
Ticker	IS	OoS
SPY	87.0	77.5
NVDA	92.0	62.5
JNJ	97.5	94.0
JPM	97.5	57.5
AAPL	85.0	86.0
QQQ	83.0	85.5
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Per-quarter MAE, static vs. rolling refit	0.264 vs. **0.186** (paired $p = 0.007$; 2×2 design, Table S28)	

4.5 Regime-Conditional Value-at-Risk Back-Test

Having shown that the per-asset marginals compose under a dependence layer, we next back-tested a regime-conditional VaR built from the CHMM one-step-ahead state forecast and observed conditional-coverage behaviour at both VaR levels (Table 5; the full four-family panel is in the appendix, Table S34). The back-test covered CHMM-N and the penalised CHMM-t at $K \in \{3, 18\}$ on the SPY out-of-sample window, under the Christoffersen joint conditional-coverage and Engle-Manganelli dynamic-quantile (DQ) tests. The back-test covers all 573 held-out forecasts, including the boundary return into the first held-out session. The four CHMM rows at VaR level $\alpha = 0.05$ were not rejected by Christoffersen-cc ($p_{\text{cc}} \in [0.39, 0.67]$), and CHMM-N at $K^{\star} = 3$ was likewise not rejected by the DQ test ($p = 0.09$); non-rejection at these sample sizes indicates compatibility with correct conditional coverage rather than a demonstration of it. Breach rates at $\alpha = 0.05$ sat close to the nominal target across the four CHMM rows, from 26 breaches (4.54%) for CHMM-N at $K = 18$ to 36 (6.28%) at $K^{\star} = 3$, and no CHMM row was rejected by Kupiec or Christoffersen-ind at that level.

The $\alpha = 0.01$ rows were also not rejected by Christoffersen-cc ($p_{\text{cc}} \in [0.10, 0.16]$) but are power-bounded at $T_{\text{OoS}} = 573$, where only about 5.7 breaches are expected; the higher-power DQ test separated the state counts, rejecting CHMM-N at $K = 18$ ($p = 0.02$) while $K^{\star} = 3$ escaped rejection only marginally ($p = 0.06$). Against two non-state-space conditional-VaR baselines (filtered bootstrap [77] and the Engle-Manganelli symmetric-absolute-value CAViaR specification [78]) the CHMM rows reached approximate parity at $\alpha = 0.05$; at $\alpha = 0.01$ CHMM-N at $K^{\star} = 3$ was the only row of the CHMM / filtered-bootstrap / CAViaR panel not rejected by DQ ($p = 0.06$), a power-bounded non-rejection rather than evidence of a significant difference between the models, since failure to reject does not establish the null and no pairwise model-comparison test was conducted. Within the CHMM family the statement is family-wide: across all sixteen family rows of the appendix panel, every other family and state count was rejected at that tier (p from 0.0005 to 0.030), as were the filtered bootstrap ($p = 0.04$) and CAViaR ($p = 0.01$) contenders. The same test harness applied to the Markov-switching GARCH benchmark of the generator panel, on the same 573-forecast origin set (MS-GARCH block of Table 5), over-breaches at the strict tail at every tested state count (11–12 breaches against 5.7 expected; exact binomial $p \in [0.017, 0.035]$) and is rejected by asymptotic DQ at that tier ($p \in [0.016, 0.019]$).

Three qualifications keep this from being read as a CHMM victory. First, a parametric-bootstrap calibration of the DQ statistic under each fitted null (Appendix A.7.11; $B = 2{,}000$, covering exactly the two rows named here and conditioning on the estimated parameters) shows the asymptotic χ^2_6 reference to be anticonservative at $T = 573$: under finite-sample nulls neither CHMM-N at $K^\star = 3$ (bootstrap $p = 0.19$ against asymptotic 0.06) nor MS-GARCH at $K = 4$ (bootstrap $p = 0.10$ against asymptotic 0.02) is rejected at the 5% level, so every 1%-tier DQ separation in this table is exploratory, and finite-sample inference extends only to those two calibrated rows. Second, a paired quantile-loss (pinball) comparison on the identical forecast days finds no significant ranking between CHMM-N at $K^\star = 3$ and MS-GARCH at $K = 4$, the filtered bootstrap, or CAViaR at either level (HAC $p \geq 0.14$ in every pair), so no model in the panel demonstrably beats another on quantile forecast accuracy. Third, non-rejection remains power-bounded at this sample size. The primary conditional-coverage evidence in this paper is therefore the 5% tier, where no model in the panel is rejected; the strict-tail evidence does not identify a single preferred conditional-risk model.

When we extended the back-test to six rolling-origin walk-forward folds, Christoffersen-cc did not reject on 19 of the 24 walk-forward rows at the 5% test level (Table S35), with the five rejections concentrated on the stress folds: all four W2 (COVID 2020) rows rejected, while W4 (2022 rate-hike onset) produced a single tail-tier rejection at $K = 18$, $\alpha = 0.01$ ($p = 0.022$) with the other three W4 rows not rejected; a Benjamini-Hochberg [79] false-discovery-rate adjustment over the 24-row panel is reported in the appendix, and we read the 16-row family panel and the 24-row walk-forward panel as multiplicity-aware descriptive evidence rather than as repeated confirmations (full walk-forward and refit-cadence panels and the four-family ablation are reported in the appendix).

We closed by comparing the four stress sources exercised across the study: the cross-ticker panel, the CRSP cross-decade transfer, a non-equity extension to the gold ETF GLD, and the walk-forward folds. All four reduced to the same three-way pattern (Table S38). A static in-sample fit held up only while the out-of-sample marginal stayed inside the range of regimes the in-sample window had already seen. On equity slices that drifted but stayed in scope, periodic refit closed the gap: the cross-ticker panel, for example, recovered from a static-fit median OoS KS of 69.1% to 84.6% under quarterly refit. On a slice whose marginal left the equity-return scope entirely, refit did not help: the GLD non-equity stress collapsed under static fitting and stayed broken at every cadence. The cross-decade transfer failed by regime introduction: the calm 2004 to 2006 slice sat outside the regime range of the 1994 to 2004 in-sample window, the static fit fell to a 3 to 5% pass rate (Table S18), and the refit question was untestable on a slice shorter than 600 trading days. The hardest case was a regime introduction landing inside the refit window itself, as in the W2 (COVID) and W4 (2022 rate-hike) walk-forward folds, where no cadence can anticipate a shift it has not yet seen; both tested walk-forward configurations (CHMM-N at $K \in \{3, 18\}$) failed the distributional test on those folds, and the conditional-VaR back-test rejected all four W2 rows and the $K = 18$ tail tier on W4. We therefore recommend periodic refit for live use, at a cadence chosen against the out-of-sample slice rather than against the in-sample fit alone.

Table 5. Regime-conditional VaR back-test on SPY OoS ($T_{\text{OoS}} = 573$ forecasts, including the boundary return into the first held-out session; seed 20260420). Median $\widehat{\text{VaR}}_t$ is in annualised log-growth units (Eq. (1)); divide by 252 for the daily log-return equivalent, e.g. the CHMM-N $K^\star = 3$ median 1% VaR of -4.56 corresponds to -1.81% daily. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The CHMM rows use the forward filter under IS-fixed parameters; the contender rows (filtered bootstrap, CAViaR) and the MS-GARCH block (in-house fits of Table S23) are scored on the same Christoffersen / DQ harness and the same 573-forecast origin set. ‡ $\alpha = 0.01$ rows are power-bounded at $T_{\text{OoS}} = 573$ (power calibration in the appendix). * DQ p-value at the 4-lag specification; CHMM-t penalised rows carry '–' in this column. The finite-sample DQ calibration of Appendix A.7.11 shows the asymptotic χ^2_6 reference is anticonservative in this design, so the 1% DQ column should be read as exploratory throughout; the test outcomes themselves are discussed in the text.

Family	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}	$p_{\text{DQ}}{}^*$
CHMM-N	3	$0.01^\ddagger$	9	1.57%	−4.56	1.61	2.37	3.97	0.14	0.06
CHMM-N	3	0.05	36	6.28%	−2.87	1.84	0.04	1.88	0.39	0.09
CHMM-N	18	$0.01^\ddagger$	9	1.57%	−5.19	1.61	2.37	3.97	0.14	**0.02**
CHMM-N	18	0.05	26	4.54%	−3.02	0.27	0.52	0.79	0.67	0.68
CHMM-t (λ=20)	3	$0.01^\ddagger$	8	1.40%	−5.57	0.81	2.81	3.62	0.16	–
CHMM-t (λ=20)	3	0.05	32	5.58%	−2.73	0.40	0.78	1.18	0.56	–
CHMM-t (λ=20)	18	$0.01^\ddagger$	10	1.75%	−6.23	2.63	1.98	4.61	0.10	–
CHMM-t (λ=20)	18	0.05	29	5.06%	−3.14	0.00	1.40	1.40	0.50	–
Non-state-space conditional VaR contenders.										
Filtered bootstrap	–	$0.01^\ddagger$	8	1.40%	−4.81	0.81	2.81	3.62	0.16	**0.04**
Filtered bootstrap	–	0.05	24	4.19%	−2.89	0.84	0.85	1.69	0.43	0.55
CAViaR (SAV)	–	$0.01^\ddagger$	6	1.05%	−4.82	0.01	3.97	3.98	0.14	**0.01**
CAViaR (SAV)	–	0.05	25	4.36%	−2.99	0.51	0.68	1.19	0.55	0.76
Regime-switching-volatility conditional VaR contender (MS-GARCH; same 573-forecast origin set as the rows above).										
MS-GARCH	2	$0.01^\ddagger$	12	2.09%	−4.05	5.27	1.36	**6.63**	**0.04**	**0.02**
MS-GARCH	2	0.05	32	5.58%	−2.61	0.40	0.03	0.42	0.81	0.79
MS-GARCH	3	$0.01^\ddagger$	11	1.92%	−4.14	3.86	1.65	5.51	0.06	**0.02**
MS-GARCH	3	0.05	29	5.06%	−2.65	0.00	0.19	0.20	0.91	0.74
MS-GARCH	4	$0.01^\ddagger$	11	1.92%	−4.16	3.86	1.65	5.51	0.06	**0.02**
MS-GARCH	4	0.05	30	5.24%	−2.67	0.07	0.12	0.19	0.91	0.86

5 Discussion

On the marginal axis, the main empirical finding is that a continuous-emission HMM trained by EM with quantile-based initialisation, at $K^\star = 3$ on daily US-equity returns, substantially improves the heavy-tailed marginal fit over the Gaussian baseline and leaves the raw growth rate serially uncorrelated, with the remaining excess kurtosis gap set by the choice of heavy-tailed emission family. The mechanism behind the residual gap is empirical rather than structural: CHMM-N at $K^\star = 3$ produced simulated IS excess kurtosis 3.83 against observed 7.68 (Table 2), so under the fitted maximum-likelihood three-state specification the Gaussian mixture understated the observed heavy tail. A three-component Gaussian scale mixture can in principle reach arbitrarily high kurtosis through a small-weight, large-variance component, but the joint ML fit, disciplined by the bulk of the data, did not select such a configuration; what a finite Gaussian mixture structurally cannot supply is a genuinely power-law tail, since it remains asymptotically light-tailed at every parameter setting. Switching to a heavy-tailed emission narrowed the gap but did not close it: the shared-ν Student-t ablation, the variant guide's heavy-tail recommendation at $K^\star = 3$ (Table 1), reached simulated IS excess kurtosis 4.68 with no penalty hyperparameter; the penalised per-state CHMM-t at $\lambda = 20$ overshot to 17.67 (with λ tuned at $K = 18$ and re-used at $K^\star = 3$ as an upper bound); CHMM-L sat between them at 5.24 without any shape parameter; and the bimodal $\hat{p}_k$ partition of CHMM-GED at $K = 18$ was the data-driven version of the Gaussian/Laplace hybrid one would otherwise assemble by hand. The Hill tail-index diagnostic bounds how much of this gap is real: the observed top-5% index $\hat{\alpha} \approx 3.15$ lies below 4, so the population fourth moment is plausibly infinite and the sample excess kurtosis of 7.68 is a draw-sensitive statistic rather than a fixed target [5]; on the threshold-local Hill estimate itself every family at $K^\star = 3$ sat inside the empirical $(2, 5)$ band (Table 3), the regime mixture reproducing the observed tail thickness at the depths the top-5% estimator sees even under Gaussian emissions. That is a finite-sample statement about the estimated threshold range, not evidence of a common power-law tail: for the Gaussian, Laplace, and exponential-type GED mixtures the estimator has no finite regular-variation index to converge to, and the in-band values are threshold-local shape diagnostics.

On the temporal axis, the same fit reduces the absolute growth-rate sample-ACF error relative to i.i.d. baselines at lags 1–63 while not beating the i.i.d. floor at lags 64–252 (0.041 against 0.036; Figure 2): the long-horizon slow decay is not reproduced, at any state count we fitted. The fitted spectra explain the shape of this outcome without explaining it away: the likelihood fits concentrate their ACF budget on one or two short-to-medium-horizon modes (dominant $|\lambda|$ between 0.93 and 0.95, half-lives around two to three weeks) at every K, so the near-band gain and the far-band failure come from the same fitted configuration. What the fitted models select is a statement about the estimation criterion as much as about the class; the realizable ACF-targeted comparison and the marginal-versus-ACF frontier of Section 4.2 measure the class-versus-criterion split directly (the marginal body comes along with the ACF under the frontier's in-sample body criterion; the deep tail is the open axis), and the appendix diagnostics keep those questions separate throughout.

For state-count selection these results are statements about fitted outcomes. In the $K = 2$ replication on SPY, the IS KS pass rate fell to 79.0% (well below the 91.5% for CHMM-N at $K^\star = 3$), while the absolute growth-rate ACF-MAE stayed essentially constant (0.0501 versus 0.0462) and the fitted $K = 2$ spectrum carried the entire absolute growth-rate ACF in a single fitted mode ($\lambda_2 = 0.942$, half-life ≈ 11.6 trading days). The short-and-medium-horizon ACF gain over the i.i.d. floor was already present at any $K \geq 2$, independent of initialisation, and moving from $K = 2$ to 3 changed the marginal fit, not the fitted temporal structure, so the operational choice of K was driven by the marginal distribution, and the emission-family ablation showed which symmetric heavy-tailed family carried the residual kurtosis gap most clearly. These are descriptions of what the likelihood fits deliver at each K, not bounds on what each class could attain under a different objective.

This reading may look like it inverts Rydén et al. [12], who found the marginal adequate and the slow absolute-return ACF the one fact a Gaussian HMM could not fit well by maximum likelihood. The difference is the data, not a contradiction: Rydén et al. read the stylized facts on series winsorised at four sample standard deviations, following Granger and Ding, which brought the kurtosis of absolute returns within reach of a Gaussian mixture; their own fits on the raw subseries already showed the other channel binding, with the two-regime models understating the absolute-return kurtosis (8.2 against an observed 27.1, and 14.0 against 33.7). We retain the heavy tail as the risk signal of interest, and on these data the fitted low-state Gaussian mixtures fell well short of the observed excess kurtosis, so the binding channel is distributional, and, by the frontier experiment of Section 4.2, specifically the deep tail rather than the distribution body. On the ACF we agree with their in-principle finite-geometric-memory limit, restated in the Conclusion: our ACF gain over the i.i.d. floor is confined to lags 1–63, the far band (64–252) is not reproduced, and the structural fix for genuinely slow decay is the semi-Markov chain of Bulla and Bulla [13].

One real limitation: a CHMM fit once in sample does not generalise across regime introductions. The GLD non-equity stress and the W2 (COVID 2020) and W4 (2022 rate-hike) walk-forward folds were regime introductions whose OoS marginal sat outside the range covered by any of the IS states; on those slices, OoS performance broke down (Table S38). Single-name equity returns are well known to carry structural breaks [60–62], and the cross-ticker concentration of failures on names that introduced a new regime (LLY, UNH, NEM) was the same pattern at the panel level. We recommend periodic refit [63, 64] for non-stress equity slices. It did not save GLD or the W2 and W4 stress folds, where the IS distribution simply did not span the OoS slice and faster refit cadences did not close the gap. Bayesian online change-point detection [65] and the particle-filter recursion of Fearnhead and Liu [66] detect a regime introduction rather than absorbing it into a static fit; integrating online detection with the CHMM is a natural follow-up. The implication for K-selection is that the choice should be driven by the marginal distribution, not the ACF. Held-out KS, log-likelihood, and BIC were the criteria that actually moved with K on held-out data, while stress-fold rejections reflected what the out-of-sample slice contained rather than a flaw in the in-sample fit. A second, data-scope limitation is the vendor feed switch inside the OoS window: the 2024 segment uses Polygon consolidated aggregates while the 2025–2026 segment uses Alpaca IEX-only bars, the raw files are date-disjoint, and feed effects cannot be separated from genuine time variation (Appendix A.5); all OoS statements should therefore be read as joint statements about the held-out period and its mixed feed.

Periodic refit handled non-stress drift but not regime introductions; two further scope limits remain, the stylized facts we evaluated against and the model size at which we compete with deep generators. The Cont [5] stylized-fact list also includes the leverage effect, the negative $\mathrm{Corr}(G_t, |G_{t+1}|)$ characteristic of equity returns; we evaluated against only the three symmetric stylized facts and did not target the leverage effect. A partial effect nonetheless emerges from the regime means: CHMM-N's per-path envelope brackets the observed IS value of -0.135 and the other variants recover about half of it (Table S25), but the dynamic effect is not modelled, and emission-level asymmetry is left to future work. On parametric complexity, CHMM-N at $K^\star = 3$ has $K(K-1) + 2K = 12$ free parameters, rising to 342 at the $K = 18$ sensitivity ceiling (CHMM-t adds K degrees-of-freedom parameters, CHMM-GED adds K shape parameters; the initial distribution is held fixed at uniform in the Gaussian fitter and is therefore not counted, while the heavy-tailed fitters update it as γ_1, adding $K-1$ to their counts), whereas the convolutional generator-critic architecture of Wiese et al. [10] sits orders of magnitude higher. The CHMM therefore competes against deep generators at a parameter budget small enough to make the fitted-transition rank and emission-separation conditions inspectable, alongside posterior diagnostics and the closed-form eigenvalue argument that motivates our $K^\star$ choice. The same budget also keeps each per-state emission's location, scale, and shape directly inspectable as an economic regime, whereas a deep generator's latent representation does not separate into interpretable regimes. These limits are addressable within the unified four-family framework: asymmetric emissions, a time-varying transition matrix, and richer dependence structures.

6 Conclusion

A continuous-emission HMM trained by Baum-Welch with quantile-based initialisation, at $K^\star = 3$ on daily US-equity returns, substantially improves the heavy-tailed marginal fit over the Gaussian baseline, leaves the raw growth rate serially uncorrelated, and reduces the absolute growth-rate ACF error relative to i.i.d. baselines at lags 1–63, while it does not beat the i.i.d. floor in the far band (lags 64–252; 0.041 against 0.036, slightly worse): the long-horizon slow decay is not reproduced by the fitted models, and exact power-law asymptotics are structurally out of reach for any finite chain (the ACF is a finite sum of geometric modes), requiring a different duration law or long-memory latent process. The fitted evidence is consistent throughout: the mixture-of-eigenvalues rank condition [14–16] behind the Rydén et al. [12] limitation was not empirically active on the SPY fits at $K \in \{3, 18\}$, and the converged multistart cross-ticker check found $K = 18$ improving the in-sample near-band ACF match at only 7/31 tickers (median difference 0.005 in $K^\star = 3$'s favour).

Over the finite lag band studied, the realizable capacity and frontier experiments show that the observed finite-band ACF gap is not a class-capacity limit at the achieved body-fit level: valid three-state chains optimized directly on the ACF track it roughly three times more closely than the published converged multistart likelihood fits do (median near-band MAE 0.0162 against 0.0497), and a weighted-objective sweep finds valid three-state fits whose ACF error is close to the best achieved in the sweep while their distribution-body fit reaches the multistart likelihood fit's level under the frontier's in-sample CDF criterion (marginal density within 0.011 nats per observation), with the joint behaviour holding at the typical ticker (24/31 tickers within a 1.5 regret factor on both axes at once). Under that criterion the ACF and the bulk marginal do not compete at three states; this rejects a necessary trade-off between the two measured targets at the achieved levels, not marginal or dependence trade-offs in general. The open axis is the deep tail: the ACF-competitive fits' 1%/99% quantile errors are roughly three times the multistart likelihood fit's, with mixture excess kurtosis 2.3 against 7.0 as the descriptive counterpart, and whether one three-state chain can carry the tail and the ACF together was not established in either direction.

A held-out caveat attaches: at the median single-name ticker the IS-fitted persistence did not transfer to the 2024–2026 window (both state counts sit behind the OoS zero-curve reference), while SPY itself did transfer. Within the unified EM framework (shared forward-backward recursion and transition update, family differences localised to the emission M-step), the shared-ν Student-t variant narrowed the excess kurtosis gap to 4.68 against observed 7.68 without a tuning hyperparameter, an exploratory recommendation selected on the OoS window.

The two applications behaved as designed within stated limits. The regime-conditional Value-at-Risk was not rejected by the Christoffersen joint conditional-coverage test at $\alpha = 0.05$ on the main OoS window or on 19/24 walk-forward rows, with rejections concentrated on the W2 COVID and W4 rate-hike stress folds; the strict-tail ($\alpha = 0.01$) tier is exploratory, because a finite-sample calibration shows the asymptotic DQ reference anticonservative at $T = 573$ and paired quantile losses rank no panel model above another. In the cross-asset composition, a suggestive 2×2 experiment on identical quarter-level targets under strict common random numbers (nine complete quarter blocks on one basket) showed that quarterly refit of the dependence layer reduces OoS dependence error by an order of magnitude more than the Gaussian-versus-Student-t family choice.

Three scope bounds temper these results: the i.i.d. bootstrap beats the CHMM on raw single-window OoS KS (a marginal-axis result), the volatility-clustering axis itself is led by the explicit-duration HSMM-N (a multistart local-EM fit whose 0.0394 per-path figure holds at its declared $D_{\max} = 200$ truncation, moving across 0.039–0.046 over the truncation-sensitivity grid) and MS-GARCH at $K = 3$ (0.0408) over CHMM-N (0.0462), both at substantially weaker marginals, and the OoS window contains a vendor feed switch that confounds feed effects with time variation. No formal privacy guarantee is claimed for the synthetic output; the scope is daily US equities under stationary OoS, and we recommend periodic refit for production use.

The primary extension points outside the model class: the mixture-of-eigenvalues identity caps the absolute growth-rate ACF at finitely many geometric decay modes, so exact non-geometric asymptotics require a different duration law or long-memory latent process: a dedicated maximum-likelihood explicit-duration HSMM with heavy-tailed sojourns [13, 80] or a long-memory latent-volatility formulation [20], directions our Gamma-sojourn sensitivity results at $K = 18$ (Table S15) already indicate. An equally direct lever inside the class is the estimation objective: the realizable capacity experiment shows a valid three-state chain can track the finite-horizon sample ACF about three times more closely than the likelihood fits do, and the frontier sweep shows the marginal body can come along at essentially no ACF cost under its in-sample body criterion, so an ACF-aware or weighted criterion (the frontier's weighted objective is a working prototype) is a natural alternative to changing the duration law, with the deep tail the axis such a criterion must be scored on. Within the framework, the natural next steps are skewed emission families (targeting the leverage effect), covariate-dependent transitions or online change-point detection for the stress folds the static fit leaves open (untested here, not demonstrated remedies), factor or vine dependence structures at larger cross-sections, an innovation-copula construction that would preserve per-asset serial dependence in the composition, and an end-to-end portfolio-level test of the regime-conditional VaR. Each targets one stated limitation of the present scope; the sojourn-law extension is the one that leaves the finite geometric-memory limit behind.

Conflict of Interest Statement. The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions. J.V. directed the study. A.A. developed the continuous model and simulation code, implemented Baum-Welch, conducted the in-sample and out-of-sample analysis, and generated all figures and tables. C.J. contributed to EM and numerical block-update methodology review and to validation of the cross-decade and cross-ticker pipelines. All authors edited and reviewed the final manuscript.

Data Availability Statement. The manuscript sources and the derived result tables it typesets are available under an MIT license from the paper repository: https://github.com/varnerlab/CHMM-Paper-Repository. All executable code (fitting, simulation, decoding, and validation) lives in the companion Julia repository `CHMM-Model-Repository`: https://github.com/varnerlab/CHMM-Model-Repository. A master script rebuilds the headline stages, and every other reported table has a dedicated runner; the manifest `results/artifacts_manifest.csv` maps each table and figure to its runner, artifact, and seed, and an automated consistency gate checks the printed values against their artifacts. The results reported here correspond to commit `7230f71` of that repository. Daily prices for the 2014–2026 windows are sourced from Polygon.io consolidated aggregates via Massive (2014-01-03 to 2024-12-31) and Alpaca Markets IEX-only bars (2025-01-03 to 2026-04-20); the raw vendor files are date-disjoint, and the feed switch falls inside the OoS window (Appendix A.5). The cross-decade panel uses CRSP data under its institutional license; those source data are not redistributed.

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A Supplementary Material

This appendix collects supplementary material referenced in the main text: the EM forward-backward recursions and algorithm pseudocode, and the validation metric definitions. It also reports the formal propositions, the state-resolution and emission-family sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

Forward-backward recursions and M-step updates. For reference, the Baum-Welch description in the main text is underpinned by the standard log-space recursions [67, 68]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log f_k(O_1), \tag{9}$$
$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log f_j(O_t), \quad t = 2, \dots, T, \tag{10}$$

$$\log \beta_T(k) = 0, \tag{11}$$
$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log f_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1, \tag{12}$$

where $\text{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, f_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, f_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{13}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{14}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < \epsilon$ with $\epsilon = 10^{-4}$, where $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T(k)$ is the standard Baum-Welch observed-data log-likelihood of the parameters entering iteration n, evaluated by that iteration's forward pass *before* any M-step can mutate them; the check runs before the update, so the returned parameters are always an evaluated iterate with a matching smoothed posterior, and the loop makes one final evaluation-only pass so that the last M-step update is itself scored. For the exact Gaussian and Laplace conditional maximisations the likelihood is monotone across iterations and the last evaluated iterate is returned; for the Student-t and GED numerical block updates ascent is not guaranteed, the trace is monitored instead (see the numerical-optimisation scope paragraph of the supplementary propositions), and the best evaluated finite iterate $\arg\max_n \mathcal{L}^{(n)}$ is returned. Implementation note on $\boldsymbol{\pi}$: the Gaussian fitter departs from Eq. (14) in one place, holding $\boldsymbol{\pi}$ fixed at its uniform initial value instead of applying $\pi_k^{\text{new}} = \gamma_1(k)$; the Student-$t$, Laplace, and GED fitters apply the γ_1 update. Parameter counts and BIC penalties for CHMM-N accordingly exclude $\boldsymbol{\pi}$. Simulation draws S_1 from the stationary distribution $\bar{\boldsymbol{\pi}}$ of $\hat{\mathbf{T}}$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM/ECME, closed form given $u_{t,k}$). For Student-t emissions $f_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [21], Liu and Rubin [22] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{15}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{16}$$

The degrees-of-freedom parameter ν_k is then refined by a 40-iteration one-dimensional golden-section search on the per-state posterior-weighted objective $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$, with the smoothed posteriors γ held fixed. Maximising this marginal state-weighted log-likelihood for ν_k, rather than the augmented-data u-functional, follows the spirit of the ECME construction of Liu and Rubin [22]; in their single-component i.i.d. setting, where all $\gamma_t(k) = 1$, that objective is the observed-data likelihood, whereas already for i.i.d. mixtures the posterior-weighted objective differs from the observed-data likelihood [21], and in the HMM the observed-data likelihood also depends on ν_k through the forward recursion, so this update is a hybrid generalised block-coordinate step on a surrogate objective and does not inherit the ECME observed-data ascent guarantee. We diagnose convergence from the observed-data log-likelihood trace rather than assert monotonicity, returning the best evaluated finite iterate (which is also restored if an update degenerates to a non-finite likelihood). We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \, \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{17}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [21], Liu and Rubin [22]-style ν update; the rate sweep and its effect on simulated IS excess kurtosis are reported in the K-selection and ν_k diagnostics appendix, Table S32).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $f_k(x) = \frac{1}{2b_k}\exp(-|x-\mu_k|/b_k)$, with the per-state Laplace scale denoted b_k to avoid collision with the backward variable β_t, the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k))\,, \qquad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\,|O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{18}$$

The weighted median is the first order statistic whose cumulative posterior weight reaches half the total, the minimiser of the weighted L^1 objective $\sum_t \gamma_t(k)\,|O_t - \mu|$; when the half-weight falls exactly between two adjacent order statistics, every point in the closed interval between them is a minimiser. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM-style update with closed-form scale). For Generalized Error Distribution emissions $f_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp\big(-(|x - \mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three block updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, the first block updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \text{GoldenSearchMin}_{\mu \in I_k} \sum_t \gamma_t(k) \left|O_t - \mu\right|^{p_k^{(n)}}, \qquad I_k \;=\; \big[\min_t O_t,\, \max_t O_t\big], \tag{19}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k) \left|O_t - \mu_k^{(n+1)}\right|^{p_k^{(n)}}\right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{20}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \text{GoldenSearchMax}_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \, \log f_k\big(O_t;\, \mu_k^{(n+1)},\, \alpha_k^{(n+1)},\, p\big), \tag{21}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. The scale update is the exact conditional maximiser. The location objective is convex when $p_k \geq 1$ but can be nonconvex when $p_k < 1$, and both bracketed searches are finite numerical approximations. Accordingly, we use "ECM-style" for this three-block procedure and assess convergence from the observed-data log-likelihood trace; the exact ECM monotonicity result [71] would require every block to attain a conditional maximum. At the boundary cases, $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$) and $p_k = 1$ recovers Laplace (with $b = \alpha$).

Algorithm descriptions. The unified EM procedure shared across all four emission families is given in Algorithm 1: a shared framework (quantile-based initialisation, log-space forward-backward, transition update) plus four family-specific branches at lines 3, 13, and 35. The simulator is given in Algorithm 2, and the cross-asset rank-reordering simulator for the multi-asset construction in Algorithm 3. Viterbi and SIM-resampling pseudocode are provided in the companion code repository; both are textbook constructions and we omit them from the appendix.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 35 branch on the emission family $F \in \{\mathcal{N}, t, L, \text{GED}\}$; every other line is identical across families. The CHMM-N and CHMM-L variants instantiate classical EM; CHMM-t and CHMM-GED use ECM-style block updates whose shape step is a finite one-dimensional golden-section search. Hyperparameter defaults used throughout this paper: convergence tolerance $\epsilon = 10^{-4}$ on $\Delta\mathcal{L}$, `max_iter` $= 60$, 40-iteration golden-section sub-step on every bracketed update, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 6$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and a μ_k search interval widened to cover the observed data range.

Require: Observations $O_{1:T}$, number of states K, tolerance ϵ, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \text{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \text{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \text{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \text{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$, $\quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \text{median}(C_k)$, $\quad b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$. {Laplace scale b_k.}
8:     $F = \text{GED}$: $\mu_k \leftarrow \text{mean}(C_k)$, $\quad \alpha_k \leftarrow \max(\text{std}(C_k), 10^{-6})$, $\quad p_k \leftarrow p^{(0)}$.
9: **end for**
10: $T_{ij} \leftarrow 1/K$; $\quad \pi_k \leftarrow 1/K$; $\quad \mathcal{L}_{\text{prev}} \leftarrow -\infty$.
11: **EM Loop.** {one final evaluation-only pass, so every returned iterate has an evaluated likelihood.}
12: **for** $n = 1$ to `max_iter` $+ 1$ **do**
13:   **E-Step, per-family emission log-likelihood.**
14:   **switch** F
15:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
16:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
17:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
18:     $F = \text{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log\Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
19:   **E-Step, shared log-space forward-backward.**
20:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
21:   **for** $t = 2$ to T, $j = 1$ to K **do**
22:     $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
23:   **end for**
24:   $\log \beta_T(k) \leftarrow 0$.
25:   **for** $t = T-1$ down to 1, $i = 1$ to K **do**
26:     $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
27:   **end for**
28:   Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (13).

Algorithm 2 CHMM simulation of synthetic excess growth rate paths (unified four-family framework). The emission draw at line 4 branches on the family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$ exactly as in Algorithm 1: $\mathcal{N}(\mu_k, \sigma_k^2)$ for CHMM-N, $t_{\nu_k}(\mu_k, \sigma_k)$ for CHMM-t, Laplace(μ_k, b_k) for CHMM-L, GED(μ_k, α_k, p_k) for CHMM-GED; the latent chain and transition step are shared.

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}, F)$ with $\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$; (μ_k, α_k, p_k) for $F = \mathrm{GED}$. Number of steps M, Number of paths P

Ensure: Simulated growth-rate paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

1: **for** $p = 1$ to P **do**
2:   Sample initial state S_1 from the stationary distribution of $\hat{\mathbf{T}}$.
3:   **for** $t = 1$ to M **do**
4:     Emit observation $\hat{G}^{(p)}_t \sim f_{S_t}(\,\cdot\,; \boldsymbol{\theta}_{S_t})$ with f_k the F-family per-state density.
5:     **if** $t < M$ **then**
6:       Transition to next state $S_{t+1} \sim \mathrm{Categorical}(\mathbf{T}_{S_t,\cdot})$.
7:     **end if**
8:   **end for**
9: **end for**
10: **return** $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

Likelihood of the current iterate. $\mathcal{L}^{(n)} \leftarrow \operatorname*{logsumexp}_k(\log \alpha_T(k))$ {observed-data log-likelihood of the parameters entering iteration n, evaluated before any update; for $F \in \{t, \mathrm{GED}\}$ record (parameters, $\mathcal{L}^{(n)}, \gamma$) as the best checkpoint whenever $\mathcal{L}^{(n)}$ improves on it.}
  if $|\mathcal{L}^{(n)} - \mathcal{L}_{\mathrm{prev}}| < \epsilon$ **or** $n = \texttt{max\_iter} + 1$ **then**
    break {the just-evaluated iterate is returned; no unevaluated update can escape the loop.}
  end if
  $\mathcal{L}_{\mathrm{prev}} \leftarrow \mathcal{L}^{(n)}$.
  Shared transition update. $T_{ij} \leftarrow \sum_{t=1}^{T-1} \xi_t(i,j) \,/\, \sum_{t=1}^{T-1} \gamma_t(i)$; $\quad \pi_k \leftarrow \gamma_1(k)$ for $F \in \{t, L, \mathrm{GED}\}$ {both sums run over $t = 1, \dots, T-1$, one fewer term than the γ sums of the emission updates below; for $F = \mathcal{N}$, $\boldsymbol{\pi}$ stays at its uniform initial value by documented convention.}
  M-Step, per-family emission update.
  switch F
    $F = \mathcal{N}$ {classical Baum-Welch [69].}
      $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
      $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
    $F = t$ {ECM-style block updates after Peel and McLachlan [21], Liu and Rubin [22]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k at fixed γ.}
      $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
      $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
      $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
      $\nu_k \leftarrow \mathrm{GoldenSearchMax}_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40 iterations.}
    $F = L$ {closed-form weighted Laplace MLE.}
      $\mu_k \leftarrow \mathrm{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {first order statistic with cumulative weight $\geq W_k/2$.}
      $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
    $F = \mathrm{GED}$ {three-stage ECM-style update: bracketed μ_k, closed-form α_k, bracketed p_k.}
      $\mu_k \leftarrow \mathrm{GoldenSearchMin}_{\mu \in I_k} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40 iterations on $I_k = [\min_t O_t, \max_t O_t]$.}
      $\alpha_k \leftarrow \max\Big(\big[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\big]^{1/p_k},\ 10^{-6}\Big)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k / \partial \alpha$.}
      $p_k \leftarrow \mathrm{GoldenSearchMax}_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \mathrm{GED}(O_t; \mu_k, \alpha_k, p)$ {40 iterations.}
end for
return $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$; (μ_k, α_k, p_k) for $F = \mathrm{GED}$. For $F \in \{\mathcal{N}, L\}$ the last evaluated iterate is returned (EM ascent is exact for these families); for $F \in \{t, \mathrm{GED}\}$, whose hybrid surrogate blocks do not guarantee observed-data ascent, the best evaluated finite checkpoint $\arg\max_n \mathcal{L}^{(n)}$ is returned, and a non-finite forward pass restores that checkpoint as well. In all cases the returned parameters carry an evaluated likelihood and a matching posterior γ.}

Algorithm 3 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

1: **for** $p = 1$ to P **do**
2: Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T\times d}$ from $\mathcal{C}(\Sigma)$:
3: $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T\times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
4: $Y \leftarrow ZL^\top$
5: **if** $\mathcal{C} =$ Gaussian **then**
6: $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
7: **else**
8: Draw $W \sim \chi^2_\nu/\nu$ i.i.d. for $t = 1, \ldots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
9: $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
10: **end if**
11: **for** $j = 1$ to d **do**
12: Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 2.
13: Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \leq \cdots \leq \tilde{g}_{j,(T)}^{(p)}$.
14: Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \ldots, T$.
15: $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ for $t = 1, \ldots, T$.
16: **end for**
17: **end for**
18: **return** $\{\hat{G}_{j,t}^{(p)}\}$

Scope remark on Algorithm 3: the copula sample $\mathbf{U}^{(p)}$ is drawn independently across the T rows, so step 4 assigns each asset's sorted CHMM values to i.i.d. time ranks. Each column of the output is a permutation of the corresponding simulated CHMM path: the per-asset empirical marginal is preserved exactly and the contemporaneous cross-asset rank dependence follows $\mathcal{C}(\Sigma)$, but the per-asset serial dependence of the underlying CHMM path (volatility clustering, regime residence) is destroyed by the reordering. The output is a cross-sectional object, and the cross-asset evaluation accordingly scores per-asset marginal fit and cross-asset correlation reproduction only.

Per-state shape diagnostics: CHMM-t and CHMM-GED. This appendix backs the main-text discussion of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

CHMM-t per-state ν_k. The per-state ν_k histogram for the $K = 18$ CHMM-t fit is plotted in Figure S1. Thirteen of the eighteen states rested at the upper ECM bracket ($\nu_k = 50$), and only two states lay near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture excess kurtosis was driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The generator comparison reports and discusses the $\hat{p}_k$ partition (Figure S4). Briefly, the $K = 18$ CHMM-GED fit on SPY IS was bimodal: eleven states attained exactly $p_k = 3.0$ (thirteen were Gaussian-like with $p_k \geq 1.85$), one state landed at intermediate $p_k = 1.6$, and four states clustered in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY was 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket was non-binding.

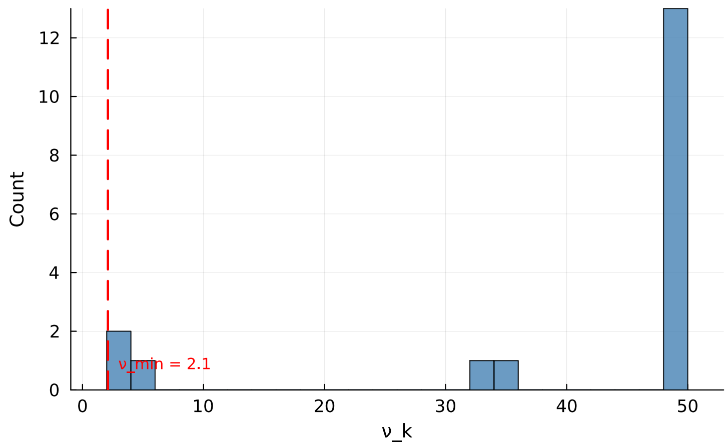


Figure S1. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

The bimodal partition replicated across 10 Monte Carlo seeds and across the cross-asset universe (full robustness panel in the cross-asset robustness appendix).

Bracket-sensitivity sweeps. A CHMM-t bracket sweep on $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0, \infty\}$ at $K = 18$ showed that the excess kurtosis overshoot is a feature of the Student-t emission at $K = 18$, not a lower-bracket artefact. The simulated IS excess kurtosis stayed in a heavy-tailed band across finite brackets and collapsed only in the full Gaussian limit. The analogous CHMM-GED sweep on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ left the non-Gaussian state count invariant, with Laplace-like states in $\{3, 4, 5\}$; tightening $p_{\max}$ from 4.0 to 3.0 cost only a few nats of LL. The $p_{\min}$ sweep on $\{0.5, 0.7, 0.85\}$ was fully degenerate: every floor produced an identical fit because the smallest observed $\hat{p}_k$ on SPY was 0.86.

Evaluation summary. Observed growth rates G_t are fit by Baum-Welch CHMM at the default $K^\star = 3$, with $K = 18$ retained as the sensitivity reference for excess kurtosis fidelity; 1,000 paths are then simulated and per-asset metrics computed (Table 2). For the multi-asset analysis we reuse the per-asset $K^\star = 3$ fits as marginals, draw a copula sample $\mathbf{U} \sim \mathcal{C}(\Sigma)$ from the fitted Student-t copula on ranks, rank-reorder the per-asset single-asset paths to match $\mathbf{U}$, and report cross-asset metrics (Table 4). The single-asset CHMM framework is shared across both; the CHMM training and simulation steps are Algorithms 1 and 2, the cross-asset rank-reordering step is Algorithm 3, and the architecture diagram for the single-asset framework is Figure 1.

A.2 Validation Metric Details

This appendix documents the eight per-path metrics used in the evaluation protocol. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [72, 73] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|, \tag{22}$$

where F_n and G_m are the empirical CDFs of the two samples and n and m their sizes. The KS pass rate is the fraction of paths whose marginal is not rejected against the reference data at $\alpha = 0.05$. We read it as a descriptive marginal-fidelity score, not a calibrated test: the asymptotic two-sample KS null assumes i.i.d. observations, whereas observed and simulated paths carry serial dependence, so the nominal p-values are not calibrated here. A stationary block-bootstrap recalibration (appendix) supplies dependence-aware critical values by drawing, per null replicate, two *independent* block resamples of the observed series and computing the KS statistic between them, so the null replicate mimics the actual observed-versus-simulated comparison; the threshold remains conditional on the empirical IS distribution (no generator refit per replicate), so the recalibrated pass rates are also read descriptively.

Anderson–Darling (AD) pass rate. The AD test [81] is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate excess kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

Autocorrelation-function MAE (ACF-MAE). Mean absolute error between the observed and simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags, computed per simulated path and averaged over the P paths,

$$\text{ACF-MAE} = \frac{1}{P}\sum_{p=1}^{P}\frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \hat{\rho}^{\text{sim},(p)}_{|G|}(\tau)\right|. \tag{23}$$

This is the direct diagnostic for the Rydén et al. [12] limitation.

Threshold-local Hill estimate. For the magnitude series $|G_t|$ with descending order statistics $x_{(1)} \geq \cdots \geq x_{(T)}$ and $k = \lceil 0.05\,T \rceil$ upper-tail exceedances, the Hill estimator [76] is

$$\hat{\xi} = \frac{1}{k}\sum_{i=1}^{k}\log\frac{x_{(i)}}{x_{(k+1)}}, \qquad \hat{\alpha} = 1/\hat{\xi}, \tag{24}$$

so that $\hat{\alpha}$ estimates the tail index of a Pareto-type (regularly varying) tail when one exists. For distributions without a regularly varying tail, including the finite Gaussian, Laplace, and exponential-type GED mixtures simulated here, the estimator has no finite population index to converge to, and $\hat{\alpha}$ at fixed k is a threshold-local shape diagnostic of the top-5% exceedance region rather than an asymptotic tail index; we use the term "tail index" for regularly varying models only and read the simulated values as threshold-local. The same estimator is applied to the observed window and to each simulated path at the same k fraction, so observed and simulated values are comparable at the depths the estimator sees. The observed confidence interval is a stationary block bootstrap (mean block length 10, $B = 2{,}000$); the simulated band is the across-path $[2.5\%, 97.5\%]$ quantile band. The per-side variant applies the estimator to gains and losses separately with equal exceedance counts.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [82],

$$W_1(F,G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|\, du. \tag{25}$$

Hellinger distance. A histogram-based overlap distance in $[0,1]$,

$$H(P,Q) = \frac{1}{\sqrt{2}}\sqrt{\sum_{i=1}^{B} (\sqrt{p_i} - \sqrt{q_i})^2}, \tag{26}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed growth-rate matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of the main text cross-asset analysis), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P}\sum_{p=1}^{P} \frac{2}{d(d-1)} \sum_{1\le i<j\le d} \left|\hat{\Sigma}^{(p)}_{\text{sim},\, ij} - \hat{\Sigma}_{\text{obs},\, ij}\right|.$$

A.2.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS sample-CRPS, reported as a tie-break alongside Table 2 in the main text, is a proper scoring rule complementing the KS pass rate [55]. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

$$\widehat{\text{CRPS}}_t = \frac{1}{N}\sum_{i=1}^{N} |x_i - y_t| - \frac{1}{N(N-1)} \sum_{1\le i<j\le N} |x_i - x_j|. \tag{27}$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout the main text; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test [57] on the per-t loss differential $d_t = \widehat{\text{CRPS}}^A_t - \widehat{\text{CRPS}}^B_t$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3} \rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The main-text panel covers exactly the five displayed CHMM rows (CHMM-N, penalised CHMM-t, CHMM-L, CHMM-GED, and shared-ν CHMM-t) over their ten pairwise loss differentials with Holm step-down correction; the supported statement is that no pairwise equal-mean-loss null is rejected after correction, together with HAC 95% confidence intervals on the mean differences of roughly ± 0.005–0.010. This is a non-rejection, not an equivalence result: no equivalence margin is pre-specified, and one pair (penalised CHMM-t vs. CHMM-L) is nominally significant before correction.

A.2.2 Christoffersen Conditional-Coverage Panel

The main-text back-test (Table 5) evaluates the regime-conditional VaR, whose threshold moves with the filtered state probabilities. This appendix reports the complementary Christoffersen [17] panel for the *unconditional* generator VaR, whose threshold is constant across t: $\mathrm{LR}_{\mathrm{uc}}$ (unconditional coverage; 74), $\mathrm{LR}_{\mathrm{ind}}$ (breach independence), and $\mathrm{LR}_{\mathrm{cc}} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α-quantile of vec(sim archive) across paths and time), then $\mathrm{br}_t = (R_t \leq \widehat{\mathrm{VaR}}_\alpha)$.

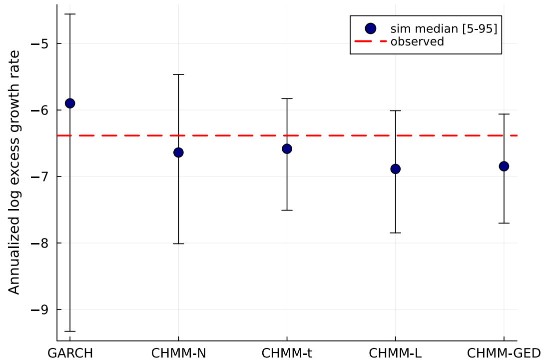
Two patterns stand out (Table S1). On the unconditional (Kupiec) leg no CHMM variant was rejected at either level on either window; the $\mathrm{LR}_{\mathrm{uc}}$ statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clustered at 3.03–3.83 (just below the χ^2_1 critical value 3.841), reflecting the integer-breach grid at $T_{\mathrm{OoS}} = 572$ noted in the main text table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejected: $\mathrm{LR}_{\mathrm{ind}}$ ranged over 4.71–56.46 across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility. The OoS-window breaches concentrate inside the high-volatility regime, and a constant-across-t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} \,|\, \mathrm{history}_t)$ through the per-state emission mixture; that is exactly the construction the main text builds and back-tests (Table 5), whose rows are not rejected by the joint conditional-coverage test that every constant-threshold generator fails here.

Table S1. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The CHMM-t row is the default unpenalised per-state-ν fit. "br rate" is the empirical breach rate; $\mathrm{LR_{uc}}$ is the Kupiec statistic ($\sim \chi^2_1$ under correct unconditional coverage); $\mathrm{LR_{ind}}$ is the Christoffersen independence statistic ($\sim \chi^2_1$ under independent breaches); $\mathrm{LR_{cc}} = \mathrm{LR_{uc}} + \mathrm{LR_{ind}}$ is the joint conditional-coverage statistic ($\sim \chi^2_2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

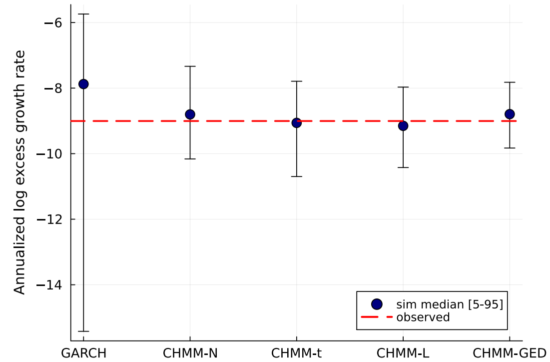
					Kupiec		Christ. ind		Christ. cc	
Model	Win	α	breaches	br rate	$\mathrm{LR_{uc}}$	p	$\mathrm{LR_{ind}}$	p	$\mathrm{LR_{cc}}$	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

A.2.3 Value-at-Risk and Expected-Shortfall Envelope Visualization (Companion to Table S16)

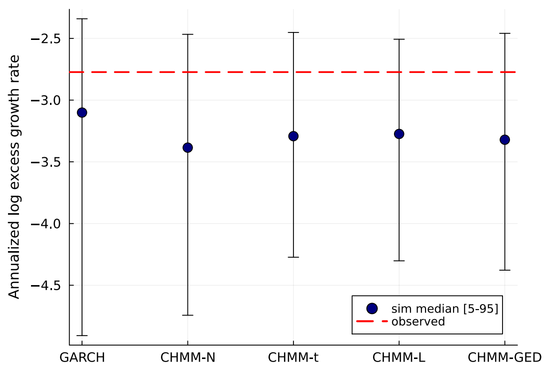
The median and $[5\%, 95\%]$ envelopes across simulated paths are visualised in Figure S2; the numeric values are reported in Table S16.



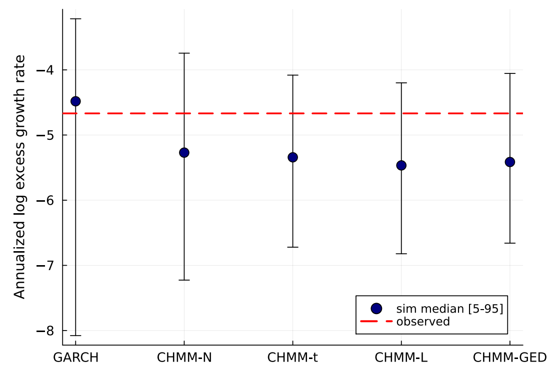
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure S2. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy under the evaluation protocol). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the conditional-coverage comparison among volatility-aware models (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Every observed red line falls inside the corresponding simulated envelope, so the observed tail statistics are visually compatible with the fitted unconditional distributions; this is a predictive diagnostic, and the formal coverage evidence is the Kupiec and Christoffersen likelihood-ratio panel of the main text VaR back-test.

A.3 Formal Theoretical Statements

The following assumptions support the spectral identity. We also state the scope of the numerical-optimisation, identifiability, and consistency claims so that those issues are separated from the empirical fitting procedure.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix* $\mathbf{T} \in \mathbb{R}^{K \times K}$ *is irreducible and aperiodic on the state space* $\{1, \ldots, K\}$*, so the chain admits a unique stationary distribution* $\bar{\boldsymbol{\pi}}$

satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$ *and* $\lambda_1 = 1$ *is a simple eigenvalue with all other eigenvalues strictly inside the unit disk [75].*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density* $f_k(\cdot;\boldsymbol{\theta}_k)$ *has finite second moment,* $\mathbb{E}_{f_k}[G^2] < \infty$*. For a non-degenerate absolute growth-rate ACF we further require the state-conditional absolute growth-rate means* $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ *to differ across at least two states; this rules out only the trivial flat-ACF case and is not needed for the ACF identity itself.*

Numerical optimisation scope. The CHMM-N and CHMM-L M-steps are exact conditional maximisations. For CHMM-t and CHMM-GED, exact maximisation of every numerical block would give the standard ECM/ECME ascent property [70, 71]. The implementation instead uses fixed-iteration bounded searches for ν_k, μ_k, and p_k. These are approximate block updates, and the GED location objective can be nonconvex when $p_k < 1$. We therefore do not claim exact likelihood monotonicity for every finite-precision iteration. Numerical convergence is assessed from the observed-data log-likelihood trace using the tolerance and iteration cap stated in Algorithm 1. Even when an ascent condition is enforced, bounded monotone likelihood values imply convergence of the likelihood sequence, not necessarily convergence of the full parameter sequence to a unique maximiser.

Numerical full-rank check on $\hat{\mathbf{T}}$. Full rank of $\mathbf{T}$ is a separate sufficient condition in the HMM identifiability results of Allman et al. [83] and Gassiat et al. [84]; it does not follow from irreducibility. As an empirical diagnostic, the fitted $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS was full rank in the numerical sense (smallest singular value $\sigma_{\min} \in [0.0007, 0.0170]$, condition numbers 63 to 1,620), and the deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ had numerical rank exactly $K - 1 = 17$ across all four families. This fitted-matrix check is not by itself a proof of population identifiability.

Proposition 1 (Conditional HMM identifiability)**.** *Suppose the number of states* K *is known,* $\mathbf{T}$ *has full rank, and the* K *per-state emission distributions are linearly independent. Then, by the nonparametric identifiability theorem of Gassiat et al. [84], the transition matrix and the emission distributions are identifiable from the joint law of three consecutive observations of the stationary chain, up to permutation of the latent-state labels.*

The theorem of Gassiat et al. [84] covers real-valued (continuous-emission) observations directly and is the result the displayed conditions come from; the earlier generic-identifiability theorem of Allman et al. [83], built on Kruskal's tensor decomposition, concerns HMMs with finitely many observed values and is cited here as the antecedent that the continuous-emission statement extends. *Scope.* Distinct Gaussian or Laplace location-scale components satisfy standard finite-mixture identifiability conditions [85], but this paper does not prove the required linear independence for the full varying-shape Student-t and GED families. In particular, the Student-t family has no moment-generating function at finite degrees of freedom, so an MGF argument is unavailable. The proposition is therefore a conditional statement, not a claim that pairwise-distinct fitted parameters alone establish identifiability for every emission family.

Consistency scope. Standard HMM maximum-likelihood consistency results require more than Assumptions 1–2: typically identifiability of the population model, continuity and integrability of the log-density, a compact or otherwise controlled parameter space, and a uniform mixing or forgetting condition [86, 87]. We do not assert that the two displayed assumptions alone verify all of those conditions for every varying-shape emission family. The empirical fitting and validation procedures in this paper do not rely on an asymptotic consistency theorem.

Proposition 2 (Marginal preservation under rank reordering)**.** *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j, and for simulated path p let the rank-reordered output be $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$, where $r^{(p)}_{j,t}$ is the rank of the path-p copula sample in column j at time t. Then for every asset j the empirical CDF of $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ equals that of $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$.*

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.4 Spectral ACF Identity: Extended Derivation

The closed-form mixture-of-eigenvalues form for the absolute growth-rate ACF, given in Eq. (8), is used throughout the main text. The bilinear cross-moment identity $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (14, 15; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 16); the underlying Markov-chain spectral decomposition is standard [75]. We restate the absolute growth-rate specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex and non-diagonalisable extensions that the main text does not. Our contribution is the explicit application of the spectral form to recast the empirical Rydén et al. [12] low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, and the empirical demonstration that the rank constraint was not what limited the fit at moderate K on equity-return data.

Theorem 1 (Mixture-of-eigenvalues identity for the absolute growth-rate ACF)**.** *Let $\{(s_t, G_t)\}_{t\in\mathbb{Z}}$ be a stationary CHMM on $\{1,\ldots,K\}$ with irreducible aperiodic transition matrix $\mathbf{T}$ (Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write $\bar{\boldsymbol{\pi}}$ for the unique stationary distribution, $\mathbf{D} = \mathrm{diag}(\bar{\boldsymbol{\pi}})$, $m_k = \mathbb{E}[|G_t| \mid s_t = k]$, $\mathbf{m} = (m_1,\ldots,m_K)^\top$, $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top\mathbf{m}$, and $\sigma^2_{|G|} = \mathrm{Var}(|G_t|)$. Assume $\mathbf{T}$ is diagonalisable with right eigenvectors $\mathbf{v}_k$ and left eigenvectors $\mathbf{w}_k$ biorthonormal, $\mathbf{w}_k^\top\mathbf{v}_l = \delta_{kl}$, indexed so that $\lambda_1 = 1 > |\lambda_2| \geq \cdots \geq |\lambda_K|$. Then for every integer lag $\tau \geq 1$,*

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} a_k\,\lambda_k^\tau, \qquad a_k \;=\; \frac{(\mathbf{m}^\top\mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top\mathbf{m})}{\sigma^2_{|G|}}.$$

Proof. By stationarity, $\rho_{|G|}(\tau) = (\mathbb{E}[|G_t|\,|G_{t+\tau}|] - \mu^2)/\sigma^2_{|G|}$, so it suffices to evaluate the cross-moment for $\tau \geq 1$. Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using HMM conditional independence ($G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \geq 1$) plus the Markov factorisation $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i(\mathbf{T}^\tau)_{ij}$ gives the bilinear form

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top\mathbf{D}\,\mathbf{T}^\tau\,\mathbf{m}, \qquad \tau \geq 1. \tag{28}$$

Perron–Frobenius gives $\lambda_1 = 1$ simple with $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$ [75]; the dyadic expansion $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k\geq 2}\lambda_k^\tau\mathbf{v}_k\mathbf{w}_k^\top$ separates the dominant projector. The $k = 1$ term contributes μ^2 and cancels in the autocovariance, leaving $\mathrm{Cov}(|G_t|, |G_{t+\tau}|) = \sum_{k\geq 2} c_k\lambda_k^\tau$ with $c_k = (\mathbf{m}^\top\mathbf{D}\mathbf{v}_k)(\mathbf{w}_k^\top\mathbf{m})$; dividing by $\sigma^2_{|G|}$ gives Equation (8). □

Lag zero and within-state variance. The identity (28) fails at $\tau = 0$ since $\mathbb{E}[G_t^2 \mid s_t = i] \neq m_i^2$. Eigenvalue completeness gives $\sum_{k\geq 2} c_k = \mathrm{Var}(m_{s_t})$, the between-state variance, so $\sum_{k\geq 2} a_k = \mathrm{Var}(m_{s_t})/\sigma^2_{|G|} \in [0, 1]$. The normalized within-state variance $\mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]/\sigma^2_{|G|} = 1 - \sum_{k\geq 2} a_k$ explains the difference between $\rho_{|G|}(0) = 1$ and the spectral expression evaluated at lag zero. The further change to lag one, $\rho_{|G|}(1) = \sum_{k\geq 2} a_k \lambda_k$, also reflects transition decay through the factors λ_k.

Squared growth-rate, complex-eigenvalue, and Jordan-block extensions. If every emission has a finite fourth moment, replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (\mathbb{E}[G_t^2 \mid s_t = i])_i$ leaves the proof termwise valid and gives $\rho_{G^2}(\tau) = \sum_{k\geq 2} \tilde{a}_k \lambda_k^\tau$ on the same non-unit spectrum. This additional condition is not automatic for CHMM-t when some $\nu_k \leq 4$. Complex-conjugate eigenvalue pairs $(re^{\pm i\theta}, \cdot)$ contribute real damped oscillations $Ar^\tau \cos(\theta\tau) + Br^\tau \sin(\theta\tau)$. For non-diagonalisable $\mathbf{T}$, a Jordan expansion instead adds polynomial-times-geometric prefactors, with exponential rate controlled by $\max_{k\geq 2} |\lambda_k|$; Theorem 1, which assumes diagonalizability, does not cover that case as stated. The dominant timescale used in the main text spectral mechanism corresponded to a real positive λ_2 on the fits of this paper.

A.5 OoS Feed-Boundary Diagnostic

The OoS series spans January 4, 2024 through April 20, 2026 (573 sessions; $T_{\mathrm{OoS}} = 572$ within-window growth rates, with the boundary return into the first held-out session additionally used by the VaR back-tests) and is sourced from two vendors: Polygon.io consolidated aggregates via Massive through December 31, 2024, and Alpaca Markets IEX-only bars from January 3, 2025. The raw vendor files are date-disjoint, so no same-date cross-vendor sample exists and no vendor-equality check is possible from the repository data. A row-equality check between the stitched OoS file and the Alpaca source file would be uninformative here: the stitched rows are copied from those Alpaca rows, so exact equality holds by construction, and we do not report such a check. Instead we report a descriptive comparison of the two OoS segments on either side of the feed switch (Table S2), which characterises the boundary but, because the segments cover different calendar periods, cannot separate a feed effect from genuine time variation.

Table S2. OoS feed-boundary diagnostic. Per-segment annualised growth-rate summaries on either side of the January 3, 2025 feed switch, from the VWAP field used throughout the paper and from the close column as a price-field robustness check. The two-sample KS statistic compares the pre- and post-switch segments and is confounded with time variation. The final block reports the stored CHMM-N $K^\star = 3$ filter-conditional VaR breaches by segment.

Field	Segment	n	mean	sd	exc. kurt
VWAP	Polygon (2024)	249	+0.228	1.640	+2.31
VWAP	Alpaca/IEX (2025–26)	322	+0.142	2.215	+5.24
close	Polygon (2024)	249	+0.229	2.004	+1.81
close	Alpaca/IEX (2025–26)	322	+0.141	2.913	+19.77

KS pre vs. post: VWAP $D = 0.099$ ($p = 0.13$); close $D = 0.065$ ($p = 0.59$).
Cross-boundary return (mixed-feed endpoints): VWAP +1.523; close +2.511 (annualised).

CHMM-N $K^\star$=3 VaR breaches, Polygon: $n = 250$, 5%: 17 (6.80%, exp. 12.5), 1%: 4 (1.60%).
Alpaca: $n = 323$, 5%: 19 (5.88%, exp. 16.2), 1%: 5 (1.55%).

The post-switch segment has a visibly larger VWAP-scale standard deviation (2.215 against 1.640) and heavier sample tails; the same ordering appears in the close column, whose post-switch excess kurtosis is dominated by the early-2025 drawdown sessions. Neither KS statistic rejects at conventional levels, and per-segment VaR breach rates are similar, but these comparisons cannot identify a feed effect: the pre-period is a single calendar year and the post-period contains a distinct volatility regime. Consolidated versus IEX-only aggregation therefore remains an unresolved scope limitation of the OoS window, stated as such in the main text. The diagnostic is reproducible from `runners/diagnostics/run_feed_boundary_check.jl` (artifact `results/diagnostics/feed_boundary_check.txt` of the companion code repository).

A.6 Variant Decision Guide

This appendix recommends a CHMM emission family by use-case priority (Table S3). All four variants share the forward-backward framework, the transition update, and quantile-based initialisation; they differ in the emission density and its M-step update. The shared-ν Student-t row is the heavy-tail recommendation: per-state heavy tails without a penalty hyperparameter at the highest IS KS pass rate of any CHMM variant in Table 2. Because the recommendation draws on OoS performance and no untouched test window is reserved behind the OoS evaluation window, it is exploratory (Table 1). The penalised CHMM-t at $\lambda = 20$ is retained as a sensitivity reference; the λ value was tuned at $K = 18$ and is reported at $K^\star = 3$ as an upper bound on the unpenalised heavy tail.

Table S3. Variant decision guide at $K^\star = 3$.

Use-case priority	Recommended variant
Simplest closed-form fit; excess kurtosis fidelity secondary	CHMM-N
Closest IS excess kurtosis match without a shape parameter	CHMM-L
Per-state heavy tails, no penalty hyperparameter *(main)*	CHMM-t shared-ν
Per-state heavy tails, IS-calibrated λ-shrinkage	CHMM-t pen. ($\lambda = 20$)
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

A.7 State-Resolution and Multi-Emission Sensitivity

This appendix reports the cross-ticker panels and their quarterly-refit rebuilds, the spectral effective-rank diagnostics, the Rydén $K = 2$ replication, the excess kurtosis bootstrap CIs and the shared-ν ablation, the VaR power calibrations and refit-cadence sweeps, and the cross-decade CRSP validation; the full K-sweep table and the EM convergence traces are companion artefacts in the code repository. The cross-ticker generalisation distribution at the main state count is summarised in Table S4.

Table S4. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420, main-paper $K^\star = 3$). The OoS distribution is wider than the IS distribution and is concentrated at the tickers that introduced a new regime (LLY, UNH, NEM). The full per-ticker panel, per-sector rollup, $K = 6$ and $K = 18$ sensitivity rebuilds, and quarterly-refit approach are reported in this appendix.

Metric	$K^\star = 3$
IS KS pass rate (%) median	96.8
IS KS pass rate (%) mean ± s.d.	95.1 ± 4.9
OoS KS pass rate (%) median	69.1
OoS KS pass rate (%) mean ± s.d.	66.2 ± 28.2
$\lvert G_t \rvert$ ACF-MAE median	0.0399
Excess kurtosis residual (sim − obs) median	7.18
Tickers OoS KS < 60% (IS-fixed)	11/30

A.7.1 Full State-Resolution Sensitivity Table (companion artefact)

The full K-sweep table for SPY CHMM-N referenced in the main text state-count selection (KS, AD, excess kurtosis, W_1, Hellinger, OoS coverage at $K \in \{3, 6, 9, 12, 15, 18, 21\}$, 1,000 paths, seed 20260420) is provided as a companion artefact in the code repository. The qualitative pattern is that distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band and ACF-MAE is essentially flat across the entire sweep, consistent with the single-dominant-mode spectral mechanism.

All models converged within the 60-iteration budget with non-decreasing observed log-likelihood traces in every run; exact per-iteration monotonicity is guaranteed for the closed-form Gaussian and Laplace updates, and was observed empirically, though it is not guaranteed, for the ECM-style Student-t and GED updates. Lower K converged in 15–25 iterations, $K = 18$ in 25–40, with a final $|\Delta\mathcal{L}| < 10^{-4}$ at every state count. Convergence-trace figures and per-iteration log-likelihood histories are provided in the companion code repository.

A.7.2 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel at the main text state count $K^\star = 3$ are reported in Table S4. The full per-ticker rollup is given at $K = 18$, the sensitivity reference for excess kurtosis fidelity, rather than at $K^\star = 3$ (Table S5). The per-ticker pattern was essentially K-robust across $\{3, 6, 18\}$, with identical 11/30 failure counts and an overlapping ticker list (Table S4), so the rollup at $K = 18$ is representative of the $K^\star = 3$ panel up to small per-ticker shifts. Each ticker is fit independently under the penalised CHMM-t framework ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 2. The per-ticker ν_k medians sat at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage was operative across the panel; minimum and maximum ν_k values are in the per-ticker CSV in the companion code repository.

Table S5. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in Table S4.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$\lvert G_t \rvert$ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	−0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	−0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	−0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	−0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	−7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	−0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	−4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	−8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	−0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	−1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	−1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	−0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.7.3 60-Ticker Sector Expansion at $n = 6$ per Sector

We expanded the sector panel to $n = 6$ per sector (60 tickers, three additional large-cap representatives per GICS sector) and re-ran the ANOVA; the expansion confirmed the $n = 3$ reading, again finding no significant sector effect on OoS KS. We report this as a non-detection rather than a demonstration of absence: no formal power analysis, sector-variance confidence interval, or equivalence margin accompanies the ANOVA, so a small sector effect below the design's sensitivity is not excluded. The aggregate OoS KS median and failure rate at $K = 18$ closely matched the corresponding 30-ticker $K = 18$ rollup (73.4%, 11/30, Table S5; a descriptive comparison, no test attached); the Table S4 aggregate at $K^\star = 3$ had the same 11/30 failure count at OoS median 69.1%. Cross-sector dispersion accounted for only a small fraction of OoS KS variance at the larger sample size; the remainder was per-ticker. The main-text reading that failures are ticker-specific rather than sector-driven therefore survived the sample-size expansion, though the expansion by itself does not establish power against small sector effects.

A.7.4 Effective Spectral Rank of the Absolute Growth-Rate ACF

The main-text spectral mechanism states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has rank at most $K-1$, so the absolute growth-rate ACF identity (8) contains at most $K-1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. Before ranking, complex-conjugate eigenvalue pairs are grouped into single real damped-oscillatory components (each contributing $2\,\mathrm{Re}(a_k\lambda_k^\tau)$; phase and cancellation therefore enter with their signs), so the unit of account is a real component rather than an eigenvalue, and the component count is at most $K-1$. Shares are defined on the *absolute component-magnitude budget* $B = \sum_c |\mathrm{contrib}_c(1)|$, which upper-bounds $|\rho_{|G|}(1)|$ and is not itself a percentage of $\rho_{|G|}(1)$, since signed contributions can cancel; the signed total $\rho_{|G|}(1) = \sum_c \mathrm{contrib}_c(1)$ is reported per panel. The signed per-component contributions at $\tau \in \{1, 5, 20\}$ for the fitted CHMM-N at $K = 18$, $K = 3$, and $K = 2$ on SPY IS are reported in Table S6; all three fits are converged (tolerance 10^{-4}, no iteration cap reached; the $K = 18$ fit uses five multistarts and the best evaluated likelihood). Per-state moments $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ are analytic folded-normal moments; the stationary vector comes from a checked left-eigenvector solve; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$; as a numerical-fidelity check, the maximum absolute error of the component-sum ACF against the direct matrix formula over $\tau \le 252$ is below 10^{-13} in every panel.

Table S6. Per-component contribution to the absolute growth-rate ACF identity, CHMM-N at $K = 18$ (top panel; top six grouped components plus tail aggregate, 12 components from 17 non-unit eigenvalues), at $K = 3$ (middle panel; full), and at $K = 2$ (bottom panel; single mode). Complex-conjugate pairs are grouped into single real components (kind "pair"); contributions are signed reals; "cum" is cumulative share of the absolute budget $B = \sum_c |\mathrm{contrib}_c(1)|$. At $K = 18$ the converged fit concentrates the budget on two persistent components ($|\lambda| = 0.931$ and 0.921, both real, jointly 94.0% of B; three components reach 95%), and at lag 20 only those two contribute non-negligibly. At $K = 3$ the dominant component ($|\lambda| = 0.953$) alone carries 96.8% of B, and at $K = 2$ the rank bound is at its tight extreme: the single non-unit eigenvalue ($|\lambda| = 0.942$) carries 100%. One to two persistent modes therefore carry the fitted temporal axis at every state count, consistent with the main-text finding that what binds the likelihood fits at low K is the joint objective's allocation to the marginal mixture rather than the decay-mode budget (Section 4.2).

rank	kind	$\lvert\lambda\rvert$	contrib(1)	contrib(5)	contrib(20)	cum
$K = 18,\ \rho_{\lvert G\rvert}(1) = 0.319,\ B = 0.325,\ \sigma^2_{\lvert G\rvert} = 2.558$						
1	real	0.931	+0.2198	+0.1651	+0.0565	0.677
2	real	0.921	+0.0853	+0.0613	+0.0177	0.940
3	pair	0.160	+0.0049	≈ 0	≈ 0	0.955
4	pair	0.311	+0.0042	+0.0001	≈ 0	0.968
5	pair	0.154	+0.0032	≈ 0	≈ 0	0.978
6	real	0.179	+0.0014	≈ 0	≈ 0	0.982
7–12	various	various	$\lvert\cdot\rvert \le 0.0014$ each	≈ 0	≈ 0	1.000
$K = 3,\ \rho_{\lvert G\rvert}(1) = 0.287,\ B = 0.287,\ \sigma^2_{\lvert G\rvert} = 2.506$						
1	real	0.953	+0.2778	+0.2296	+0.1123	0.968
2	real	0.866	+0.0093	+0.0052	+0.0006	1.000
$K = 2,\ \rho_{\lvert G\rvert}(1) = 0.252,\ B = 0.252,\ \sigma^2_{\lvert G\rvert} = 2.380$ *(rank bound tight)*						
1	real	0.942	+0.2515	+0.1983	+0.0813	1.000

The empirical effective rank is consistent with the main-text framing: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in the companion K-sweep artefact in the code repository is the macroscopic signature of the *fitted* temporal axis being driven by one or two decay modes at every state count in the sweep, a description of what the likelihood fits select, not of what each class could attain under a different criterion. The K-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), which the fitted $K = 2$ spectrum illustrates: a single mode ($\lambda_2 = 0.942$ Gaussian, geometric half-life ≈ 11.6 trading days, a short-to-medium-horizon mode, not the observed slow far-band decay; 0.954 shared-ν Student-t) carries 100% of the lag-1 $|G|$ ACF. The failure mode at $K = 2$ is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (8) therefore plays a pedagogical role in this paper rather than serving as a direct K-selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, excess kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.7.5 Cross-Ticker Spectral Effective-Rank Diagnostic

To test whether the SPY-only result of Table S6 generalises across tickers, we computed the grouped-component diagnostic on the 30-ticker sector-balanced panel plus SPY at *both* $K^\star = 3$ and $K = 18$, under two ranking norms: the lag-1 budget $B = \sum_c |\text{contrib}_c(1)|$ and a horizon-aware budget $B_\int = \sum_c \sum_{\tau=1}^{252} |\text{contrib}_c(\tau)|$ (a component small at lag 1 can matter at later lags, so a lag-1 budget alone is not an effective-rank diagnostic for the full curve). Every fit is a *converged multistart* estimate: tolerance 10^{-4} on the log-likelihood with an iteration cap of 4,000 that no fit reached, three starts at $K^\star = 3$ and five at $K = 18$ (canonical quantile start plus jittered starts), with per-ticker optimizer evidence (evaluation counts, final increments, converged flags, cross-start likelihood spreads) retained in the artifact (`spectral_rank_cross_ticker_fits.csv`). All 62 fits converged on the tolerance rule ($K = 18$: median 571 evaluations, maximum 1,400; cross-start log-likelihood spread median ≈ 15 nats, so the multistart is not redundant). Alongside the shares, the same run computes the per-ticker MAE between the fitted population $|G|$ ACF and the sample ACF on the near band (lags 1–63) and far band (64–252), on both the IS fit window and, as the declared out-of-sample criterion (held out from fitting, though the window itself has been used in earlier analyses of this paper, so the comparison is not confirmatory), the OoS window, each with its zero-curve (i.i.d.) reference (Table S7).

Table S7. Cross-ticker spectral diagnostic at $K^\star = 3$ and $K = 18$ (converged multistart fits), two ranking norms, and the model-vs-sample ACF criterion. Distribution across 31 tickers (30-ticker sector panel plus SPY control). "lag-1" ranks grouped components by $|\text{contrib}_c(1)|$; "horizon" by $\sum_{\tau \leq 252} |\text{contrib}_c(\tau)|$. Per-ticker ACF reconstruction errors against the direct matrix formula are all below 10^{-11}. Bottom block: median model-vs-sample $|G|$ ACF MAE, near band (lags 1–63), on the IS fit window and the held-out OoS window, with the zero-curve reference; far-band (64–252) medians in the text.

	$K^\star = 3$		$K = 18$	
Statistic	lag-1	horizon	lag-1	horizon
median dom share	0.721	0.891	0.654	0.880
$[Q_1, Q_3]$	[0.676, 0.851]	[0.808, 0.946]	[0.504, 0.777]	[0.761, 0.943]
minimum	0.542	0.592	0.405	0.489
median n_{95} (max)	2 (2)	2 (2)	5 (8)	2 (5)
median ACF MAE 1–63, IS	0.0497		0.0551	
median ACF MAE 1–63, OoS	0.0441		0.0430	
zero-curve reference (IS / OoS)	0.0901 / 0.0354			

Realizable ACF-targeted fits. The fitted comparison above shows what maximum-likelihood estimation delivers at each state count; it does not show what the class *could* deliver on the ACF axis under a different criterion. That is measured directly with actual HMMs: for each ticker we optimize a valid stationary Gaussian-emission K-state chain (row-stochastic transition matrix from a row-wise softmax parametrization, exact stationary law, analytic folded-normal moments) so that its population $|G|$ ACF matches the observed IS sample ACF (SSE over lags 1–252; multistart finite-difference Adam, 20 starts at $K^\star = 3$ and 8 at $K = 18$, one start seeded at a likelihood fit, a fresh single-start Baum-Welch refit capped at 1,000 iterations (the internal single-start reference), *not* the published converged multistart fit; its own population-ACF error ships in the artifact and the seeded start can only improve on it). Every reported error is *attained by a valid HMM*: the numbers are attainability certificates (upper bounds on the class's best error), not certified global optima, and each winner's full model (transition matrix, stationary law, emissions, fitted and target curves, every start's diagnostics) is persisted and reload-verified by the test suite (row sums, stationarity residual, recomputed population ACF). Cross-ticker medians:

	near band (1–63)	far band (64–252)
likelihood fit, $K^\star = 3$	0.0497	0.0245
likelihood fit, $K = 18$	0.0551	0.0224
ACF-targeted HMM, $K^\star = 3$	0.0162	0.0162
ACF-targeted HMM, $K = 18$	0.0140	0.0154

A valid three-state chain tracks the sample ACF roughly three times more closely than the converged likelihood fit at the same state count: the class is not what limits the fitted ACF. The paired per-ticker gap between the $K^\star = 3$ and $K = 18$ arms is small (median of differences +0.0019 near band, +0.0008 far band), but this is *achieved accuracy under the declared optimizer*, not a class-level equivalence statement: all 31 of the $K = 18$ best-start fits stopped at the iteration cap (at $K^\star = 3$, 25/31 stopped on objective stall, an improvement-stall rule, not a first-order stationarity test), the $K = 18$ stationary laws are near-degenerate (median $\min_k \bar{\pi}_k \approx 3.9 \times 10^{-5}$), and no practical-equivalence margin was declared. The best achieved ACF-only fit has a poor marginal (median mixture excess kurtosis 0.32 against the observed median 8.86 across the panel; SPY -0.14 against 7.69); whether that reflects a genuine trade-off is answered by the frontier experiment below. On SPY the best achieved ACF-targeted three-state chain uses eigenvalue magnitudes $(0.9889, 0.9480)$ (half-lives of roughly 62 and 13 trading days, materially more persistent than the converged likelihood fit's dominant 0.953), which is exactly the persistent-mode configuration the likelihood fit declines to select.

Marginal-versus-ACF frontier. A single ACF-only fit cannot identify whether the marginal and ACF axes compete inside the three-state class. A weighted-objective sweep estimates the trade-off directly: $J_\lambda = \text{ACF-SSE} + \lambda \cdot s \cdot \text{CvM}$, where CvM is the Cramér–von Mises distance between the stationary mixture CDF and the empirical CDF on a 500-point in-sample empirical-quantile grid, s is a fixed balance scale set once by the internal single-start SPY reference fit (a declared constant that defines the arm objectives), $\lambda \in \{0, 0.1, 0.3, 1, 3, 10, 30, 100\}$ plus a pure-marginal arm, six starts per arm, with a regret-based reading rule declared in the runner before the run (per ticker, regret = metric over the best value attained across arms; the axes compete when every arm's larger median regret is ≥ 2; joint attainability is indicated when some arm has both median regrets ≤ 1.5). Because two cross-ticker medians need not be attained by the same tickers, the primary panel summary, added at review, is the per-ticker joint statistic: the median over tickers of the larger of the two regrets, with the count of tickers holding both regrets at or below 1.5. The comparator for all reported metrics is the *published converged multistart likelihood fit* (the exact fits of the spectral cross-ticker panel, recomputed deterministically and re-evaluated on the same metrics in the artifact; the internal single-start reference remains a labeled secondary row). The outcome is *joint attainability of the ACF and the marginal body under this in-sample CDF criterion*: the $\lambda = 0.1$ arm attains median near-band ACF MAE 0.0165 (ACF regret 1.01) with CvM 7.2×10^{-6} (regret 1.17), *better* than the multistart likelihood fit's own CvM (5.2×10^{-5}), and stationary-mixture marginal log-likelihood -2.608 against that fit's -2.597 nats per observation, with central (5%/95%) quantile errors below it at 21/31 tickers; the median per-ticker maximum regret is 1.17, and 24/31 tickers hold both regrets at or below 1.5 with one and the same chain (25/31 at $\lambda = 0.3$ and at $\lambda = 1$). Where the ACF-competitive fits fall short is the *deep tail*: the 1%/99% quantile errors improve on the multistart likelihood fit at only 5/31 and 6/31 tickers (99%-quantile error 0.83 against 0.23), with mixture excess kurtosis 2.3 against 7.0 (observed median 8.86) as the descriptive counterpart. Because the CvM criterion underweights tails, tail-and-ACF joint attainability is left open in both directions. All frontier points are valid achieved weighted-sweep fits, not certified optima: the $\lambda = 0.1$ winners hit the 4,000-iteration cap at 29/31 tickers and the pure-marginal winners at 19/31 (so that endpoint is a weak proxy for the best achievable three-Gaussian-mixture CDF fit); the achieved points bound the true frontier from above, and the full per-arm models with per-start diagnostics and both comparators' parameters are persisted alongside the capacity certificates.

Unrestricted exponential diagnostic (exploratory). As an out-of-class reference, a heuristic free-coefficient exponential-mode fit of the same sample curves (budget-aware matching pursuit over a dictionary of positive-real, negative-real, and damped-oscillatory atoms, the pair shapes costing two modes, with exact least-squares refit and local λ refinement) gives cross-ticker median MAEs of 0.0196/0.0169/0.0156/0.0142/0.0124 (near band) and 0.0176/0.0164/0.0160/0.0159/0.0155 (far band) at $m = 1, 2, 4, 8, 17$ modes, with a paired $m = 2$ minus $m = 17$ gap of 0.0045 (near) and 0.0006 (far) and negligible dictionary-resolution sensitivity (median band-MAE change 10^{-5} at half or double resolution). The dictionary holds *representative* signed real and fixed-angle damped-oscillatory exponential shapes for diagonalizable chains; it does not contain every K-state HMM ACF curve (oscillation angles vary continuously and only the decay rates are locally refined, and non-diagonalizable transition matrices contribute polynomial-times-geometric Jordan terms that the dictionary omits), and most curves it does contain are not HMM-realizable. It is therefore a descriptive diagnostic with no bounding role in either direction: an earlier draft presented a positive-real-only version of this fit as an attainable-accuracy ceiling on HMM error, an inference that was invalid and is retracted in favour of the realizable experiment above; and indeed the realizable three-state result (0.0162 near band) lands *below* this heuristic's two-mode fit (0.0169), an ordering a genuinely nested optimum-based bracket could not produce.

Reading. Three facts, in decreasing order of strength. *First*, attainability and its scope: a valid $K^\star = 3$ HMM can track the finite-horizon sample ACF about three times more closely than the likelihood fits do, and the frontier shows this is compatible with a bulk marginal at the published multistart likelihood fit's level under its in-sample CDF criterion, so neither the decay-mode budget nor the marginal body accounts for the fitted ACF gap; the likelihood-trained fits do not exhaust the class's joint ACF-plus-bulk-marginal capability, and the axis on which they do lead is the deep tail, whose joint attainability with the ACF remains untested. *Second*, consistent with that, the converged $K = 18$ likelihood fits do not materially improve the model-vs-sample ACF over $K^\star = 3$: in-sample $K = 18$ is better at only 7/31 tickers (median $\Delta = -0.005$ in $K^\star = 3$'s favour), and on the held-out window the difference is small in the other direction (22/31, median $\Delta = +0.002$), a sign reversal that reads as instability of a small difference across windows, not as corroboration of either direction. *Third*, a genuine scope limit surfaced by the held-out criterion: at the median *single-name* ticker, both fitted models sit behind the OoS zero-curve reference (0.043–0.044 against 0.0354) because the 2024–2026 OoS sample ACF is much flatter than the IS one; IS-fitted persistence does not transfer at the typical single name, the ACF-axis analogue of the regime-drift pattern in the cross-ticker KS panel. On SPY itself, the headline instance, the fitted models do beat the OoS zero curve (0.0502 at $K^\star = 3$ and 0.0378 at $K = 18$, against 0.0654). Under both norms the $K = 18$ budget spreads over more components than before (lag-1 median $n_{95} = 5$) but the horizon-aware n_{95} median remains 2; none of the shares carries the earlier concentration claim, which the realizable comparison now carries instead. This analysis is descriptive across a dependent panel and does not establish a general power-law or multi-scale approximation.

A.7.6 Rydén K = 2 Replication

Replication of the Rydén et al. [12] $K = 2$ Gaussian setting on SPY IS under quantile and random-seed initialisation gave an essentially constant absolute growth-rate ACF-MAE across six initialisations (one quantile + five random seeds, drawn from a moderate perturbation of sample moments), while the IS KS pass rate fell well short of the heavier-tailed CHMM variants. The setting is transported rather than exact: the original fixes the state means at zero and maximises the likelihood by direct search on 1928 to 1991 S&P 500 subseries, whereas our fit estimates per-state means by EM on the SPY window. Concretely, CHMM-N at $K = 2$ reached IS KS 79.0% (against 91.5% at $K^\star = 3$), $|G_t|$ ACF-MAE 0.0501 (against 0.0462), and simulated excess kurtosis 2.57 (against 3.83): the $K = 2 \to 3$ change is almost entirely distributional (KS, excess kurtosis) and negligible on the ACF. The spectral decomposition explains why the fitted temporal axis is unaffected: at $K = 2$ a single non-unit eigenvalue ($\lambda_2 = 0.942$ Gaussian, half-life ≈ 11.6 trading days; 0.954 shared-ν Student-t) already carries 100% of the lag-1 $|G|$ ACF, and this dominant mode is of comparable persistence to the dominant mode at higher K (converged fits: Gaussian 0.942, 0.953, 0.931 at $K = 2, 3, 18$; shared-ν Student-t 0.954, 0.935, 0.908), all short-to-medium-horizon modes, none reproducing the far-band decay. A companion four-family fit at $K = 2$ (paper-canonical seed 20260420, 1,000 paths) isolates the lever: holding the two states fixed, the Gaussian emission gives IS KS 76.8% and simulated excess kurtosis 2.56 (within the replication's init range above), a per-state Student-t gives 91.6% and 6.24, and CHMM-GED 90.0% and 4.11, at an unchanged $|G_t|$ ACF-MAE (≈ 0.050); the marginal, not the state count, moves the fit. The constant ACF-MAE and the saturated single-mode spectrum describe what the likelihood fits select: moving from $K = 2$ to 3 changes the fitted marginal, not the fitted temporal structure. This is evidence about *fitted outcomes* under the likelihood criterion: it does not by itself establish what the $K = 2$ class could attain on the ACF axis under a different objective (Appendix A.7.5 reports that comparison directly).

A.7.7 Cross-Ticker Per-State-Resolution Sensitivity ($K^\star = 3$ vs $K = 6$ vs $K = 18$)

The cross-ticker aggregate distribution at all three state counts used in this paper is reported in Table S4: the default $K^\star = 3$, the BIC/CAIC parsimony choice (rolling-origin CV did not separate $K = 3$ from $K = 6$) and robust across state resolutions; the sensitivity reference $K = 6$; and the extended sensitivity reference $K = 18$.

The comparison uses three axes: KS, excess kurtosis residual, and absolute growth-rate ACF-MAE. The KS distributions were within a few pp of each other on OoS median across $K^\star = 3, 6, 18$ and at identical 11/30 failure count, with the same regime-introduction tickers driving the failures at each state resolution (NVDA, UNH, LLY, JPM, WFC, HD, NFLX, VZ, NEE, WMT, NEM at

OoS KS < 60% in Table S5), so the cross-ticker KS axis was essentially K-robust on this universe. The excess kurtosis-residual axis was not K-robust: the per-ticker median residual was smallest at $K = 18$ and larger at the smaller state counts, because at lower state resolution the per-state ν_k shrinkage at uniform λ cannot be smoothed across as many regimes, so the residual heavy tails leak into the simulated excess kurtosis. The $|G_t|$ ACF-MAE was essentially K-robust across the three state counts. In practice the three state counts are interchangeable on KS; $K = 18$ gives the best excess kurtosis fidelity; $K^\star = 3$ remains the default.

A.7.8 Block-Bootstrap CIs on Observed Excess kurtosis (IS vs OoS)

A natural question on the IS / OoS excess kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is whether it is itself statistically distinguishable. Raw excess kurtosis, however, is not a functional this paper can test directly: the observed top-5% Hill estimate is $\hat{\alpha} \approx 3.15 < 4$, so the population fourth moment is plausibly infinite (Section 4), in which case raw excess kurtosis has no stable population value to estimate and an n-out-of-n percentile bootstrap is not a calibrated interval for it. We therefore report two layers from the same stationary block bootstrap [88] ($L \in \{5, 10, 20, 50\}$, $B = 5{,}000$ replicates per L per window): the raw-kurtosis resampling bands of Table S8 as a *descriptive sensitivity display*, and, as the inferential statement, the same difference construction applied to *winsorized* excess kurtosis at trim q per tail (each tail clamped at its own q / $1 - q$ empirical quantile, thresholds recomputed inside every resample). The winsorized functional has finite moments of all orders for any tail index, so the block bootstrap is a standard tool for it; in both layers the object is the distribution of the IS − OoS *difference* across independent resample pairs, since overlap of two marginal intervals is not by itself a test of the difference.

Table S8. Stationary block bootstrap resampling bands on observed raw excess kurtosis (descriptive). SPY IS ($T = 2{,}516$) and OoS ($T = 572$) windows, $B = 5{,}000$ replicates per L. Because the fourth moment is plausibly infinite (Hill $\hat{\alpha} \approx 3.15 < 4$), these are sensitivity displays of resampling spread, *not* calibrated 95% confidence intervals; the inferential comparison is the winsorized construction in the text. "Diff 95% band" is the percentile band of the bootstrap distribution of IS − OoS across independent resample pairs; it covers zero at every block length. Pr(IS > OoS) is a *descriptive* bootstrap probability (the fraction of resample pairs with the IS draw above the OoS draw), not a p-value: each window is resampled around its own observed kurtosis, so no equal-kurtosis null is imposed.

L	IS median	IS 95% band	OoS median	OoS 95% band	Diff 95% band	Pr(IS > OoS)
5	7.21	$[2.99, 12.83]$	5.07	$[1.09, 8.97]$	$[-4.04, 9.54]$	0.748
10	7.34	$[2.49, 12.59]$	4.91	$[0.99, 8.66]$	$[-4.24, 9.67]$	0.756
20	7.30	$[2.17, 12.40]$	4.91	$[0.90, 8.26]$	$[-4.24, 9.55]$	0.739
50	7.18	$[1.92, 12.42]$	4.99	$[0.96, 7.70]$	$[-4.27, 9.65]$	0.733

Descriptively, the raw-difference percentile band covered zero at every block length (e.g. $[-4.24, 9.67]$ at $L = 10$), and the descriptive probability Pr(IS > OoS) at $L = 10$ was 0.756. The inferential result comes from the winsorized target. At trim $q = 0.01$ the observed winsorized excess kurtosis is 1.449 IS against 1.017 OoS (difference 0.432), and the percentile 95% CI for the IS − OoS winsorized difference covers zero at every block length ($[-1.26, 1.83]$ at $L = 10$; $[-1.32, 1.62]$ at $L = 5$; $[-1.20, 2.10]$ at $L = 20$; $[-1.19, 2.31]$ at $L = 50$); at $q = 0.05$ the observed difference is 0.250 with $L = 10$ CI $[-0.15, 0.51]$, again covering zero. The data therefore do not exclude equal IS and OoS *winsorized* excess kurtosis at the 5% level: a non-rejection on a well-defined functional, not evidence of equality, and not a statement about the (possibly undefined) raw population kurtosis or about the extreme tails beyond the clamp points. Three assumptions scope this inference: within-window stationarity and mixing strong enough for the block bootstrap (the walk-forward stress folds and the mixed-vendor OoS window are reasons to read it as conditional on the windows as observed), regularity of the marginal at the clamp quantiles (continuous, positive density, so the empirical thresholds are consistent), and nondegenerate variance of the winsorized functional. At $q = 0.01$ the 572-observation OoS window carries only about six observations per tail beyond each empirical clamp, so finite-sample quantile uncertainty is material at that trim; the $q = 0.05$ row, where the clamp quantiles are well-populated, is the sensitivity check and reaches the same non-rejection. Operationally, the IS-OoS excess kurtosis disagreement invoked in the main text (the descriptive analysis and the Table S3 variant-decision caveat) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by closeness of simulated to observed excess kurtosis should similarly be interpreted as point-comparisons inside a wide resampling envelope.

Bootstrap-CI placement of the penalised CHMM-t IS excess kurtosis. A natural follow-up on the main text's penalised CHMM-t at $\lambda = 20$ is whether its simulated IS excess kurtosis distribution sits inside the descriptive bootstrap band on observed: the aggregate-mean simulated IS excess kurtosis (17.67 at $K^\star = 3$, above the $L = 20$ band upper bound 12.40) could be paired with a per-path distribution that mostly sits inside the band, or with one that mostly sits above.

Table S9. Per-path simulated IS excess-kurtosis distribution under penalised CHMM-t at $\lambda = 20$, against the $L = 20$ descriptive block-bootstrap band on observed. 1,000 paths per row, $T_{\text{IS}} = 2{,}516$, seed 20260420. The band is $[2.17, 12.40]$ at $L = 20$ from Table S8 (descriptive; see that table's caption); the observed IS excess kurtosis is 7.68. "in CI %" is the fraction of simulated paths whose per-path excess kurtosis falls inside the band; "$\leq$ CI_HI %" is the fraction below the upper bound (the relevant one-sided placement check for overshoot). The agg. kurt column is this diagnostic's own simulation draw (21.92 at $K^\star = 3$, 9.86 at $K = 18$, against 17.67 and 8.56 in the main-text tables): the aggregate fourth moment is draw-sensitive when a state sits at the ν bracket, as documented in the main text, and the object here is the CI placement of the per-path distribution, not the aggregate point value.

K	agg. kurt	median	sd	Q_5	Q_{95}	≤ 12.40 (%)	in CI (%)
3	21.92	8.52	106.7	3.79	53.13	70.6	70.6
6	10.26	6.52	24.9	3.58	23.92	83.4	83.3
18	9.86	5.99	32.2	3.55	21.59	88.4	88.4

The per-path *median* simulated IS excess kurtosis sat essentially at the observed value at every K (Table S9). The medians were 8.52 at $K^\star = 3$, 6.52 at $K = 6$, and 5.99 at $K = 18$, all within ~ 1.7 units of observed 7.68 and well inside the bootstrap CI. The aggregate-mean simulated IS excess kurtosis sat above the CI upper bound at $K^\star = 3$ because the per-path distribution had a heavy right tail of paths at $Q_{95} \in \{22, 24, 53\}$ that dragged the mean up, but the bulk of the path-level mass (71–88%) sat inside the bootstrap CI at every state count. The main text reports this row as a sensitivity reference whose aggregate-mean IS excess kurtosis overshoots; the per-path distribution locates that overshoot: a small number of heavy-right-tailed paths under the per-state ν_k ECM produced an aggregate-mean overshoot, while the path-level distribution was centred near observed, with 71–88% of paths falling inside the block-bootstrap CI on the observed value at every state count.

A.7.9 Shared-ν Student-t HMM Ablation

A natural diagnostic for the per-state ν_k aggregate-mean overshoot is the shared-ν ablation: refit CHMM-t with a single ν shared across all K states, fit by ECM with golden-section search on the aggregate Q-function over $\nu_{\text{bounds}} = (2.1, 50)$. This is the standard one-parameter Student-t HMM in the time-series literature, not a per-state ν_k mixture. The shared-ν fitter is a library routine under the same convergence contract as the four per-family fitters (every iterate scored on the observed-data likelihood before mutation; the best evaluated finite iterate returned), covered by the same returned-likelihood regression test.

Table S10. Shared-ν Student-t HMM ablation against the main text per-state ν_k row (1,000 paths, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, seed 20260420, no $1/\nu$ penalty). Aggregate sim-kurt columns are mean-of-paths. Compare against the main text penalised CHMM-t ($\lambda = 20$): $K^\star = 3$ row 17.67 IS / 10.28 OoS (Table 2); $K = 18$ row 8.56 IS / 7.07 OoS (from the same penalised ECM run; consistent with the rate-sweep value 8.43 at $\lambda = 20$).

		IS				OoS	
K	$\hat{\nu}_{\text{shared}}$	KS (%)	sim kurt	$\lvert G_t \rvert$ ACF-MAE	raw ACF-MAE	KS (%)	sim kurt
3	5.81	91.9	4.68	0.0531	0.0236	82.1	4.46
6	4.67	92.7	9.38	0.0518	0.0235	82.6	6.95
18	5.80	**95.8**	**6.25**	0.0542	0.0234	**88.0**	**5.00**

Reading. The aggregate-mean IS overshoot disappeared under shared-ν at every K: the largest IS sim kurt was 9.38 at $K = 6$ (vs. observed 7.68, well inside the bootstrap CI), versus 14.4 for the unpenalised per-state ν_k fit at $K = 18$ (Table S23; consistent with the $\lambda = 0$ entry of Table S32). The $K = 18$ shared-ν row was the cleanest single-row IS / OoS heavy-tail match in the entire panel: 6.25 IS / 5.00 OoS, against observed 7.68 / 5.29 and against 8.56 / 7.07 for the penalised CHMM-t at $\lambda = 20$. IS / OoS KS pass rates were within $\sim$ 1pp of the penalised row, and the $|G_t|$ ACF-MAE was identical to it at $K = 18$. The per-state ν_k design is therefore the binding constraint behind the aggregate-mean overshoot that motivates the λ-shrinkage and bracket-lift discussion in the main text: a single shared ν at $K = 18$ produced the best joint KS / excess kurtosis row in the panel without any penalty. The analysis retains the per-state $\nu_k + \lambda = 20$ construction for direct comparability with the Peel and McLachlan [21], Liu and Rubin [22] tradition; for heavy-tail fidelity without a penalty hyperparameter, the shared-ν fit at $K = 18$ is the better choice.

A.7.10 Christoffersen-cc Monte Carlo Power Calibration at $T_{\text{OoS}} = 573$

A Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test [17] at the $T_{\text{OoS}} = 573$ length used for Table 5 simulates breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ, where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5{,}000$ replicates per cell.

Reading. At $\rho = 0$ the empirical rejection rates sat within about one point of the nominal 5% for all three statistics, confirming that the asymptotic χ^2 calibration is approximately correctly sized at $T = 573$. Power was asymmetric across α. At $\alpha = 0.05$ ($\sim$ 28.6 expected breaches), Christoffersen-cc reached 80% power against $\rho \geq 0.20$ (83.8%) and essentially full power at $\rho = 0.50$. At $\alpha = 0.01$ ($\sim$ 5.7 expected breaches) it reached 80% power only at $\rho \geq 0.50$ (88.2%); at $\rho = 0.20$ the rejection

Table S11. Christoffersen-cc power at $T_{\mathbf{OoS}} = 573$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ. $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Zero-breach replicates contribute their exact Kupiec $\mathrm{LR}_{\mathrm{uc}}$ (the $0 \log 0$ terms taken at their limit) rather than being scored as $\mathrm{LR}_{\mathrm{uc}} = 0$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	5.7	1.8	3.8
0.01	0.05	6.9	15.3	11.9
0.01	0.10	7.7	27.9	21.5
0.01	0.20	11.4	51.7	43.6
0.01	0.30	16.4	67.2	65.1
0.01	0.50	28.4	75.6	88.2
0.05	0.00	5.9	4.7	4.0
0.05	0.05	6.5	18.5	15.0
0.05	0.10	8.1	46.5	39.4
0.05	0.20	11.6	87.2	83.8
0.05	0.30	15.8	97.5	97.9
0.05	0.50	27.3	99.7	100.0

rate was 43.6%, well below conventional power thresholds. The reading for Table 5 is therefore tier-dependent. The uniform $\alpha = 0.05$ non-rejections carried strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carried strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ non-rejections should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the main-text VaR back-test should be read with this calibration in mind.

A.7.11 Engle-Manganelli Dynamic Quantile Test

As a higher-power conditional-coverage alternative to Christoffersen-cc on the same OoS window, we ran the Engle-Manganelli Dynamic Quantile (DQ) test in the standard four-lag specification [78],

$$\mathrm{Hit}_t - \alpha = \beta_0 + \sum_{i=1}^{4} \beta_i \left(\mathrm{Hit}_{t-i} - \alpha\right) + \beta_5 \, \widehat{\mathrm{VaR}}_t + u_t, \tag{29}$$

on the IS-fixed regime-conditional VaR series of the main text VaR back-test, with $\mathrm{Hit}_t = \mathbf{1}\{R_{\mathrm{OoS},t} < \widehat{\mathrm{VaR}}_t(\alpha)\}$. The DQ test statistic is $\widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \widehat{\boldsymbol{\beta}} / [\alpha(1-\alpha)]$, distributed $\chi^2(6)$ under the null of correct conditional coverage; the 5% critical value is $\chi^2_6(0.95) = 12.59$.

Reading. Neither conditional-coverage test rejected at $\alpha = 0.05$ at either state count (Table S12), so the $\alpha = 0.05$ non-rejections survive the higher-power alternative. The p-values were cc 0.391 and DQ 0.094 at $K = 3$, and cc 0.674 / DQ 0.676 at $K = 18$. At $\alpha = 0.01$ the DQ test rejected conditional coverage at $K = 18$ ($p = 0.017$) where Christoffersen-cc did not ($p = 0.137$), and was borderline at $K = 3$ ($p = 0.056$). This is consistent with the power-calibration result of Table S11: at $\alpha = 0.01$ Christoffersen-cc has only 43.6% power against moderate breach-clustering eigenvalues $\rho = 0.20$, so the higher-power DQ test detects miscalibration that Christoffersen-cc misses. The DQ result therefore reinforces the reading that $\alpha = 0.05$ is the operationally informative tier; the $\alpha = 0.01$

Table S12. Engle-Manganelli DQ test on the regime-conditional VaR, alongside Christoffersen-cc on the same breach series, for CHMM-N at $K \in \{3, 18\}$, $\alpha \in \{0.01, 0.05\}$; $T_{\text{OoS}} = 573$, $q = 4$ lagged Hit indicators, χ^2_6 critical at 5% is 12.59.

K	α	breach %	cc p	DQ	DQ p
3	0.01	1.57	0.137	12.27	0.056
3	0.05	6.28	0.391	10.82	0.094
18	0.01	1.57	0.137	**15.47**	**0.017**
18	0.05	4.54	0.674	4.00	0.676

tier is not decided by the panel and should be read as power-limited rather than as established calibration, with the DQ rejection at $K = 18$ the concrete instance.

Finite-sample calibration of the strict-tail DQ, exact binomial coverage, and paired quantile loss. The asymptotic χ^2_6 reference for the DQ statistic has no finite-sample guarantee at $T = 573$ with ≈ 5.7 expected breaches, so we calibrated it by parametric bootstrap under each fitted null: simulate $B = 2{,}000$ synthetic OoS paths of length 573 from the fitted model, rerun the identical one-step-ahead VaR filter and DQ regression on each, and score the observed statistic against that null distribution. The calibration covers exactly two rows (CHMM-N at $K^\star = 3$ and MS-GARCH at $K = 4$) and conditions on the estimated parameters (no per-replicate re-estimation), so finite-sample inference extends only to those two rows; the other strict-tail rows of the family panel carry asymptotic p-values only. The finite-sample null is far heavier-tailed than χ^2_6 (null 95% quantiles 22.2 for CHMM-N at $K^\star = 3$ and 19.9 for MS-GARCH at $K = 4$, against the asymptotic 12.59), so the asymptotic test is anticonservative in this design: the bootstrap p-values are 0.19 (CHMM-N $K^\star = 3$, asymptotic 0.056) and 0.10 (MS-GARCH $K = 4$, asymptotic 0.016), and neither model is rejected at the 5% level under its finite-sample null. Every strict-tail DQ separation in the main-text table is therefore exploratory. As a distribution-free complement, exact two-sided binomial unconditional-coverage p-values were computed for every row of the family panel plus the MS-GARCH block: all CHMM rows are compatible with nominal coverage at both levels (smallest $p = 0.087$), while the MS-GARCH 1% rows over-breach (11–12 breaches, $p_{\text{exact}} \in [0.017, 0.035]$). Finally, a paired pinball-loss comparison on the identical 573 forecast days (HAC Newey-West on the daily loss differentials) finds no significant ranking between CHMM-N at $K^\star = 3$ and MS-GARCH at $K = 4$, the filtered bootstrap, or CAViaR at either level ($|t| \le 1.48$, $p \ge 0.14$ throughout): specification tests aside, no model in the panel demonstrably beats another on quantile forecast accuracy.

A.7.12 Quarterly-Refit Regime-Conditional VaR Back-Test

Re-fitting CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window and running the forward-filter through the OoS window under each refit's parameters tests whether periodic refit improves the Christoffersen-cc statistic on the main OoS window.

The quarterly-refit construction was not rejected by Christoffersen-cc on any of the four rows, though the two $K = 18$ rows sat at $p_{\text{cc}} = 0.089$ (marginal). Refit tightened VaR coverage at $K = 18$, $\alpha = 0.05$ (3.49% vs. 4.54% IS-fixed) but did not strictly improve the cc-statistic at this T_{OoS}.

Table S13. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\text{OoS}} = 573$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 5 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}
3	0.01	6	1.05%	−5.10	0.013	3.97	3.98	0.136
3	0.05	29	5.06%	−3.12	0.004	0.19	0.20	0.906
18	0.01	5	0.87%	−5.64	0.098	4.75	4.84	0.089
18	0.05	20	3.49%	−3.10	3.06	1.78	4.84	0.089

A.7.13 Walk-Forward Refit-Cadence Sweep (Monthly / Weekly)

A within-fold refit-cadence sweep at monthly (21 trading days) and weekly (5 trading days) cadence on the six walk-forward folds at $K^\star = 3$, $\alpha = 0.05$, did not close the W2 (COVID) Christoffersen-cc rejection at any cadence; monthly refit instead introduced new failures on other folds via refit-cycle parameter drift. The mitigation for W2 / W4 is therefore not refit cadence but a model class with explicit regime-introduction handling.

A.7.14 Quarterly-Refit 30-Ticker Cross-Ticker Panel

For a quarterly-refit version of the 30-ticker cross-ticker panel (Table S4), we re-fitted the penalised CHMM-t at $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total per state resolution), simulated the next quarter under each refit, and concatenated the per-quarter simulations into a single OoS path matrix per ticker, at the default $K^\star = 3$ and at the sensitivity references $K = 6$ and $K = 18$. Same $N_{\text{paths}} = 1{,}000$ and seed 20260420 as Table S4.

Table S14. Quarterly-refit cross-ticker panel at $K^\star = 3$, $K = 6$, and $K = 18$. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol at all three state counts, vs. the IS-fixed $K = 18$ protocol of Table S4. Refit cadence = 63 OoS trading days; train window = 1,260 obs ($\sim$ 5y rolling). Per-ticker detail in the results files.

Metric	IS-fixed ($K = 18$)	Refit ($K^\star = 3$)	Refit ($K = 6$)	Refit ($K = 18$)
IS KS% median	99.5%	96.7%	98.8%	99.4%
IS KS% mean ± sd	99.3 ± 0.6%	95.1 ± 4.5%	97.4 ± 4.4%	99.2 ± 0.6%
OoS KS% median	73.4%	84.6%	**85.8**%	83.0%
OoS KS% mean ± sd	66.8 ± 29.5%	75.0 ± 23.0%	$\mathbf{76.0 \pm 21.7}$%	77.2 ± 21.2%
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]$%	$[56.3, 94.5]$%	$[57.6, 94.3]$%	$[62.0, 95.4]$%
Tickers OoS KS < 60%	$11/30 = 36.7$%	$8/30 = 26.7$%	$9/30 = 30.0$%	$\mathbf{7/30 = 23.3}$%

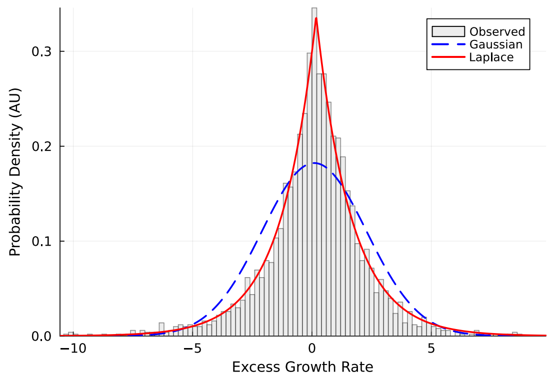
Reading. Quarterly refit shifted the OoS KS distribution materially at all three state resolutions. At $K^\star = 3$: median +15.5pp (69.1 → 84.6% vs the static-fit baseline in Table S4), mean +8.8pp (66.2 → 75.0%), failure count down by 27% (11 → 8). At $K = 18$: median +9.6pp (73.4 → 83.0%), mean +10.4pp, IQR tighter (45.2pp → 33.4pp), failure count down by 36% (11 → 7). At $K = 6$: median +10.7pp (75.1 → 85.8%), mean +9.5pp, failure count down by 18% (11 → 9). The three state resolutions were within $\sim$ 3pp of each other on the refit median (84.6 to 85.8%); the default $K^\star = 3$ was at parity with the sensitivity references under the refit protocol. The largest individual improvements concentrated on the regime-introduction tickers that the IS-fixed Table S4 flagged:

at $K = 18$, LLY 7.6% → 83.7%, UNH 14.5% → 47.3%, NEM 5.6% → 15.4%, HD 41.4% → 82.2%, NFLX 45.1% → 61.5%. The mechanism is direct: as the regime introduced new σ levels in the OoS window, each quarterly refit picked up the most recent 5y including the new regime and the simulator could match the OoS distribution.

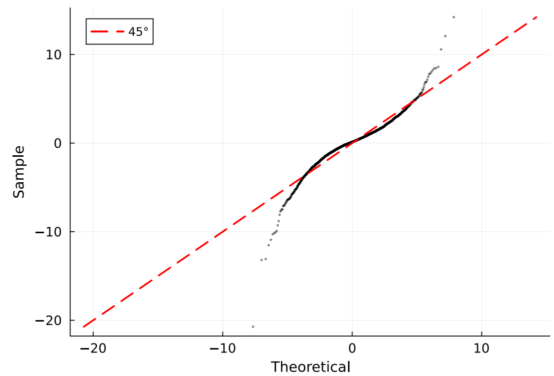
A small number of tickers regressed: DIS 96.4% → 48.4%, BAC 84.8% → 55.8%, AAPL 95.1% → 66.5%. The pattern is that tickers whose 2024–2026 OoS distribution is well-spanned by the full 2014–2024 IS lost information when the train window shrank to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The conclusion is that the periodic-refit mitigation invoked in the main-text Discussion is exercised here: the universe-scale failure rate dropped from 36.7% to 23.3% under quarterly refit, with the largest gains on the single-name regime-introduction tickers.

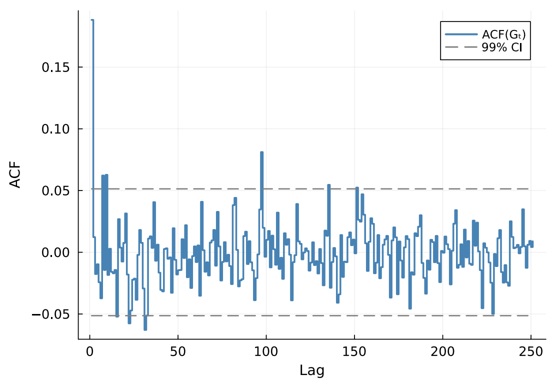
A.7.15 Stylized-Facts Figure (Empirical SPY IS)



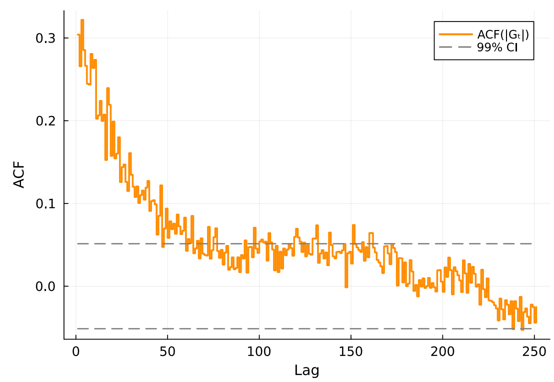
(a) Marginal density.



(b) Normal Q-Q plot.



(c) ACF of raw G_t.

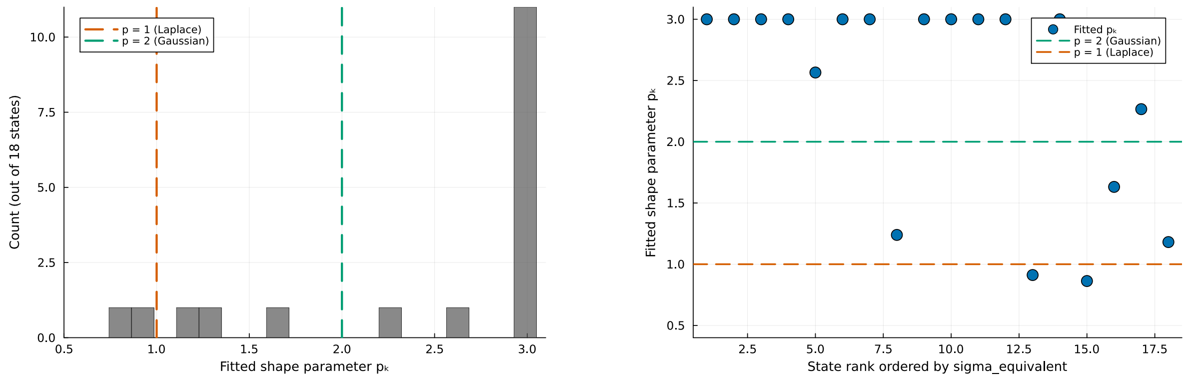


(d) ACF of $|G_t|$.

Figure S3. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

A.7.16 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

The GED variant carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitioned states bimodally (Figure S4): eleven states at the upper bracket $p_k = 3.0$ (thirteen Gaussian-like at $p_k \geq 1.85$), one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrated in the high-volatility tail of the state ordering. The partition replicated across 10 Monte Carlo seeds and across the cross-asset universe; we read it as the most distinctive empirical finding of the four-emission framework.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure S4. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.7.17 Explicit-Duration HSMM: Estimator Convention, Optimizer Evidence, and Truncation Sensitivity

The main-table HSMM-N is the Yu [80] explicit-duration EM with Gaussian emissions, off-diagonal transitions, and truncated discrete Pareto sojourns $p_\alpha(d) \propto d^{-(\alpha+1)}$ on $d = 1, \ldots, D_{\max}$ with the declared specification $D_{\max} = 200$. Three estimator properties matter for reading the row. *First, the finite-window convention*: the terminal segment is right-censored; the likelihood marginalizes an ongoing final sojourn through the duration survival function $S_\alpha(d) = \Pr(D \geq d)$, so the estimator and the simulator (which right-truncates the last sojourn at T) describe the same model; the duration M-step correspondingly maximizes $\sum_d w_d \log p_\alpha(d) + \sum_d c_d \log S_\alpha(d)$ with expected completed counts w_d and censored terminal counts c_d (without censoring the problem is a concave one-parameter exponential family solved by golden section; with censoring, a grid-bracketed golden section). The censored forward-backward recursions are pinned by a brute-force enumeration test in the code repository. The complementary initial-segment convention is that the first sojourn starts at $t = 1$ (no equilibrium left-censoring); an arbitrary data window need not begin at a renewal boundary, so this is a stated approximation, plausibly minor at $T = 2{,}516$. *Second, optimizer status*: the row is a *multistart local-EM* fit, not a certified global maximum. At $K = 3$, five starts (canonical quantile plus jittered) all converge under the per-observation 10^{-5} rule within a 0.14-nat likelihood spread (best start: 42 evaluations); at $K = 18$, three starts converge with a 30.0-nat spread (best: 198 evaluations), so multistart is material there. Traces are monotone and the evaluate-before-update contract holds; per-start diagnostics ship with the artifact. *Third, the boundary/truncation scope*: the fitted $K = 3$ duration exponents are $\alpha = (0.05, 0.75, 0.82)$ with the persistent crash state ($\mu = -0.31$, $\sigma = 3.79$) at the lower search bound $\alpha = 0.05$, and at that exponent the truncated law's mean duration is genuinely $D_{\max}$-sensitive ($\mathbb{E}[D] \approx 17.8$, 30.8, 54.2 at $D_{\max} = 100$, 200, 400). A $D_{\max} \in \{100, 200, 400\}\times$ exponent-floor $\in \{0.02, 0.05, 0.2\}$ sensitivity grid (single canonical start each) quantifies the consequence: the floor itself is nearly immaterial (metrics shift by well under one percentage point across floors at fixed $D_{\max}$, and at $D_{\max} = 400$ the fitted exponent leaves the floor, $\alpha_1 = 0.155$), but $D_{\max}$ moves the row along the marginal-clustering trade-off, with IS KS 84.6% / clustering 0.0451 at $D_{\max} = 100$, 78.0% / 0.0386 at 200, 60.9% / 0.0411 at 400 (single-start grid cells, which is why the $D_{\max} = 200$ cell differs slightly from the multistart main-table row), while the per-observation likelihood mildly prefers the smallest truncation (-1.9815, -1.9822, -1.9825). The panel-leading clustering figure of the main table therefore holds at the declared $D_{\max} = 200$ specification and is not truncation-invariant; the direction (heavier sojourn tail buys clustering at marginal cost) is stable across the grid. The off-diagonal jump matrices of the headline fits are full rank at both state counts (3 and 18 at tolerance 10^{-10}; smallest singular values 0.007 and 0.028), and neither fit is degenerate. Two earlier versions of this estimator are superseded: the continuous untruncated update $\alpha = 1/\mathbb{E}_w[\log d]$ (not the maximizer of the declared likelihood; it left the $K = 3$ fit at the i.i.d. ACF floor and collapsed the $K = 18$ fit) and the exact-boundary finite-window convention that conditioned every segmentation on a sojourn ending exactly at T while the simulator right-truncated (an estimator/generator mismatch; correcting it moved the row only marginally, e.g. OoS KS 67.7% $\to$ 68.7% at $K = 3$, but the $K = 18$ fit improved materially once multistart was added).

A.7.18 Gamma-Sojourn HSMM as an Alternative Sojourn Family

Replacing the truncated discrete Pareto sojourn pmf with a discretised continuous Gamma fit by method-of-moments at every M-step gave a moment-updated Gamma-sojourn HSMM, run on the same right-censored EM core as the Pareto fit (the moment update uses the expected completed-duration counts only, with censored terminal counts excluded, a disclosed approximation); because the duration block is a moment update rather than a maximisation of the expected complete-data log-likelihood, the procedure is not maximum-likelihood EM and does not inherit the EM ascent guarantee, and we treat it as a single-start sensitivity experiment. Against the censored multistart Pareto fits, the sojourn-family trade-off is clean at $K^\star = 3$: the Gamma sojourn leads the marginal (IS/OoS KS 78.3%/80.1% against the Pareto's 76.0%/68.7%, most visibly out of sample), while the Pareto leads volatility clustering ($|G_t|$ ACF-MAE 0.0394 against the Gamma's 0.0531; Table S15). At $K = 18$ the Gamma retains the marginal lead (87.2%/81.4% against 80.7%/73.3%) but the clustering figures are essentially tied (0.0461 Gamma, 0.0466 Pareto), so the sojourn-tail mechanism matters most at the low state count, where the single heavy-tailed (boundary-α) crash-state sojourn carries the clustering. There is still no single best-HSMM row: heavy-tailed Pareto durations buy clustering at marginal cost, light-tailed Gamma durations the reverse: the same KS-versus-ACF tension as the CHMM-vs-HSMM choice.

Table S15. Moment-updated Gamma-sojourn HSMM on SPY. Explicit-duration HSMM with discretised continuous-Gamma sojourn densities whose shape and scale are updated by method-of-moments at each M-step (a sensitivity experiment, not maximum-likelihood EM); KS pass rate, simulated excess kurtosis, and $|G_t|$ ACF-MAE on the SPY in-sample / out-of-sample windows. Against the censored multistart truncated-Pareto fits, the Gamma sojourn leads the marginal (KS) axis while the Pareto sojourn leads volatility clustering at $K^\star = 3$ (essentially tied at $K = 18$); see the text.

	KS (%) ↑		Exc. Kurt		$\lvert G_t\rvert$ ACF-MAE ↓	
K	IS	OoS	IS	OoS	IS	OoS
3	78.3	80.1	2.52	2.52	0.0531	0.0510
18	87.2	81.4	4.31	4.03	0.0461	0.0553

A.7.19 Unconditional VaR / ES Envelope Panel

Table S16. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5, 95]\%$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block. This ES envelope is unconditional; the same one-step-ahead state-mixture predictive supports a regime-conditional ES through closed-form per-state tail expectations weighted by the forward-filter state probabilities, the coherent counterpart of the regime-conditional VaR of Table 5, which we leave as an extension.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.88 [−7.83, −6.00]	−9.17 [−10.46, −7.97]	−3.30 [−3.78, −2.89]	−5.54 [−6.17, −4.90]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.67 [−8.61, −4.92]	−8.82 [−11.62, −6.86]	−3.28 [−4.30, −2.51]	−5.44 [−6.86, −4.19]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.7.20 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

To verify that the main-text penalty rate $\lambda = 20$ is not selected on data that overlaps the OoS evaluation window, we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1{,}130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$ growth-rate observations with the series sliced by date; the supplement's K-selection validation computes growth within the price slice, hence its $n = 377$ on the same dates; both slices fully pre-COVID and pre-2022-rate-hike).

Table S17. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. "val LL/obs" is per-observation held-out log-likelihood; "val KS" is two-sample Kolmogorov–Smirnov pass rate against the validation series; "sim K" columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	**−2.075**	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading. The held-out validation log-likelihood was maximised at $\lambda^\star = 20$ at $K = 18$, matching the state count of Table 2. Validation KS picked $\lambda^\star = 0$ but the band $[0, 200]$ spanned only 2.2pp (89.8 to 92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The $\lambda = 20$ specification at $K = 18$ is therefore held-out-validated and not driven by the OoS window.

Re-tuning at $K^\star = 3$. Repeating the same held-out CV at the default $K^\star = 3$ on the same pre-2020 slices gave a distinct optimum. The per-observation validation log-likelihood was monotone in λ on the same grid, with $\lambda^\star = 50$ minimising the negative log-likelihood, and validation KS also picking $\lambda^\star = 50$. The simulated excess kurtosis at $\lambda^\star = 50$ fell well below the observed 7.68 IS / 5.29 OoS targets, confirming that the penalty over-shrank at $K^\star = 3$. The analysis retains $\lambda = 20$ at $K^\star = 3$ as a sensitivity reference; the main heavy-tail recommendation remains the shared-ν fit (Table S10).

A.7.21 Cross-Decade Validation: 1994-2004 IS / 2004-2006 OoS via CRSP

To test whether the main stylized-fact reproduction is decade-specific, we ran a 1994–2004 vs. 2014–2024 SPY split via a CRSP day-pass covering daily prices for SPY plus 28 of the 30 cross-ticker panel members (NEE and APD missing from the CRSP query) from 1994-01-03 to 2006-04-28, with adjusted close = `DlyPrc/DlyFacPrc` from the CRSP CIZ-format daily stock file. The slices are SPY IS 1994-01-03 to 2004-01-02 (2,519 obs) and SPY OoS 2004-01-05 to 2006-04-28 (583 obs), under the same main protocol: 1,000 simulated paths per fit, $\alpha = 0.05$ for KS.

Table S18. Cross-decade validation on SPY 1994–2004 IS / 2004–2006 OoS (CRSP daily). Comparison rows reproduced from Table 2 (and Table S23 for the $K = 18$ row) for the 2014–2024 IS / 2024–2026 OoS window. Observed excess kurtosis on the 1994–2004 IS = 3.05 versus 2014–2024 IS = 7.68; observed excess kurtosis on the 2004–2006 OoS = 0.06 (essentially Gaussian) versus 2024–2026 OoS = 5.29. The IS-OoS excess kurtosis gap is therefore much wider on the cross-decade slice than on the main window.

		KS pass rate (%) ↑		Sim excess kurtosis		$\lvert G_t\rvert$ ACF-MAE ↓	
Model	K	IS	OoS	IS	OoS	IS	OoS
Cross-decade: 1994–2004 IS / 2004–2006 OoS (CRSP, this appendix)							
CHMM-N	3	84.9	3.3	2.20	2.24	0.0696	0.0491
CHMM-N	18	89.8	3.4	1.84	1.74	0.0720	0.0499
CHMM-t pen. ($\lambda = 20$)	3	89.1	5.4	4.97	4.18	0.0682	0.0501
Main window: 2014–2024 IS / 2024–2026 OoS (Polygon / Alpaca, Table 2)							
CHMM-N ($K^\star = 3$)	3	91.5	78.0	3.83	3.62	0.0462	0.0544
CHMM-N ($K = 18$)	18	94.1	81.8	5.04	4.44	0.0509	~ 0.054
CHMM-t pen. ($\lambda = 20$, $K^\star = 3$)	3	91.9	79.7	17.67	10.28	0.0538	0.0496

Reading. The cross-decade IS fit transferred: CHMM-N at $K = 3$ attained 84.9% IS KS on 1994–2004 vs. 91.5% on 2014–2024, and $K = 18$ attained 89.8% vs. 94.1%. The penalised CHMM-t at $\lambda = 20$, $K = 3$ attained 89.1% IS KS on 1994–2004 vs. 91.9% on 2014–2024. The IS axis of the stylized-fact reproduction claim is therefore decade-robust within ~ 7pp on KS and consistent on $\lvert G_t\rvert$ ACF-MAE (0.068–0.072 vs. 0.046–0.054, a moderate but not pathological widening).

The OoS axis is the issue. The 2004–2006 OoS slice had excess kurtosis 0.06 (essentially Gaussian) vs. the 1994–2004 IS excess kurtosis of 3.05; the IS-OoS excess kurtosis gap on this slice was much wider than on the main 2014–2024 / 2024–2026 window (where IS = 7.68 and OoS = 5.29). The

CHMM trained on the 1994–2004 IS distribution simulated paths with excess kurtosis 1.74–4.97 that were heavier-tailed than the calm 2004–2006 bull-market OoS, so the static IS-fit could not reproduce the OoS marginal and KS rejected at a 3–5% pass rate on all three rows. This is the same regime-introduction failure mode the walk-forward analysis already documented at the W2 (COVID) and W4 (2022 rate-hike onset) stress folds (Table S33, both with OoS KS $< 10\%$); the 2004–2006 OoS window is structurally a low-volatility, low-excess kurtosis slice where the IS-fixed CHMM's tail mass overshot the observed.

The conclusion is that the stylized-fact reproduction claim is decade-robust on the IS axis: the CHMM framework transferred across decades on the IS-fitting side, so the four-emission ECM framework is not specific to the 2014–2024 window. The single-window OoS pass rate, by contrast, is OoS-slice-specific (2024–2026 is a moderate-stress slice where the OoS distribution is close to the IS distribution; 2004–2006 is a calm slice where it is not). The main text already reads the walk-forward median, not the single-window OoS pair, as the operationally informative summary; the cross-decade result is consistent with that framing and adds an independent decade to the walk-forward evidence base. Periodic refit remains the deployment recommendation under either decade.

A.8 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panels referenced from the main text Results: the KS test-power calibration, the IS- and OoS-anchored block-bootstrap KS recalibrations, the QuantGAN deep-generative baseline, the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the main table, the stochastic-volatility, multifractal, and jump-diffusion baselines, the leverage-effect diagnostic, and the iterated pseudo-likelihood Student-t copula estimator; bin-count and multi-seed sensitivity checks are companion artefacts in the code repository.

A.8.1 OoS KS Test-Power Calibration

We calibrated the two-sample KS test power to anchor the OoS KS pass rates of the main generator comparison (Table S19). For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\mathrm{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-text pass-rate metric.

Table S19. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\mathrm{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\mathrm{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\mathrm{IS}}, \hat{\sigma}_{\mathrm{IS}}$) at length T_{IS}, tested against R_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in the main generator comparison (63.6 to 82.1% across the CHMM rows of Table 2); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.8.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the main IS KS pass-rate metric. The asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all

Table S20. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$; the CHMM rows are this diagnostic's own fits at $K = 18$ (CHMM-t is the unpenalised per-state-ν fit), so the pass rates are not the $K^{\star} = 3$ main-table values. "asymp pass" uses the metric definition of Table 2 (two-sample KS p-value ≥ 0.05), recomputed within this run. "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the two-sample stationary-block-bootstrap 95% critical value at mean block length L (two independent IS resamples per null replicate). Critical values: 0.0433 ($L = 5$), 0.0469 ($L = 10$), 0.0497 ($L = 20$); asymptotic i.i.d. value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	100.0	100.0	100.0
GARCH(1,1)	26.8	0.048	0.000	0.227	42.4	54.4	63.0
CHMM-N	94.8	0.546	0.047	0.976	98.4	98.8	99.2
CHMM-t	**95.8**	**0.571**	0.060	0.987	98.2	99.4	99.6
CHMM-L	94.4	0.504	0.047	0.948	98.0	99.0	99.6
CHMM-GED	95.6	0.547	0.064	0.968	**98.8**	**99.6**	**100.0**

carry volatility clustering, so the asymptotic null distribution may not apply. The binary "pass/fail at $\alpha = 0.05$" also reduces a continuous p-value to a one-bit summary. We computed two recalibrations addressing each concern (Table S20).

To address the dependence concern, we generated the empirical null distribution of the two-sample KS statistic under a two-sample stationary-block-bootstrap null [88]: each of $B = 1{,}000$ replicates draws *two independent* block resamples of the SPY IS series (mean block lengths $L \in \{5, 10, 20\}$) and computes the KS statistic between them, so the null replicate has the same structure as the actual observed-versus-simulated comparison: two independent length-n series sharing the observed marginal and short-range dependence. The resulting 95% critical values were 0.0433, 0.0469, and 0.0497, all *larger* than the asymptotic i.i.d. value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$: accounting for serial dependence widens the null distribution of the two-sample KS statistic, so the dependence-aware test is *less* strict than the asymptotic one, and the asymptotic pass rates reported in the main table are conservative in this respect. The threshold remains conditional on the empirical IS distribution (no generator refit per replicate), so all pass rates in this appendix are descriptive. For each generator we report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

To address the binarisation concern, we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

Under the dependence-aware null every CHMM variant passes on essentially all paths (98–100% across the three block lengths), the i.i.d. bootstrap passes on all, and GARCH(1,1) rises from 26.8% asymptotic to 42–63% block-aware while remaining clearly the weakest entry (Table S20). The cross-generator ordering is unchanged from the asymptotic metric; the level shift reflects the wider dependence-aware null rather than any change in the fits.

OoS-anchored block-bootstrap recalibration. The block-bootstrap recalibration of Table S20 is anchored on the IS series; we report here whether the same construction holds on the OoS window. We repeated the procedure (two independent block resamples per null replicate) re-anchored on R_{OoS} ($n_{\text{OoS}} = 572$, $B = 1{,}000$, $L \in \{5, 10, 20\}$, $N_{\text{paths}} = 500$). The critical values were 0.0857, 0.0909, and 0.0927 at $L = 5, 10, 20$ respectively, again above the asymptotic 0.0804. The OoS-anchored panel is reported in Table S21; the source CSV is in the companion code repository.

The cross-generator OoS ordering matched the IS ordering (bootstrap > CHMM-t > CHMM-

Table S21. OoS-anchored block-bootstrap KS recalibration. Same construction as Table S20 (CHMM rows at $K = 18$, this run's own fits) but with the two-sample stationary-block-bootstrap null anchored on R_{OoS} ($n = 572$). "asymp pass" is the main OoS metric (two-sample KS p-value ≥ 0.05); "blkL%" is the OoS block-bootstrap pass rate at mean block length L. The cross-generator ordering is preserved under the dependence-aware recalibration on both windows, with all pass rates shifted upward by the wider null.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑
i.i.d. bootstrap	90.4	0.437	0.030	0.912	94.2	96.8
GARCH(1,1)	59.2	0.164	0.000	0.645	65.8	75.6
CHMM-N	81.0	0.318	0.010	0.876	87.0	91.4
CHMM-t	**84.8**	**0.355**	0.018	**0.911**	**90.8**	**94.0**
CHMM-L	77.4	0.280	0.008	0.791	84.2	90.0
CHMM-GED	81.0	0.327	0.010	0.837	86.4	90.8

Table S22. Block-aware OoS KS recalibration at $L = 20$ (two-sample stationary-block-bootstrap null of 88, two independent OoS resamples per replicate, $B = 1{,}000$, 500 OoS-length simulated paths per generator, seed 20260422). The asymp column recomputes the Table 2 OoS metric at 500 paths under this table's seed and RNG stream, so its values differ from the 1,000-path main-table column by path noise ($\leq$ 3pp); the block-bootstrap 95% null KS critical value at $L = 20$ is 0.0927 vs the asymptotic 0.0804. State count $K^\star = 3$.

Model	OoS KS asymp. (%)	OoS KS block $L = 20$ (%)
Bootstrap	90.4	97.4
GARCH(1,1)	59.2	75.8
CHMM-N	79.8	90.8
CHMM-t pen. ($\lambda = 20$)	**82.4**	**92.8**
CHMM-L	64.4	83.6
CHMM-GED	79.2	90.2

N/GED > CHMM-L ≫ GARCH), and on both windows the dependence-aware pass rates sit above the asymptotic ones, so the asymptotic metric's cross-generator ranking is the robust comparison while its absolute levels are conservative.

Default-state-count summary at $L = 20$. The OoS asymptotic pass rate alongside the OoS block-bootstrap pass rate at $L = 20$ (two-sample block-bootstrap 95% critical value 0.0927 vs. asymptotic 0.0804) at the default $K^\star = 3$ is reported in Table S22. Under the dependence-aware null every $K^\star = 3$ family's pass rate rises by 10–19pp relative to the asymptotic value; the ranking across families is unchanged.

A.8.3 QuantGAN Deep-Generative Baseline

The QuantGAN deep-generative baseline is a repo-native approximation to the convolutional WGAN of Wiese et al. [10]: a 1D convolutional generator and critic (each three layers, channels 32, kernel size 5; latent dim $L = 8$, leaky-ReLU on the critic), Wasserstein loss with critic weight clipping at ± 0.01 [89], Adam at $\eta = 10^{-4}$, batch 64, 15 epochs $\times\, 120$ steps with four critic updates per generator update, trained on rolling windows of length $W = 64$ on standardised SPY growth rates and stitched end-to-end at synthesis. The implementation is Julia with `Flux.jl` [90]; seed 20260422. In the expanded panel (Table S23) this configuration reached 0% IS / OoS KS and IS excess kurtosis 2.1 vs. observed 7.68; the QuantGAN row in the main generator comparison (Table 2) is a separate temporal-convolutional run, also at 0% KS, with IS excess kurtosis 0.56. Both fall far short of the observed heavy tail, consistent with documented GAN failure modes on volatility-clustering benchmarks [39, 40]; the runner is materially smaller than the reference seven-block TCN with Lambert-W pre-processing in Wiese et al. [10], so the row is the panel's deep-generative negative control rather than a verdict on the QuantGAN literature.

A.8.4 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We added five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline to the single GARCH(1,1) Gaussian row of Table 2. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\text{IS}} = 2{,}516$); simulation follows the same recursion used in estimation.

Baselines fitted in this panel. The conditional-volatility rows comprise EGARCH(1,1) Gaussian [26], GJR-GARCH(1,1) [27, 28], GARCH(1,1) with Student-t innovations [25], HAR-RV [32] on daily squared demeaned growth rates as the RV proxy, and Markov-switching GARCH(1,1) [29] at $K \in \{2, 3, 4, 6\}$ with regime triples $(\omega_k, \alpha_k, \beta_k)$ plus softmax-normalised transition logits, fitted by grid-initialised Nelder-Mead and canonicalised by ascending unconditional variance. Higher-K MS-GARCH did not close the gap to the CHMM family on KS (IS KS plateaued at 27–37% across $K \in \{2, 3, 4, 6\}$ against CHMM-N's 94.1% at $K = 18$); the binding constraint is the unimodal-innovation per-regime structure (each regime's emission is Gaussian rescaled by the regime's conditional variance), which adding regimes does not relax. The CHMM decouples this constraint by giving each state its own emission-mixture component. Implementation details for every conditional-volatility row are in the companion code repository.

MS-GARCH reference Bayesian re-run via `MSGARCH`. As a Bayesian counterpart to the in-house Nelder-Mead MS-GARCH fit, we re-ran the same Markov-switching GARCH(1,1) specification through the reference `MSGARCH` R package of Ardia et al. [30] (version 2.51) at $K \in \{2, 3, 4\}$, driven from the Julia harness via `RCall.jl`. The fit is fully Bayesian (DEMC sampler with the package's default proper priors, 12,500 MCMC draws, 2,500 burn-in, thin 10, single chain); all 1,000 simulated paths are posterior-predictive (one path per retained posterior draw), so the simulated marginal integrates parameter uncertainty path-by-path. The reference Bayesian re-run reported lower KS pass rates than our in-house frequentist Nelder-Mead fit: 0.0% ($K = 2$), 0.1% ($K = 3$), 0.0% ($K = 4$) IS, and 5.8% ($K = 2$), 5.1% ($K = 3$), 5.3% ($K = 4$) OoS, against the in-house plateau of 27–37% IS / 33–39% OoS (Table S23). The gap is methodological rather than estimator-quality: the in-house implementation generates all 1,000 paths from one MLE point estimate, so the simulated marginal is tighter; the reference Bayesian implementation samples one set of parameters per path from the posterior, so the simulated marginal is wider and accordingly fewer paths fall within the asymptotic two-sample KS critical band. The posterior-mean log-likelihood at the in-sample data improved modestly from $K = 3$ to $K = 4$ but did not translate to a higher KS pass rate. The posterior-mean transition matrix was diagonally dominant at $K = 3$ and less so at $K = 4$, indicating the higher-K posterior was searching for additional structure that the data do not strongly support. The reference Bayesian re-run is therefore consistent with the in-house frequentist plateau at the same conclusion under either estimator: vanilla Gaussian-innovation MS-GARCH at $K \leq 4$ did not match the CHMM family on the main KS metric for this universe. Reproducibility: R 4.6.0, MSGARCH 2.51, all transitive R dependencies pinned via `renv` (`r_msgarch/renv.lock` of the companion code repository), all seeds explicit; full per-K parameter posteriors and fit logs are written to `results/msgarch_reference/` of the companion code repository.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The SM-CHMM is the conceptual foil of the present paper, in the spirit of the explicit-duration HSMM of Yu [80]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by raw fitted log-likelihood from {Pareto, negative binomial, geometric}. The plug-in estimator takes the Viterbi state path on the IS series under the converged CHMM fit, then refits per-state AR(1) emissions and a sojourn family chosen by fitted log-likelihood. On SPY IS at $K = 18$ the selected sojourn family was Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. This plug-in construction (single-shot Viterbi decoding under the converged CHMM fit, then AR(1)-residual + sojourn-family refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [80]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The plug-in's per-state variance estimates degenerate at $K = 18$, where several low-occupancy states collapse to near-zero-variance point masses, so we do not report its distributional pass-rates and retain only its VaR back-test behaviour in Table S23, where the heavy-tailed sojourns sharpen tail calibration. We do not claim that a jointly estimated HSMM would do the same; the multistart local-EM explicit-duration fit of the main table (Appendix A.7.17) covers SPY at $K \in \{3, 18\}$, and extending it across the ticker universe is left as a companion-paper direction. The low-state ML HSMM result of Bulla and Bulla [13] at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

The IS / OoS distributional fidelity, excess kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the QuantGAN deep-generative row, and the main CHMM panel, together with the SM-CHMM VaR back-test rows, are reported in Table S23.

No extended GARCH variant occupied the joint (KS, ACF) corner: GARCH(1,1)-t was the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS excess kurtosis of 15.1 overshot observed 7.68 by 2×; EGARCH and GJR-GARCH narrowed the asymmetry but not the main gap. The SM-CHMM rows under the Viterbi-AR(1) plug-in estimator contributed only their VaR back-test columns, where the heavy-tailed sojourns sharpened Kupiec tail calibration for the Gaussian and Student-t variants relative to the flat CHMM; the jointly estimated explicit-duration HSMM on SPY is in the main table, and its cross-ticker extension is left as a companion-paper direction. The QuantGAN row is the panel's deep-generative negative control: 0% IS / OoS KS, simulated excess kurtosis 2.1 vs. observed 7.68, ACF-MAE comparable to the i.i.d. baseline. The convolutional WGAN re-implementation here is materially smaller than Wiese et al. [10]'s reference seven-block TCN with Lambert-W pre-processing; a faithful reference re-run is a deferred follow-up.

Table S23. Extended baseline panel. SPY IS / OoS, $T_{\mathrm{IS}} = 2{,}516$, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. *Convention note:* this panel's $|G_t|$ ACF-MAE is the *pooled* population-curve estimand (simulated ACF averaged across paths before the MAE), whereas Table 2 uses the per-path estimand (mean of per-path MAEs, which includes per-path sampling noise); the same canonical grid-initialised GARCH(1,1) fit scores 0.0309 pooled and 0.0490 per-path (one fit, two estimands), so values must not be compared across the two tables. The CHMM-t row is the unpenalised per-state-ν fit. The SM-CHMM rows report "–" for the KS, AD, excess kurtosis, and ACF columns because the plug-in estimator's per-state variances degenerate at $K = 18$, where several low-occupancy states collapse to near-zero-variance point masses; only their VaR back-test columns are retained. The QuantGAN raw-ACF-MAE is "–" pending a full re-run of that baseline script.

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\lvert G_t\rvert$	G_t	br%	$\mathrm{LR}_{\mathrm{uc}}$	br%	$\mathrm{LR}_{\mathrm{uc}}$
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-t	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
MS-GARCH ref. B. $K = 2$	0.0	5.8	–	–	4.0	0.0465	0.0240	–	–	–	–
MS-GARCH ref. B. $K = 3$	0.1	5.1	–	–	4.5	0.0433	0.0241	–	–	–	–
MS-GARCH ref. B. $K = 4$	0.0	5.3	–	–	6.0	0.0446	0.0241	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	88.2	74.2	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	**91.9**	79.5	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	94.3	81.6	91.1	**82.0**	6.7	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	91.1	79.1	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	–	–	–	–	–	–	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	–	–	–	–	–	–	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	–	–	–	–	–	–	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

A.8.5 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

For completeness alongside the Markov-switching family of the main generator comparison, we added the canonical state-space and jump baselines that Table 2 does not exercise: a stochastic-volatility (SV-AR(1)) row in the lognormal log-AR(1) tradition [33, 91]; a Markov-switching multifractal row [92] at $\bar{k} = 8$ multipliers; and a Poisson-Gaussian jump-diffusion row [93]. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\mathrm{paths}} = 1{,}000$ pipeline used for the main panel, under the same global seed. Implementation lives in the companion `CHMM-Model` repository.

Reading. The stochastic-volatility row reproduced the IS excess kurtosis target almost exactly (7.52 vs. observed 7.68) and matched the CHMM-N $|G_t|$ ACF-MAE within 0.0004 (0.0466 vs 0.0462 at $K^\star = 3$); its KS pass rate was poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. The Merton jump-diffusion row was the inverse trade-off:

Table S24. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed $= 20260420$ as Table 2. Observed excess kurtosis: 7.68 IS, 5.29 OoS.

	KS (%) ↑		Excess kurtosis		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t \rvert$	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but excess kurtosis 3.12 undershot 7.68 by half and the absolute growth-rate ACF-MAE regressed to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 2) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. The multifractal row under the moment-matching fit reported here (closed-form ACF-of-log$|r|$ matching with $\bar{k} = 8$ binomial multipliers) did not converge to a viable generator on this $T_{\text{IS}} = 2{,}516$ window: the marginal excess kurtosis collapsed to 0.62 and KS dropped to 0%. A full HMM-style filter on the $2^{\bar{k}}$-state latent multiplier chain [92] is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM was dominated.

The main conclusion survived the addition of these three rows: no single competitor occupied the joint-fit corner that the CHMM family at $K = 18$ spans (KS $\geq 80\%$ and $|G_t|$ ACF-MAE ≤ 0.06 on every family row, with CHMM-L closing the excess kurtosis gap to within 1 unit of observed; Table S23). The SV-AR(1) model was the closest competitor on excess kurtosis and ACF but lost on KS; Merton-JD won on KS but lost on excess kurtosis and ACF; MSM under this fit was dominated.

A.8.6 Leverage-Effect Diagnostic

This appendix reports the Cont [5] leverage-effect column: the lag-1 correlation $\mathrm{Corr}(G_t, |G_{t+1}|)$ between today's signed growth rate and tomorrow's absolute growth rate, which is negative on equity returns. We computed the per-path distribution under each generator at $K = 18$, $N_{\text{paths}} = 500$, seed $= 20260420$.

Table S25. Leverage effect $\mathrm{Corr}(G_t, |G_{t+1}|)$**, per-path distribution.** Observed values: SPY IS -0.135, OoS -0.214. Per-generator median plus $[Q_5, Q_{95}]$ across $N_{\text{paths}} = 500$. A generator captures the leverage effect when its $[Q_5, Q_{95}]$ envelope brackets the observed value. Bold marks generators whose $[Q_5, Q_{95}]$ envelope brackets the observed IS value; the OoS observed value of -0.214 is more negative than every generator's Q_5 in the panel, indicating the OoS regime carries leverage stronger than any generator (including asymmetric GARCH) reproduces under static IS-fitted parameters.

	IS (observed -0.135)		OoS (observed -0.214)	
Generator	median	$[Q_5, Q_{95}]$	median	$[Q_5, Q_{95}]$
i.i.d. bootstrap	$+0.000$	$[-0.034, +0.030]$	$+0.004$	$[-0.069, +0.068]$
GARCH(1,1) Gaussian	-0.004	$[-0.066, +0.064]$	-0.003	$[-0.112, +0.107]$
EGARCH(1,1)	$\mathbf{-0.093}$	$[\mathbf{-0.147}, \mathbf{-0.047}]$	-0.087	$[-0.199, -0.002]$
GJR-GARCH(1,1)	$\mathbf{-0.079}$	$[\mathbf{-0.153}, \mathbf{-0.017}]$	-0.073	$[-0.174, +0.023]$
CHMM-N ($K = 18$)	$\mathbf{-0.089}$	$[\mathbf{-0.138}, \mathbf{-0.037}]$	-0.087	$[-0.190, +0.015]$
CHMM-t ($K = 18$)	-0.067	$[-0.116, -0.020]$	-0.067	$[-0.166, +0.019]$
CHMM-L ($K = 18$)	-0.065	$[-0.109, -0.020]$	-0.064	$[-0.159, +0.021]$
CHMM-GED ($K = 18$)	-0.072	$[-0.113, -0.029]$	-0.076	$[-0.163, +0.025]$

Reading. The CHMM family at $K = 18$ produced a non-trivial negative leverage signal despite symmetric per-state emissions: state-mixing on $\hat{\mathbf{T}}$ asymmetrically couples high-σ_k states to negative-μ_k states, carrying $\sim$ 65% of the IS observed leverage magnitude. The CHMM-N IS interval $[Q_5, Q_{95}] = [-0.138, -0.037]$ bracketed the IS observed value -0.135 at the lower boundary, as did the two asymmetric GARCH variants (EGARCH and GJR-GARCH), which carry an explicit leverage term; CHMM-GED was the second-strongest CHMM entry by median without bracketing. The OoS observed leverage of -0.214 exceeded every generator's Q_5 (including asymmetric GARCH); closing this residual gap requires skew-emission CHMM extensions or refit to the OoS distribution, both deferred to companion work.

A.8.7 Iterated Pseudo-Likelihood Student-t Copula Estimator

A natural concern with the main text two-step copula estimator (Kendall's-τ inversion for $\hat{\Sigma}$, then profile MLE on ν with $\hat{\Sigma}$ held fixed) is that it may be biased toward the Gaussian limit when the marginal excess kurtosis differs across assets. We addressed this with an iterated pseudo-likelihood estimator that alternates over (Σ, ν) on the same six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T_{\text{IS}} = 2{,}516$, seed 20260420): a bracketed golden-section maximisation of the copula log-likelihood over ν at the current Σ, alternating with a method-of-moments refit of Σ at the current ν on the t-quantile-transformed pseudo-uniform sample with PSD projection. Iteration terminates at $|\Delta \log L| < 10^{-3}$. The ν step is a likelihood maximisation, but the Σ step is a moment substitution rather than an argmax, so the alternation is not a joint MLE and need not increase the likelihood at every step; we report it as a robustness check on the two-step estimator rather than as a gold-standard joint maximiser.

Table S26. Two-step versus iterated pseudo-likelihood Student-t copula estimation on the six-asset US-equity universe. The off-diagonal correlation differences are reported as the maximum absolute difference $\max_{ij} |\Delta\rho_{ij}|$ across the $\binom{6}{2} = 15$ pairs and the median absolute difference; both are sub-0.025, so the main text $\nu^\star = 6$ choice on $\hat{\Sigma}$ obtained by Kendall's-τ inversion is robust to the estimator switch.

Estimator	$\hat{\nu}$	log-likelihood
Two-step construction	6.00	6157.47
Iterated pseudo-likelihood	**6.40**	**6163.02**
log-L improvement		+5.55
$\max \lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.024
median $\lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.011
$\hat{\nu}_{\text{full}}$ inside the Wilks 95% CI $[6, 7]$?		Yes

The full MLE moved $\hat{\nu}$ from 6.00 to 6.40 (+5.55 log-L) with all $|\Delta\rho_{ij}| \leq 0.025$ across the 15 pairs (Table S26); the two-step estimator is therefore not detectably biased toward the Gaussian limit and $\nu^\star = 6$ is robust to the estimator switch.

A.9 Cross-Asset Dependence: Full Multi-Asset Panel

This appendix collects the full multi-asset detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.9.1 Cross-Asset Dependence Models: Derivations and Diagnostics

This appendix provides the technical background for the SIM and copula constructions used in the main text cross-asset analysis. These constructions are the multi-asset layer of the empirical study: per-asset marginals come from the independent CHMMs of the single-asset cross-ticker analysis, and joint dependence is overlaid on top of those marginals.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [47, 48] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so each column's empirical marginal is preserved exactly [51]. The same permutation property is the construction's temporal limit: because $\mathbf{U}$ is sampled independently across rows, the reordering permutes each asset's simulated CHMM path in time, so the composed columns do not retain the serial dependence (volatility clustering, regime residence) of the underlying single-asset paths. Every cross-asset result in this appendix is therefore a statement about per-asset marginals and contemporaneous dependence, not about joint temporal dynamics.

Kendall's τ inversion. Given IS growth rates $\mathbf{G} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi \tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [50, 94]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density [49] at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\begin{aligned} \log c_t(\mathbf{u}; \Sigma, \nu) \;=\; & \log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\,\log \Gamma\!\left(\tfrac{\nu}{2}\right) - d\,\log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2}\log|\Sigma| \\ & - \tfrac{\nu+d}{2}\,\log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\,\log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \end{aligned} \tag{30}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (30) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In the main text cross-asset analysis the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we compute the Cholesky factor $\Sigma = LL^\top$, draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, then return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. The fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in the main text cross-asset analysis are reported in Table S27. The high $R^2 = 0.840$ for QQQ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while the lower $R^2 = 0.260$ for JNJ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table S36.

Table S27. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.9.2 Student-t Copula Profile Log-Likelihood

The profile log-likelihood of the Student-t copula used in the multi-asset construction, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in the main text cross-asset analysis, is plotted in Figure S5. The curve was smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum. The separation from the Gaussian copula limit ($\nu \to \infty$) is established by the boundary-appropriate likelihood-ratio test with a simulated Gaussian-copula null reported in the half-unit-grid analysis (LR = 697.2, Monte-Carlo $p = 0.002$).

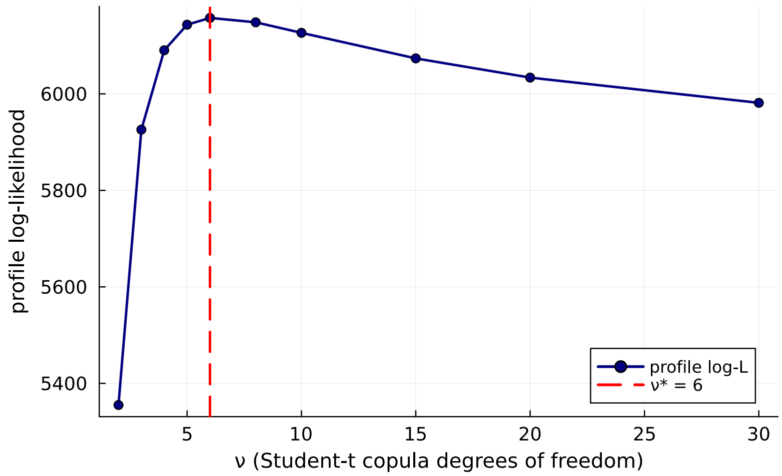
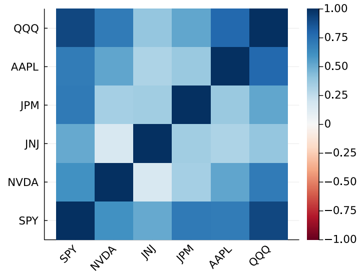


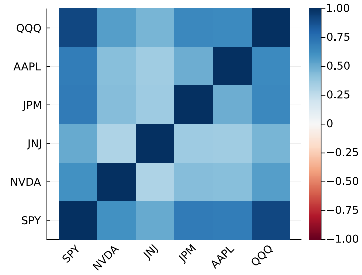
Figure S5. Profile log-likelihood of the Student-t copula used in the multi-asset construction on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The profile uses rank-PIT pseudo-observations while ν is varied; fitted CHMM marginals do not enter this calculation.

A.9.3 Cross-Asset Correlation Heat Maps (Companion to Table S36)

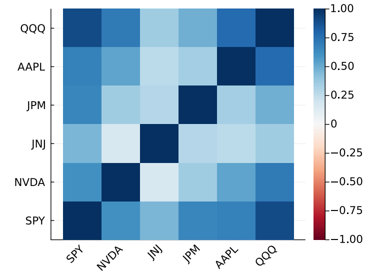
The observed and path-averaged simulated correlation matrices for the dependence models of the cross-asset analysis are visualised in Figure S6. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table S36. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K^\star = 3$, so any visible differences between the simulated panels reflect only the dependence construction.



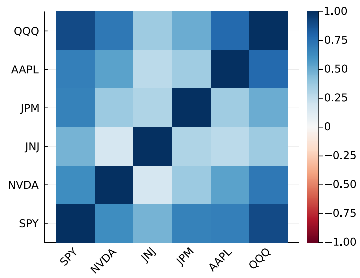
(a) Observed.



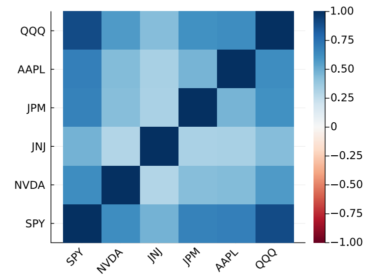
(b) Single Index Model.



(c) Gaussian copula.



(d) Student-t copula, $\nu^* = 6$.



(e) Truncated C-vine.

Figure S6. Cross-asset IS correlation reproduction across dependence models (multi-asset construction, companion to Table S36). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess growth rates for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\text{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K^\star = 3$ from Table 4; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.9.4 Truncated C-vine Copula on CHMM Marginals

The truncated C-vine variant in panel (e) of Figure S6 uses SPY as the root and selects each pair-copula edge from Gaussian and Student-t families; every selected pair was Student-t. On the same CHMM-N marginals at $K = 3$ and 200 paths the truncated C-vine attained an off-diagonal MAE of 0.068 on IS and 0.236 on OoS (Table S36), strictly above both elliptical copulas. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, a stronger restriction than either elliptical copula. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d.

A.9.5 Family-Versus-Refit Copula Experiment on the OoS Window

The static IS fit gave OoS off-diagonal MAE 0.209 against IS 0.027 (Table 4); we attributed the gap to the stationarity-scope limit and recommended periodic refit. Two distinct design choices could drive the OoS dependence error, the copula *family* (Gaussian versus Student-t) and the *refit policy* (static IS fit versus quarterly rolling refit), and comparing their effects requires measuring both on the same estimand. This experiment therefore crosses the two factors: four arms, $\{\text{static}, \text{rolling}\} \times \{\text{Gaussian}, \text{Student-}t\}$, all scored on the *identical* sequence of per-quarter targets. The OoS span is partitioned into nine complete blocks of 63 trading days plus the final 5-day partial block; every block's simulated panel (200 paths, strict common random numbers across all four arms: identical base normals and identical per-asset marginal draws per block, with the Student-t chi-square mixing variates drawn from a separate component stream the Gaussian arms never consume) is scored against that block's realised correlation matrix. The partial block is scored and reported but excluded from the paired inference: a six-asset sample correlation matrix estimated from five observations is rank-deficient and target-noise-dominated, and equal-weighting it against 63-day blocks would distort the paired comparison. The per-asset CHMM-N marginals at $K^\star = 3$ are held fixed at the Table 4 IS fits in *all four* arms, so only the dependence layer moves, and quarter-to-quarter target difficulty cancels in the paired per-quarter differences. Rolling arms refit on the trailing 252 days ending the session before the block starts; each Student-t refit yields a Kendall's-τ correlation matrix and a profile-MLE-selected $\nu^\star$ on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. The per-quarter detail is reported in Table S28.

Table S28. Family-versus-refit copula experiment on the OoS window: off-diagonal MAE per quarter for the four arms $\{\text{static}, \text{rolling}\} \times \{\text{Gaussian (G)}, \text{Student-}t\}$ (six-asset universe; CHMM-N marginals at $K^\star = 3$ held fixed at IS fits in all arms; 200 paths per simulation under strict per-quarter common random numbers across arms; seed root 20260422). All arms are scored against the identical realised per-quarter correlation targets; per-quarter MAE levels are higher than the full-window static value of Table 4 because 63-day realised correlation matrices are noisier targets than the single full-window matrix. The 5-day partial block ($*$) is scored but excluded from the mean row and the paired inference. For reconciliation, the full-window static construction of Table 4 recomputed inside this harness gives 0.208 (Student-t) and 0.204 (Gaussian); those anchors take no part in the paired comparison.

Quarter	Forecast block	Days	Refit $\nu^\star$	t static	t rolling	G static	G rolling
1	$[2517, 2579]$	63	15	0.221	0.198	0.217	0.197
2	$[2580, 2642]$	63	15	0.341	0.212	0.337	0.209
3	$[2643, 2705]$	63	10	0.309	0.237	0.303	0.236
4	$[2706, 2768]$	63	6	0.285	0.136	0.285	0.125
5	$[2769, 2831]$	63	6	0.243	0.249	0.238	0.243
6	$[2832, 2894]$	63	6	0.235	0.236	0.235	0.231
7	$[2895, 2957]$	63	5	0.273	0.143	0.271	0.131
8	$[2958, 3020]$	63	5	0.273	0.121	0.270	0.108
9	$[3021, 3083]$	63	6	0.195	0.144	0.190	0.133
10*	$[3084, 3088]$	5	10	0.513	0.556	0.497	0.557
mean (complete)	–	–	–	0.264	0.186	0.261	0.179

Measured on the same quarter-level estimand, the refit effect is an order of magnitude larger than the family effect. The quarterly refit reduced the mean per-quarter off-diagonal MAE from 0.264 to 0.186 within the Student-t family (mean paired difference 0.077, paired $t(8) = 3.62$, two-sided $p = 0.007$, 95% interval $[0.028, 0.127]$, 7/9 complete quarters favouring the refit) and from 0.261 to 0.179 within the Gaussian family (mean difference 0.082, $t(8) = 3.63$, $p = 0.007$, interval $[0.030, 0.133]$, 8/9 quarters). The family effect on the identical targets is small and slightly *favours the Gaussian* copula: Student-t minus Gaussian is 0.003 in the static arms ($t(8) = 3.86$, $p = 0.005$, 7/9 quarters) and 0.007 in the rolling arms ($t(8) = 4.42$, $p = 0.002$, 9/9 quarters). Because the strict common-random-number design eliminates simulation noise from the within-quarter contrasts, the tiny family effect is precisely resolved (nominally significant despite being 11–27 times smaller than the refit effect), which is exactly the reading the design supports: the family choice moves the error by a few thousandths, the refit policy by nearly a tenth. With only nine time blocks, overlapping 252-day estimation windows across consecutive refits, and no dependence correction, these paired-t results are suggestive rather than confirmatory; the per-quarter differences in Table S28 are the primary evidence. The refit improvement was concentrated in the mid-to-late complete quarters (Q4, Q7, Q8 improve by ≥ 0.13 in both families) where the rolling window incorporates more recent OoS data, while Q5 and Q6 were flat and the excluded partial block Q10 is target-noise-dominated in every arm. The refitted $\nu^\star$ sequence $(15, 15, 10, 6, 6, 6, 5, 5, 6)$ tracked the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the quarterly-refit recommendation in the main-text Discussion. The residual level (rolling Student-t mean 0.186 against IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table S30: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter's start, only track it once it has unfolded.

A.9.6 Non-Overlapping Six-Name Comparison Basket

The main six-asset universe contains overlapping ETFs and constituents (SPY holds NVDA, JNJ, JPM, and AAPL; QQQ holds AAPL and NVDA), which concentrates the dependence structure in strong positive co-movement: the observed mean absolute off-diagonal correlation on the IS window is 0.529. To test the composition away from that overlap we repeated the pipeline on six single names, one representative from each of six selected GICS sectors, with no ETFs and no cross-holdings: MSFT (Information Technology), UNH (Health Care), BAC (Financials), CAT (Industrials), PG (Consumer Staples), and XOM (Energy), whose IS mean absolute off-diagonal correlation is 0.361. Per-asset CHMM-N marginals at $K^\star = 3$, Kendall's-τ inversion, rank reordering, 200 paths, seed 20260422 (runner `runners/cross_asset/run_nonoverlap_basket.jl` of the companion code repository; artifact `results/robustness/nonoverlap_basket.csv`).

Table S29. Non-overlapping six-name basket (one large-cap from each of six selected GICS sectors, no ETFs; CHMM-N marginals at $K^\star = 3$; 200 paths; seed 20260422). Per-asset KS pass rates at $\alpha = 0.05$ and off-diagonal correlation MAE for the Student-t ($\nu^\star = 6$) and Gaussian copulas on the IS and OoS windows.

Asset	t IS	t OoS	Gauss IS	Gauss OoS
MSFT	90.5%	91.0%	91.5%	89.5%
UNH	99.5%	21.5%	98.5%	18.0%
BAC	96.5%	82.5%	98.5%	84.5%
CAT	97.5%	82.5%	98.5%	86.5%
PG	100.0%	97.5%	99.5%	95.5%
XOM	78.5%	57.0%	78.5%	65.0%
Off-diag MAE	0.031	0.253	0.032	0.248

Three observations. First, the composition is not an artefact of the overlap-easy universe: on the weaker-dependence basket the in-sample off-diagonal MAE (0.031 Student-t, 0.032 Gaussian) is comparable to the main basket's 0.027, and profile MLE again selected $\nu^\star = 6$. Second, the static-fit OoS degradation is worse here (0.253 against 0.209 on the main basket), consistent with the stationarity-scope reading: six independent single names carry more idiosyncratic regime drift than an ETF-anchored cluster. Third, the per-asset failure pattern reproduces the single-name regime-introduction mechanism documented in the cross-ticker panel: UNH, one of the named regime-introducing tickers of the 30-ticker analysis, collapses OoS (21.5% KS) while the other five names stay between 57% and 97.5%. The Student-t versus Gaussian OoS gap on this basket, like the main basket's, is small but stable across the paired replicates of the next paragraph (mean +0.0080 in favour of the Gaussian); the in-sample ordering is the reverse (0.031 against 0.032) and is reported descriptively.

Across-seed uncertainty of the Student-t versus Gaussian OoS gap. The main-text statement that the Gaussian copula's static OoS off-diagonal MAE sits slightly below the Student-t's is a paired Monte-Carlo measurement rather than a single-draw comparison. The marginal and copula fits are deterministic given the data (the quantile-initialised EM and the Kendall-τ/profile-MLE estimators consume no random draws, which we verified by fitting under deliberately perturbed generator states and comparing all parameters exactly), so the replication holds exactly the headline parameterisation fixed and repeats only the 200-path simulation over 20 fixed seeds. Within each replicate the two copulas are simulated under strict common random numbers: both consume identical base normals and identical marginal-path draws, and the Student-t's chi-square mixing variates come from a component stream the Gaussian never touches (runner `runners/cross_asset/run_seed_uncertainty.jl` of the companion code repository; the seed-level output is not vendored in this repository). On the main basket the Student-t mean was 0.2089 (sd 0.0016) against 0.2034 (sd 0.0011) for the Gaussian, a paired gap of +0.0055 (sd 0.0011) with 95% t-interval $[+0.0050, +0.0060]$; on the non-overlapping basket the corresponding gap was +0.0080 (sd 0.0015) with interval $[+0.0073, +0.0088]$. Both intervals exclude zero, so the gap is stable across replicates at this N_{paths} rather than path-to-path noise. These intervals are Monte-Carlo precision for this fixed-fit simulation design, not sampling uncertainty about copula performance on new data; the gap is in any case a fraction of the paired static-versus-refit difference (0.077–0.082 mean per-quarter improvement on the main basket; Table S28) and does not alter the reading that the quarterly refit, not the dependence family, dominates the static OoS error.

A.9.7 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extended the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, a non-equity commodity asset class.

Single-asset univariate fidelity of GLD. We fitted CHMM-N, CHMM-t, and CHMM-L on the GLD growth-rate series at $K = 18$; in sample all three families passed cleanly, while out of sample all three collapsed. All three emission families passed in sample, but the OoS KS pass rate collapsed for every family as the observed marginal moved outside the in-sample regime range. Inspection of the GLD price path on the 2024–2026 OoS window showed the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in the main text cross-ticker analysis, on a non-equity asset; the limitation is not equity-specific.

Quarterly-refit follow-up on GLD. A quarterly-refit version of GLD CHMM-N at $K^\star = 3$ (5y rolling estimation window, 63-day refit cadence, 200 paths; `runners/cross_asset/run_non_us_asset_quarterly_refit.jl`) lifted the OoS KS pass rate only marginally and did not recover the failure mode. Periodic refit therefore did not recover the GLD failure mode in any meaningful sense: the model is architecturally mis-specified for the persistent demand shock that re-priced gold

Table S30. Per-pair off-diagonal correlation MAE on the seven-ticker universe (multi-asset construction; Student-t copula on CHMM-N marginals at $K^\star = 3$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports the absolute deviation $|\Sigma_{\text{sim,avg}} - \Sigma_{\text{obs}}|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\lvert\Delta\rvert$	OoS $\lvert\Delta\rvert$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

across the OoS window, and the failure is not just staleness of the static fit. The periodic-refit recommendation of the main-text Discussion does not extend to non-equity assets under regime introductions of this magnitude.

Multi-asset: 7-ticker copula MAE. Refitting the Student-t copula on the augmented seven-ticker universe left the IS fit essentially unchanged and *improved* the OoS fit relative to the six-ticker baseline. At $\nu^\star = 6$ the IS off-diagonal MAE stayed close to the six-ticker baseline (0.029 against 0.027; Table 4), and the OoS off-diagonal MAE fell below it (0.179 against 0.211 as recomputed inside this run's harness; the main-text static value is 0.209). Adding the gold ETF, with its low equity correlation, reduced the OoS off-diagonal MAE because the equity-cluster correlation degradation is the dominant component; pairs involving GLD sat at moderate OoS deviations (between 0.04 and 0.19) while the largest equity-cluster pairs (JNJ–QQQ, SPY–JNJ, NVDA–JNJ) sat above 0.48 on OoS (Table S30).

Per-pair OoS gap localisation. The per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$, is reported in Table S30. The OoS gap concentrated in pairs involving JNJ (the defensive single-name proxy): the JNJ IS correlation with the broad equity factor (QQQ, SPY) collapsed on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sat in the middle of the ranking with OoS deviations between 0.10 and 0.19. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

A.9.8 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for $\nu^\star$

To confirm that the upper bound of the unit-grid Wilks CI on $\nu \in [3, 12]$ is not grid-induced, we computed a half-unit-grid refinement of the profile MLE plus a parametric bootstrap CI. We refit the Student-t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \ldots, 12.0\}$ (19 grid points) and ran a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student-t copula at the fitted $(\hat{\Sigma}, \nu^\star)$, refitting the Kendall's-τ

correlation and the profile MLE on each replicate.

The half-unit grid moved the optimum from $\nu^\star = 6.0$ (unit grid) to $\nu^\star = 6.5$ at profile log-likelihood 6,158.0 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid was $[\nu_{\mathrm{lo}}, \nu_{\mathrm{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI was $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit-$\nu^\star$ distribution; bootstrap median $\nu^\star = 6.5$). This bootstrap quantifies the sampling variability of $\nu^\star$ conditional on the Student-t family: every replicate is generated at the fitted finite $\nu^\star = 6.5$ and refit on the finite grid, so the interval $[6.0, 7.0]$ is not, by itself, a test against the Gaussian-copula limit ($\nu \to \infty$), which lies outside both the generating model and the refit grid. *Wilks-theorem regularity caveat.* Wilks's theorem likewise requires the parameter under test to lie in the interior of the parameter space and the model to be regular [95, 96]; for the Gaussian-copula limit the regularity condition fails (the limit lies on the boundary of the natural parameter space), so the Wilks 95% CI is an interior-grid CI for finite ν, not a hypothesis test against the Gaussian limit. The Gaussian-vs-Student-t question is instead settled by a boundary-appropriate likelihood-ratio test with a simulated null: with Σ estimated by Kendall's τ under both hypotheses, the observed statistic $\mathrm{LR} = 2\,[\max_\nu \log L_t(\nu) - \log L_{\mathrm{Gauss}}] = 697.2$ ($\hat{\nu} = 6.5$ on a grid extended to $\nu = 200$), while $B = 500$ parametric simulations from the fitted Gaussian copula gave null quantiles $q_{50}/q_{95}/q_{99} = -1.1/2.8/4.3$ (with 92.8% of null replicates maximising at the grid top), for a Monte-Carlo p-value of 0.002. On the IS window the Student-t copula's fit improvement over the Gaussian copula is therefore far outside Gaussian sampling variability; this in-sample distinction coexists with the OoS off-diagonal MAE comparison (0.209 vs. 0.204), whose gap sits inside the *unpaired* path-to-path simulation-noise floor at $N_{\mathrm{paths}} = 200$ but is resolved as a stable $+0.0055$ paired difference (95% t-interval $[+0.0050, +0.0060]$) under the common-random-numbers replication of Section A.9.6; both statements concern the downstream correlation metric rather than copula fit, and they are compatible because pairing cancels the shared path noise that dominates the unpaired comparison.

A.10 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. A multi-objective selection combining information-criterion (IC) rank with IS and OoS distributional pass rates would overlap the 2024–2026 window later used for VaR (and, at face value, favours $K = 18$). The K-selection design used here therefore decouples the two windows: we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2{,}013$) and scored every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 503$, including the boundary return into the first validation session); the 2024–2026 window is not touched, and all K-selection designs in this block use the session-VWAP growth-rate series of the headline analysis. BIC and CAIC selected $K^\star = 3$ over both the smaller $K = 2$ and all larger state counts (BIC 7745 at $K = 3$ vs. 7832 at $K = 2$ and 7848 at $K = 6$); AIC selected $K = 6$, and HQC selected $K = 6$ by a narrow margin (7699 vs. 7703 at $K = 3$). The held-out per-observation log-likelihood was nearly flat across the $K = 3$ to $K = 9$ band, nominally peaking at $K = 9$ (-2.2840, against -2.2999 at $K = 3$ and -2.2917 at $K = 6$, a spread of 0.016 nats per observation); held-out two-sample KS nominally selected $K = 2$, but this validation slice is a structural-break period on which every state count scores poorly (2–5%), so its KS ranking is not discriminating (Table S31). No criterion selected $K = 18$. Extending the search down to $K = 2$ does not change the selection: on this window $K = 2$ loses the held-out log-likelihood and every information criterion to $K = 3$, because its estimation log-likelihood is 66 nats worse (-3893 vs. -3827), far exceeding the six-parameter penalty saving. $K = 2$ minimises the penalised criteria only on the calmer pre-2020 sub-window, where the lighter tail lets two Gaussian components nearly suffice, itself consistent with K-selection tracking the marginal tail; on the full window $K = 2$ reproduces neither the heavy tail (simulated excess kurtosis 2.56 vs. observed 7.68) nor the IS KS (76.8% under the canonical-seed four-family fit vs. 91.5% at $K^\star = 3$; the six-initialisation replication ensemble of the sensitivity appendix puts the same $K = 2$ fit at 79.0%). The held-out re-selection therefore supports the main text's $K^\star = 3$ state count as the parsimony choice inside a near-flat $K = 3$ to $K = 9$ band; $K = 18$ is retained in the main text as an excess kurtosis-fidelity sensitivity reference rather than as the state count, because its earlier multi-objective justification weighted tail fidelity (excess kurtosis, AD) and downstream VaR behaviour alongside likelihood. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K-selection (no rate-hike confound). To check whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022–2023 slice) selects the same $K^\star$, we re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1{,}130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 378$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). The per-observation held-out log-likelihood was -1.9200 at $K = 3$, -1.8953 at $K = 6$, -1.9074 at $K = 9$, -1.9390 at $K = 12$, -2.0795 at $K = 18$, and -2.1027 at $K = 21$. **The held-out log-likelihood selected $K = 6$.** The held-out KS pass rate was high and flat on this calm slice (87–93% across the sweep, nominal peak at $K = 18$), so it is not discriminating here. AIC also selected $K = 6$, while BIC, HQC, and CAIC selected $K = 2$ with $K = 3$ immediately behind (BIC 4034 at $K = 2$ vs. 4053 at $K = 3$), reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out log-likelihood rose from $K = 3$ to a peak in the $K = 6$ to $K = 9$ band and then degraded, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favoured a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022–2023 slice.

k-fold rolling-origin pre-2020 K-selection. The single-fold pre-2020 result is one observation; to confirm that the $K^\star$ choice is not a sampling artefact, we report mean ± s.d. of held-out per-observation log-likelihood and held-out KS at $K \in \{3, 6, 9, 12, 18\}$ across multiple folds. We implemented four expanding-window rolling-origin folds (a five-fold design forces one fold to have only ~ 1 year of training, below the practical floor for $K = 18$ EM convergence on this dataset; averaging is per-observation so fold-length differences are not an issue). Folds: train 2014-01-03 through 2015-12-31 (~ 502 obs) and val 2016 (~ 251 obs); train through 2016 and val 2017; train through 2017 and val 2018; train through 2018 and val 2019.

Across the four folds the held-out log-likelihood was flat from $K = 3$ to $K = 6$ and degraded beyond. The aggregate per-observation held-out log-likelihood (mean / s.d. across folds; Table S31) was -1.7930 / 0.289 at $K = 3$, -1.7997 / 0.313 at $K = 6$, -1.8550 / 0.263 at $K = 9$, -1.9177 / 0.269 at $K = 12$, and -2.0614 / 0.213 at $K = 18$. The held-out KS pass rate (mean / s.d. across folds, very wide because Fold 2's 2017 validation year produced near-zero KS for every K as a low-variance "calm year" artefact) was 57.0/39.5% at $K = 3$, 60.4/37.3% at $K = 6$, 61.5/37.5% at $K = 9$, 58.7/38.9% at $K = 12$, and 61.6/41.4% at $K = 18$. Descriptive sampling-error reads on the held-out per-observation log-likelihood gave $K = 6$ vs $K = 3$ a mean diff of -0.0067 (pooled SE 0.151, approximate $z = -0.04$) and $K = 18$ vs $K = 6$ a mean diff of -0.262 (pooled SE 0.134, approximate $z = -1.95$); with four folds these are descriptive checks, not formal tests.

Robustness check at half-year cadence. To verify that the result is not an artefact of the one-year fold cadence, we re-ran the same diagnostic with six expanding-window folds at half-year validation cadence (covering 2017–2019 in non-overlapping six-month chunks; train windows ~ 3.0y to ~ 5.5y). The aggregate per-observation held-out log-likelihood (mean / s.d. across six folds) was -1.7670 / 0.329 at $K = 3$, -1.7398 / 0.335 at $K = 6$, -1.7857 / 0.293 at $K = 9$, -1.8320 / 0.273 at $K = 12$, and -1.9644 / 0.194 at $K = 18$. Descriptive sampling-error reads gave $K = 6$ vs $K = 3$ a mean diff of $+0.027$ (pooled SE 0.136, approximate $z = +0.20$) and $K = 18$ vs $K = 6$ a mean diff of -0.225 (pooled SE 0.112, approximate $z = -2.01$). The half-year design's $K = 6$ vs $K = 3$ sign flipped relative to the full-year design ($+0.027$ vs -0.0067), and at half-year cadence the paired picture is one-sided: $K = 6$ edged $K = 3$ on all six fold differences, by 0.009 to 0.060 nats per observation. The magnitudes remain small relative to fold-to-fold dispersion, the two cadences disagree in sign, and with $n = 4$ or 6 fold differences no formal significance claim is supportable in either direction; the cadence disagreement itself indicates that the $K = 6$ / $K = 3$ comparison is unstable at this fold count. The $K = 18$ vs $K = 6$ deterioration replicated at both cadences ($z = -2.01$ at half-year vs $z = -1.95$ at full-year, read descriptively).

Reading. The held-out per-observation log-likelihood comparison between $K = 3$ and $K = 6$ is unresolved by these designs: the mean differences are small, the sign flips between fold cadences (with the half-year cadence uniformly but mildly favouring $K = 6$), and four or six fold differences cannot support formal significance claims in either direction. The single-fold result that initially selected $K = 6$ is one realisation. The held-out KS pass-rate criterion is not informative on either design (between-fold s.d. is dominated by 2017 calendar-year artefacts that hit every K uniformly). We therefore fix the main state count at $K^\star = 3$ as a parsimony choice among empirically similar held-out scores rather than as a statistically established winner, retain the $K = 6$ block as a sensitivity reference, and note that every design agrees on the substantial $K = 18$ held-out disadvantage. The default $K^\star = 3$ also carries the lower state count for the risk-management use cases of Table 5.

HAC-corrected sampling-error checks on the paired diff series. A Newey–West Bartlett HAC variance ($h_{\mathrm{NW}} = \lfloor n^{1/3} \rfloor$) on the per-fold validation-log-likelihood difference series gave $z_{\mathrm{HAC}} = -0.24$ for $K = 6$ versus $K = 3$ at the four-fold cadence and $z_{\mathrm{HAC}} = +5.10$ at the six-fold cadence (the latter reflecting that all six half-year fold differences favoured $K = 6$ by a small margin), and $|z|_{\mathrm{HAC}} = 3.68$ and 2.95 for $K = 18$ versus $K = 6$. With only four or six fold differences, however, the normal and HAC approximations are too small-sample to support formal significance claims, and the sign disagreement between cadences on $K = 6$ versus $K = 3$ underlines that instability. We therefore treat these statistics as descriptive checks: the $K = 3$ / $K = 6$ separation is small and cadence-dependent, while both cadences showed a substantially larger held-out log-likelihood disadvantage for $K = 18$.

Penalised-ECM rate sweep. The bracket sensitivity is reported in the per-state ν_k diagnostics block of the algorithms appendix; here we report the complementary $1/\nu_k$ shrinkage rate sweep that underlies the main text's $\lambda = 20$ specification. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduced simulated excess kurtosis monotonically (Table S32): $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brought simulated excess kurtosis to ≈ 8 (close to the observed 7.68) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moved $(2.1 \to 4.89)$; upper bracket and median remained at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We defined six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold's train slice,

Table S31. Pre-OoS held-out K-selection for CHMM-N. Held-out per-observation log-likelihood (ℓ_{val}, higher is better) and two-sample KS pass rate by state count K, under four validation designs that never touch the 2024–2026 OoS window: a single-fold 2022–2023 slice ($n = 503$), a strictly pre-2020 single-fold slice ($n = 378$), four expanding-window rolling-origin folds, and six half-year-cadence folds (rolling designs report the across-fold mean, with s.d. in parentheses); all designs use the session-VWAP growth-rate series and include the boundary return into each validation window. No criterion selects $K = 18$; the $K = 3$ versus $K = 6$ comparison is small and cadence-dependent (mean diff -0.0067 four-fold, $+0.027$ six-fold; Newey–West HAC $z = -0.24$ and $+5.10$, the latter with all six paired fold differences mildly favouring $K = 6$), and with four or six fold differences these reads are descriptive, while $K = 18$ versus $K = 6$ is a clear log-likelihood disadvantage at both cadences (HAC $|z| = 3.68$ and 2.95). The main text adopts $K^\star = 3$ as a parsimony choice among empirically similar held-out scores; $K = 18$ is retained only as an excess kurtosis-fidelity reference.

	2022–23 ($n = 503$)		Pre-2020 ($n = 378$)		4-fold rolling		6-fold $\frac{1}{2}$y
K	ℓ_{val}	KS%	ℓ_{val}	KS%	ℓ_{val}	KS%	ℓ_{val}
2	−2.3075	5.4	−1.9247	91.0	−1.7928 (0.29)	61.3 (41.0)	−1.7741 (0.33)
3	−2.2999	4.4	−1.9200	90.4	−1.7930 (0.29)	57.0 (39.5)	−1.7670 (0.33)
6	−2.2917	3.0	−1.8953	87.2	−1.7997 (0.31)	60.4 (37.3)	−1.7398 (0.34)
9	−2.2840	3.2	−1.9074	90.8	−1.8550 (0.26)	61.5 (37.5)	−1.7857 (0.29)
12	−2.3062	2.2	−1.9390	92.0	−1.9177 (0.27)	58.7 (38.9)	−1.8320 (0.27)
15	−2.3029	2.2	−1.9698	90.4	–	–	–
18	−2.3136	2.4	−2.0795	93.0	−2.0614 (0.21)	61.6 (41.4)	−1.9644 (0.19)
21	−2.3508	3.0	−2.1027	90.8	–	–	–

Table S32. Penalised-ECM $1/\nu_k$ shrinkage-rate sweep (CHMM-t, $K = 18$, SPY in-sample). An exponential prior on $1/\nu_k$ at rate λ shrinks the lower degrees-of-freedom bracket toward the Gaussian limit; only the smallest per-state ν ($\nu_{\min}$) moves, while the median and upper bracket stay pinned at 50. The recommended $\lambda = 20$ brings simulated excess kurtosis close to the observed 7.68 at a $\approx$ 1pp IS KS cost.

Rate λ	$\nu_{\min}$	Sim. exc. kurtosis	IS KS%
0	2.10	14.30	95.7
5	3.31	11.19	95.8
20	4.89	8.43	94.7
50	6.46	5.41	96.3
100	9.90	3.96	95.6
200	50.0	3.67	95.7

and reported KS pass rate, simulated excess kurtosis, and $|G_t|$ ACF-MAE on the fold's test slice ($N_{\text{paths}} = 500$). Across the six folds (Table S33), CHMM-N at $K^\star = 3$ attained median (IQR) KS 62.1% $[7.2, 78.4]$ and $|G_t|$ ACF-MAE 0.0563 $[0.0552, 0.0592]$; the $K = 18$ excess kurtosis-fidelity sensitivity reference attained 67.7% $[8.2, 75.0]$ and 0.0542 $[0.0508, 0.0571]$. Two folds carried sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attained KS 61–83%, consistent with the main OoS pass rate. The main ranking on the four non-stress folds was window-robust: the two state counts sat within 5.6pp on median walk-forward KS and within 0.005 on $|G_t|$ ACF-MAE; on stress folds (COVID, rate hike) all CHMM rows dropped sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The main-text interpretation, that the CHMM is the joint-fit row in the panel under stationary OoS conditions with periodic refit recommended for production deployment, is unchanged under the walk-forward stress test.

Table S33. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. The KS pass rate is reported at $\alpha = 0.05$. Fold W2 covers the COVID drawdown; fold W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

	KS (%)		Kurt sim		ACF-MAE $\lvert G_t \rvert$	
Fold (test window)	$K = 3$	$K = 18$	$K = 3$	$K = 18$	$K = 3$	$K = 18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	$[7.2, 78.4]$	$[8.2, 75.0]$	$[2.51, 4.72]$	$[2.50, 4.35]$	$[0.0552, 0.0592]$	$[0.0508, 0.0571]$

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of the main text to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yielded sixteen $(K, \alpha, \text{family})$ rows on the OoS window of 573 forecasts (Table S34). No row was rejected by Kupiec, Christoffersen-ind, or Christoffersen-cc: $p_{\text{cc}} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. Thus the main-window conditional-coverage result was not specific to the Gaussian emission family; these non-rejections do not by themselves identify the latent-state mechanism as the cause of the result. The same sixteen rows under the higher-power Engle-Manganelli DQ test reproduce the main-text tail-tier reading: at $\alpha = 0.01$ every row except CHMM-N at $K^\star = 3$ ($p = 0.056$) is rejected, with p from 0.0005 to 0.030, while at $\alpha = 0.05$ no row is rejected ($p \geq 0.094$). The CHMM-N DQ rows are vendored at `results/robustness/engle_manganelli_dq.csv`; the remaining family rows of the sixteen-row DQ panel are not vendored in this repository and are regenerated by the companion code repository's back-test runner.

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table S33, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table S35. The conditional Christoffersen-cc was not rejected at $\alpha = 0.05$ on ten of twelve fold-K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrated on W2 (COVID, $p_{\text{cc}} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction escaped rejection on nine of twelve combinations: W2 failed at both K ($p_{\text{cc}} < 10^{-3}$) and W4 at $K = 18$ failed at $p_{\text{cc}} = 0.022$ on a LR_{ind} tail-event clustering rather than a LR_{uc} coverage miss ($\text{LR}_{\text{ind}} = 7.49$ at $\text{LR}_{\text{uc}} = 0.11$). The pass-rate was therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table S33. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 / W4 magnitude (a regime introduction the IS distribution does not span), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction was window-robust, supporting the main-text framing that the regime-conditional VaR construction extends beyond the main single-window result.

Table S34. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{\mathrm{OoS}} = 573$ forecasts, including the boundary return into the first held-out session; seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. No $(K, \alpha, \text{family})$ row is rejected by Kupiec, Christoffersen-ind, or Christoffersen-cc at the 5% test level.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$
CHMM-N	3	0.01	9	1.57%	−4.56	1.61	2.37	3.97
CHMM-N	3	0.05	36	6.28%	−2.87	1.84	0.04	1.88
CHMM-N	18	0.01	9	1.57%	−5.19	1.61	2.37	3.97
CHMM-N	18	0.05	26	4.54%	−3.02	0.27	0.52	0.79
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	−5.57	0.81	2.81	3.62
CHMM-t ($\lambda = 20$)	3	0.05	32	5.58%	−2.73	0.40	0.78	1.18
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	−6.23	2.63	1.98	4.61
CHMM-t ($\lambda = 20$)	18	0.05	29	5.06%	−3.14	0.00	1.40	1.40
CHMM-L	3	0.01	5	0.87%	−5.43	0.10	4.75	4.84
CHMM-L	3	0.05	26	4.54%	−2.62	0.27	2.23	2.50
CHMM-L	18	0.01	9	1.57%	−6.17	1.61	0.29	1.89
CHMM-L	18	0.05	32	5.58%	−2.82	0.40	0.78	1.18
CHMM-GED	3	0.01	7	1.22%	−5.72	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.41%	−2.54	0.20	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	−6.53	1.61	2.37	3.97
CHMM-GED	18	0.05	25	4.36%	−3.09	0.51	2.57	3.08

Benjamini-Hochberg adjustment across the back-test panels. Because the study reports many back-tests, we apply a Benjamini-Hochberg step-up adjustment [79] within each panel, treating the panel as the family (artifact `results/robustness/bh_fdr.csv`; runner `runners/var_backtest/run_bh_fdr.jl` of the companion code repository). On the 24-row walk-forward Christoffersen-cc panel the five raw rejections reduce to three at $q = 0.05$: both W2 rows at $\alpha = 0.01$ ($q < 10^{-3}$) and W2 at $K = 18$, $\alpha = 0.05$ ($q = 0.014$) survive, while W2 at $K = 3$, $\alpha = 0.05$ ($q = 0.065$) and the W4 tail-tier rejection at $K = 18$ ($q = 0.107$) do not. On the sixteen-row four-family Christoffersen-cc panel there are no raw rejections and therefore none after adjustment. On the sixteen-row four-family DQ panel, six of the seven raw rejections survive at $q = 0.05$ (the CHMM-L $K = 18$, $\alpha = 0.01$ row drops to $q = 0.069$); CHMM-N at $K^\star = 3$ is not rejected before or after adjustment. The multiplicity-adjusted picture therefore sharpens rather than weakens the descriptive reading: the W2 COVID fold is the robust failure, the single W4 tail-tier rejection is not BH-significant, and the tail-tier DQ separation across emission families survives the correction.

Per-ticker $\hat{\lambda}^\star$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six focal tickers (KS-degradation tolerance ≤ 1.5pp from $\lambda = 0$) gave per-ticker grid minimizers $\lambda^\star_{\mathrm{SPY}} = 10$, $\lambda^\star_{\mathrm{NVDA}} = 20$, $\lambda^\star_{\mathrm{JNJ}} = 20$, $\lambda^\star_{\mathrm{JPM}} = 10$, $\lambda^\star_{\mathrm{AAPL}} = 10$, $\lambda^\star_{\mathrm{QQQ}} = 0$: $\lambda = 20$ is a reasonable uniform default but per-ticker tuning is recommended whenever a ticker's residual excess kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Cross-asset summary. The full per-construction multi-asset panel is in the cross-asset appendix; for reference, the main ordering is summarised in Table S36.

Emission-family lookup. The four CHMM variants' per-state densities and parameter ranges are listed in Table S37 as a quick reference complementing the inline definitions in the main text.

Table S35. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t, conditional $\widehat{\mathrm{VaR}}_t(\alpha)$ is the α-quantile of the K-component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} \mid \mathcal{F}_t = \text{train} \cup \text{test}[1{:}t-1])$. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}
W1	3	0.01	2	0.80%	−4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	−2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	−5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	−2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	−6.54	32.93	0.87	33.80	**0.00**
W2	3	0.05	23	9.13%	−4.59	7.34	1.71	9.05	**0.01**
W2	18	0.01	14	5.56%	−6.67	25.59	1.56	27.15	**0.00**
W2	18	0.05	25	9.92%	−4.54	10.11	2.56	12.68	**0.00**
W3	3	0.01	6	2.39%	−4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	−2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	−5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	−3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	−7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	−4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	−9.62	0.11	7.49	7.60	**0.02**
W4	18	0.05	10	4.00%	−6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	−6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	−3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	−7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	−3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	−4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	−2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	−5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	−2.98	3.06	0.40	3.46	0.18

Table S36. Cross-asset dependence summary (multi-asset construction; CHMM-N marginals at $K = 3$; 200 paths; seed 20260422). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner in the off-diagonal MAE columns only: the three copulas preserve the same CHMM-N marginals by rank reordering, so their per-asset KS medians differ only by Monte-Carlo path noise and no KS winner is marked. The Student-t copula at $\nu^\star = 6$ attains the lowest IS off-diagonal MAE; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.204 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in the cross-asset appendix.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	71.0	84.2	0.077	0.252
Gaussian copula	88.8	84.2	0.029	**0.204**
Student-t copula ($\nu^\star = 6$)	89.5	81.5	**0.027**	0.209
Truncated C-vine (root SPY)	90.0	81.0	0.068	0.236

Stationarity-scope summary. Static-fit OoS performance and periodic-refit mitigation across the four independent stress sources tested in this paper are summarised in Table S38.

Table S37. The four CHMM emission families. Per-state density f_k with parameter ranges. The Gaussian and Laplace variants are nested special cases of CHMM-GED: $p_k = 2$ for all k gives Gaussian, $p_k = 1$ gives Laplace, both interior to the implemented range.

Variant	Per-state density f_k	Per-state parameter range
CHMM-N	$\mathcal{N}(\mu_k, \sigma_k^2)$	$\sigma_k > 0$
CHMM-t	$t_{\nu_k}(\mu_k, \sigma_k)$	$\sigma_k > 0$, $\nu_k \in [\nu_{\min}, \nu_{\max}]$
CHMM-L	$\mathrm{Laplace}(\mu_k, b_k)$	$b_k > 0$
CHMM-GED	$\mathrm{GED}(\mu_k, \alpha_k, p_k)$	$\alpha_k > 0$, $p_k \in [p_{\min}, p_{\max}]$

Table S38. Stationarity-scope summary. Static-IS-fit OoS performance and periodic-refit mitigation across the four independent stress sources tested in this paper.

Stress source	Static-fit OoS KS	Refit mitigation
Cross-ticker ($K^\star = 3$, 30 panel)	$11/30 < 60\%$	Quarterly: $8/30 < 60\%$
Cross-decade (2004 to 2006)	3 to 5%	slice < 600 d, untested
GLD non-equity stress	0%	Quarterly: 2.5%
Walk-forward W2, W4	$< 10\%$	no cadence closes